

Project:
**Analyze Data in a Model Car
Database with MySQL
Workbench**

Getting started – Overview of the Project

Skills You'll Showcase

- Import an existing database using MySQL Workbench.
- Familiarize yourself with a business and its data by reviewing a relational model diagram and exploring tables of data in MySQL Workbench.
- Analyze inventory data using SQL queries in MySQL Workbench that retrieve data from a multiple-table relational database using SQL commands such as: select, order by, where, group by, and having.
- Develop recommendations and suggestions for solving a business need/problem based on data analysis.
- Support your recommendations and suggestions for inventory reduction in the form of scripted queries.

What You'll Create

- By the end of this project, you'll have a file containing the following artifacts ready to upload:
- Detailed descriptions of a minimum of two suggestions for reducing inventory to allow for the closing of a storage facility.
- One or more SQL queries/scripts that support each of your suggestions.
- We'll take these outputs and help you create a detailed project overview, like the one shown below, to share with potential employers.

Step1: Required Files & Set-up

I started with setting up MySQL Workbench and importing the MySQL database in Power BI for visualizations:

MySQL set-up:

First thing I need to do was to install the MySQL Workbench in order to have a proper environment to work with. After MySQL Workbench is installed, configured, launched and connected to a local MySQL server instance, I had to download the “mintclassicsDB.sql” file, which is the Mint Classics relational database, and then import it to use the script (the script includes commands to create the schema) to create the Mint Classic database on my MySQL Workbench platform.

Importing the Database into Power BI

During this step, you might encounter a message stating, **"This connector requires one or more additional components to be installed before it can be used."** This means an extra component is needed for the connection. Simply follow the provided link, which will likely direct you to:

 [MySQL Connector Download](#)

Download the required component and, before attempting to import the database again, try refreshing Power BI or restarting the application. Once that's done, you can follow the detailed instructions in this video on **How to Connect MySQL Database to Power BI**, where the process is explained thoroughly:

 [YouTube Tutorial](#)

Project Scenario & Business Problem

Project Scenario:

Mint Classics Company, a retailer of classic model cars and other vehicles, is looking at closing one of their storage facilities. The main question is which one to close? Closing which one is more beneficial or less costly for the business?

To support a data-based business decision, they are looking for suggestions and recommendations for reorganizing or reducing inventory, while still maintaining timely service to their customers. For example, they would like to be able to ship a product to a customer within 24 hours of the order being placed.

Business problem:

It is important to understand the business problem before starting the analysis. Because it is not really possible to ask questions related to the data (which is the first step of analysis, at least for me) without knowing what problem we are trying to solve.

Understanding the Business Problem

Mint Classics Company sells classic model cars and other vehicles. They are considering closing one of their storage facilities (warehouses). However, they want to ensure that:

- Customers still receive their orders quickly (within **24 hours** of ordering). This is very important because we don't want to lose customers, which => losing sales.
- Inventory is **organized efficiently** to reduce storage costs.
- The company **does not lose sales** due to stock shortages or delays.

Understanding the Database

Understanding the Data

Another important step before analysis is the data we are working with. We need to be familiar with the database, what tables contain what information, which tables are more important for solving the business problem, how the tables are related (in the case of relational database, which is our case), etc.

After understanding the database, it is time to focus on solving the business problem. What can be analysed and what kind of insight might be interesting (ask questions, analyse, get answers, give business recommendations).

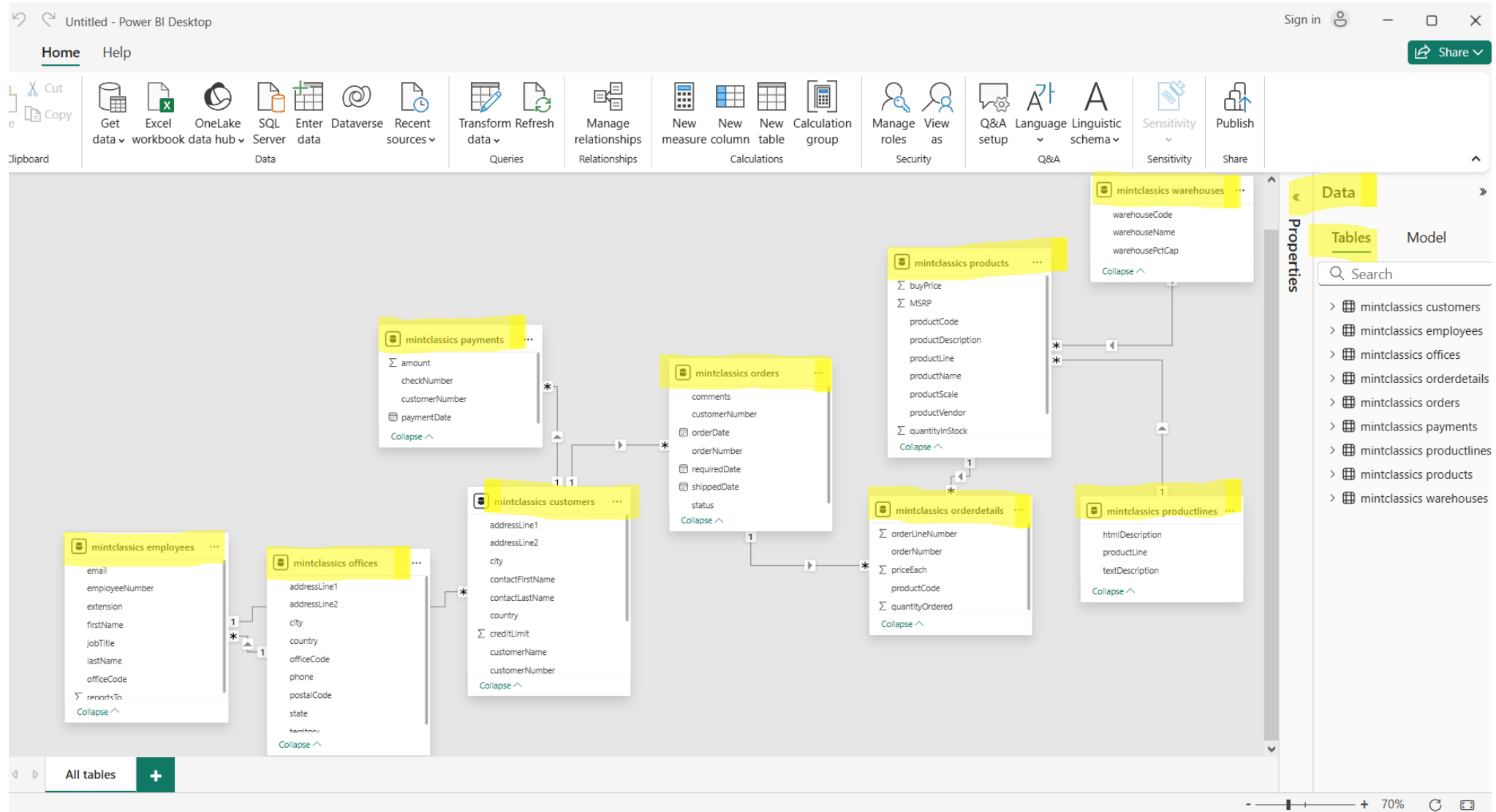
This step will continue with asking questions. Questions always help me go deeper, and this is helpful for analysing the data - seeking answers leads to continuous learning and deeper understanding.

Since we have 9 tables and an **ER diagram(check below)**, let's start by:

- Checking **how they are related** (primary keys, foreign keys)
- Understanding **what each table represents**
- Identifying **which tables are most relevant** to solving the problem

My goal -> Familiarize myself with the database and the business process: Before focusing on the business problem, I will focus on analysing the dataset, finding relationships between tables, answer some questions. This is, I will do general EDA (exploratory data analysis), answer general questions that might pop up for the dataset we have. This will help me to get familiar with the mintclassics dataset and to find relationships. Later, I can focus on solving the business problem.

Before moving into Step 2, I want to provide the snapshot of how our relational database or in other words the **EER (Extended Entity-Relationship)** diagram looks like on **Power BI** tool (table names are highlighted):



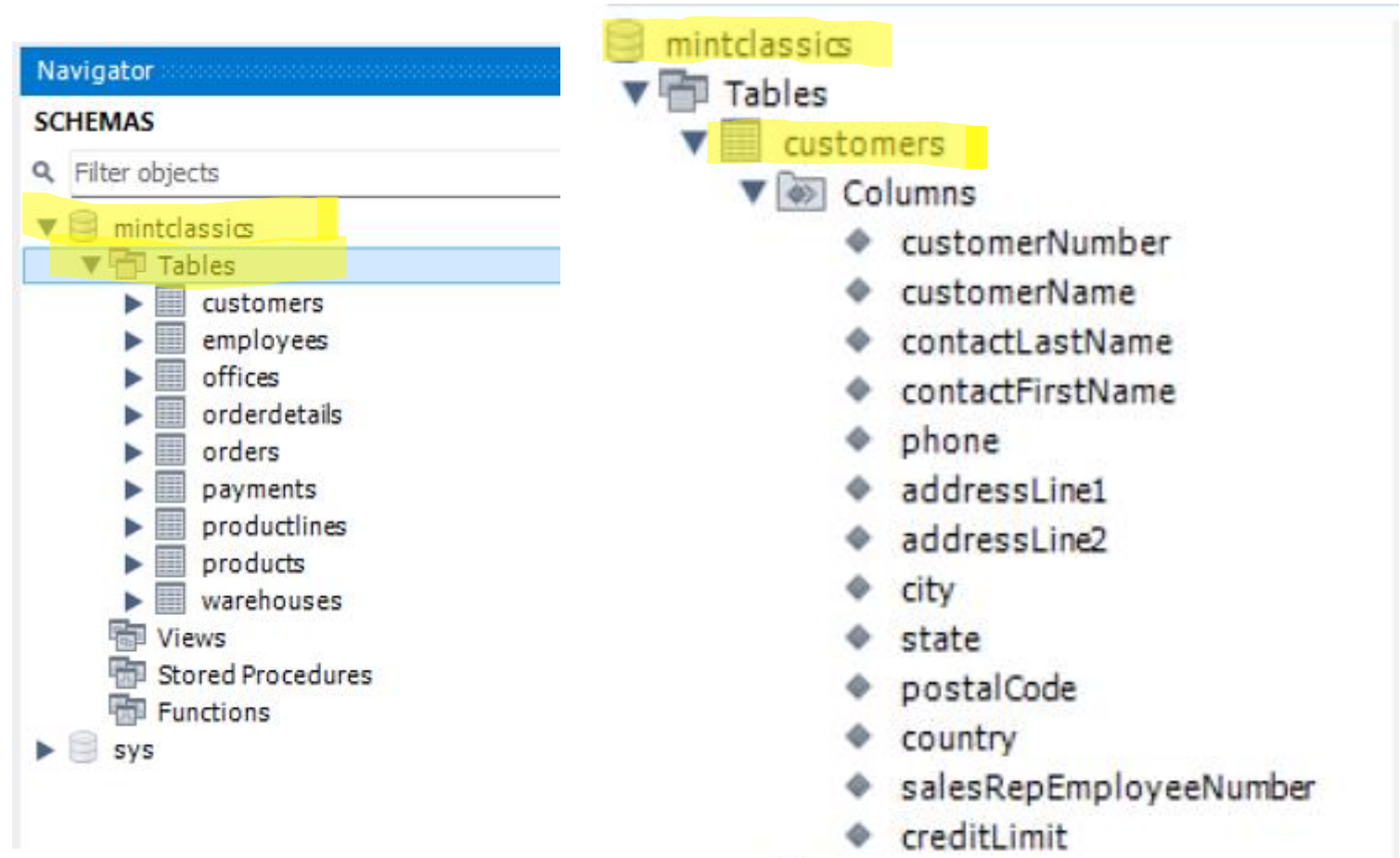
Step 2: Understanding Table Relationships

In this step, before starting the analysis, I usually take time to understand the project database, what kind of data we are working on, what tables do I have in this database, how they are related, what data they contain, are there any common information (column names might be different but the data it covers can be the same).

Our database is called **mintclassics** (it comes from the hypothetical company name **Mint Classics**), which contains 9 tables (customers, employees, offices, orderdetails, orders, payments, productlines, products, warehouses).

I will go through all the tables to understand what information they contain.

The first one is the **customers** table: this table contains data about the customer demographics. Note that our customers are companies but not individual person, but our contacts are people, of course. This table has 13 columns, which are given in the snapshot on the left:



customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1
475	West Coast Collectables ...	Thompson	Steve	3105553722	3675 Furth Circle
459	Warburg Exchange	Ottlieb	Sven	0241-039123	Walserweg 21
144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555	Berguvsvägen 8
181	Vitachrome Inc.	Frick	Michael	2125551500	2678 Kingston Rd.
298	Vida Sport, Ltd	Holz	Mihael	0897-034555	Grenzacherweg 237
201	UK Collectables, Ltd.	Devon	Elizabeth	(171) 555-2282	12, Berkeley Gardens Blvd
205	Toys4GrownUps.com	Young	Julie	6265557265	78934 Hillside Dr.
186	Toys of Finland, Co.	Karttunen	Matti	90-224 8555	Keskuskatu 45
259	Toms Spezialitäten, Ltd	Pfalzheim	Henriette	0221-5554327	Mehrheimerstr. 369
398	Tokyo Collectables, Ltd	Shimamura	Akiko	+81 3 3584 0555	2-2-8 Roppongi
450	The Sharp Gifts Warehouse	Frick	Sue	4085553659	3086 Ingle Ln.
328	Tekni Collectables Inc.	Brown	William	2015559350	7476 Moss Rd.
161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle

When checking the data in the columns of the **customers** table, the first 6 of them is all about the customer names, addresses, contacts and contact persons, and no Null values exist in these columns.

Whereas the other 7 columns contains some Null values. I will see later if these has any impact on my analysis.

About the information it contains, they are countries, cities and states where the customers/companies locate, their postal codes, companies credit limits (I need to check what this actually refers to), and salesRepEmployeeNumber (I guess this means Representative Employee in Sales (?))

addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
NULL	Nantes	NULL	44000	France	1370	21000.00
NULL	Las Vegas	NV	83030	USA	1166	71800.00
Level 3	Melbourne	Victoria	3004	Australia	1611	117300.00
NULL	Nantes	NULL	44000	France	1370	118200.00
NULL	Stavern	NULL	4110	Norway	1504	81700.00
NULL	San Rafael	CA	97562	USA	1165	210500.00
NULL	Warszawa	NULL	01-012	Poland	NULL	0.00
NULL	Frankfurt	NULL	60528	Germany	1504	59700.00
NULL	San Francisco	CA	94217	USA	1165	64600.00
NULL	NYC	NY	10022	USA	1323	114900.00
NULL	Madrid	NULL	28034	Spain	1370	227600.00
NULL	Luleå	NULL	S-958 22	Sweden	1504	53100.00
NULL	Kobenhavn	NULL	1734	Denmark	1401	83400.00

This table contains information about employee demographics, their email addresses, offices, job titles and the employee number of whom they report to. For example, Murphy Diane is the President with employee number 1002, and he has a NULL values in reportsTo column. This is because he reports to nobody, but others report to him, apparently, since he is a President of the company. I am not sure about the **extension** column for the moment.

I need to do the same check for other 8 tables (okaaay). The second table is the **employees** table, and contains 8 columns:

employees
Columns
employeeNumber
lastName
firstName
extension
email
officeCode
reportsTo
jobTitle
Indexes
Foreign Keys
Triggers

```
4 • SELECT * FROM mintclassics.employees;
```

Result Grid

Filter Rows:

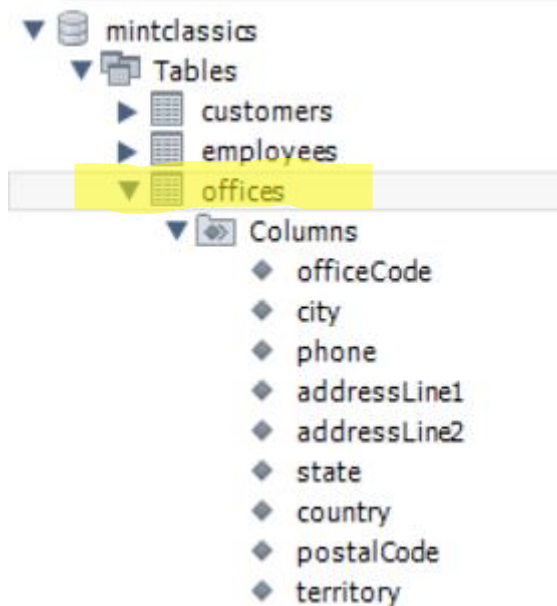
Edit:

Export/Import:

Wrap Cell Content:

	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
	1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
	1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
	1076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1002	VP Marketing
▶	1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
	1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)
	1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056	Sales Rep
	1611	Fixter	Andy	x101	afixter@classicmodelcars.com	6	1088	Sales Rep
	1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1088	Sales Rep
	1619	King	Tom	x103	tking@classicmodelcars.com	6	1088	Sales Rep
	1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep
	1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102	Sales Rep
	1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102	Sales Rep
	1501	Bott	Larry	x2311	lbott@classicmodelcars.com	7	1102	Sales Rep
	1504	Jones	Barry	x102	bjones@classicmodelcars.com	7	1102	Sales Rep
	1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1102	Sales Rep
	1165	Jennings	Leslie	x3791	ljennings@classicmodelcars.com	1	1143	Sales Rep

employees 1 ×



Offices table : contains 9 columns and 8 rows.

When we check the table, we can see that the business has total 7 offices in different countries; 3 offices in USA, the others in France, Japan, Australia, UK, 1 office for each.

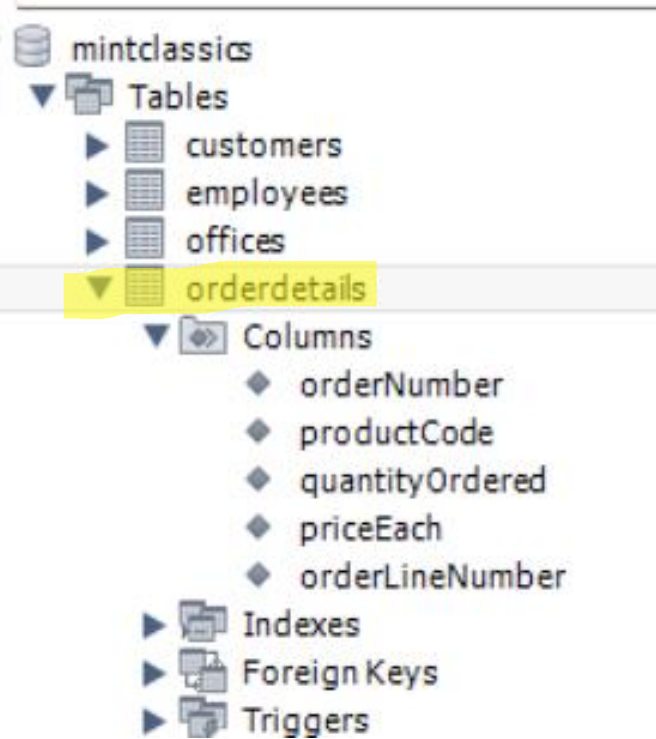
By having information from the **employees** and the **offices** table, we can find out in which **country** or **city** each employee works in.

```
4 • SELECT * FROM mintclassics.offices;
```

[illegible]

orderdetails table covers the information about the product code, quantity ordered, price for each product/productCode, order number, which seems like not unique. Need to check this out. Also, what orderLineNumber is about?

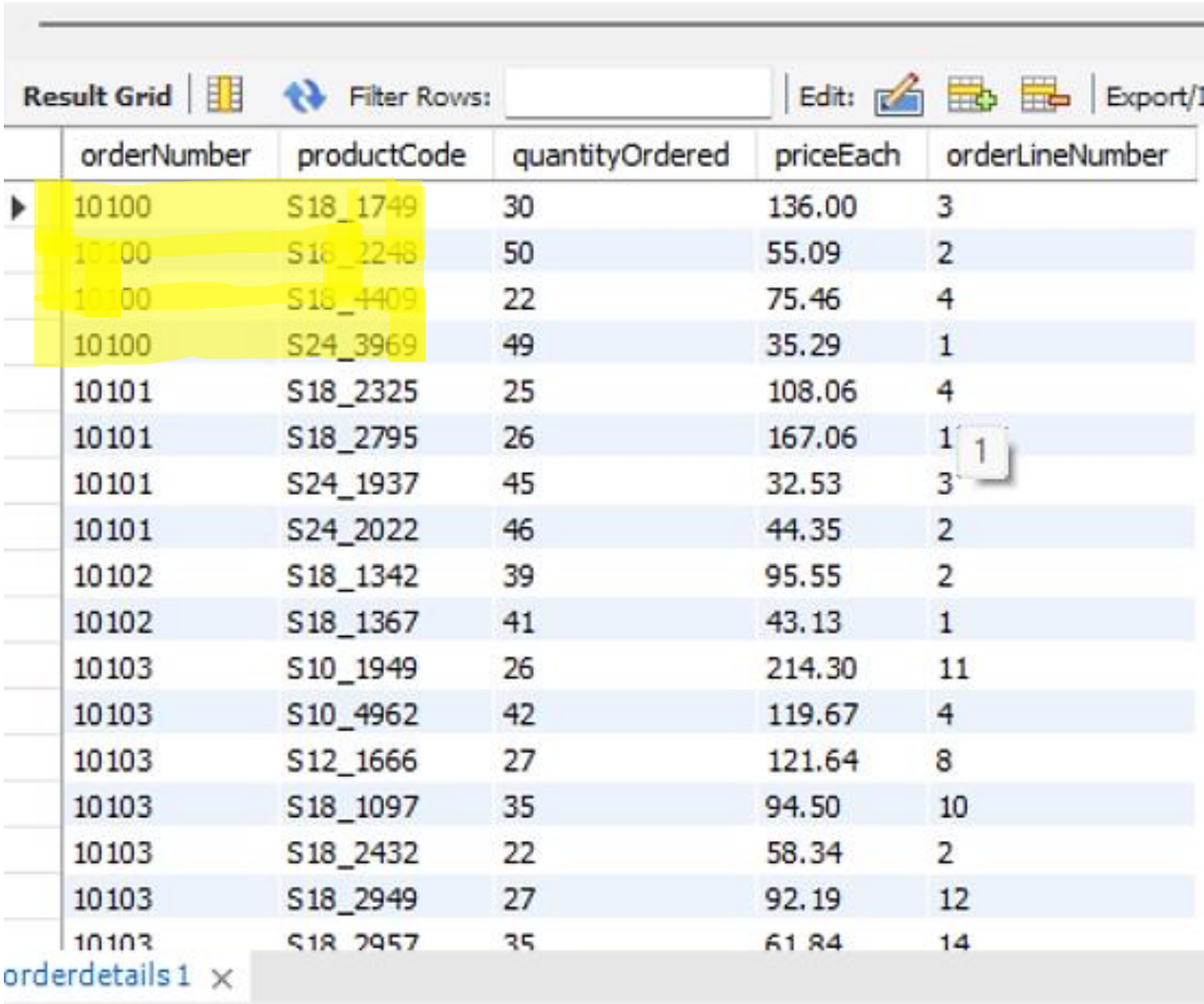
It turned out that the orderNumber is actually unique. The reason why it appears multiple times in the table is because multiple products can be ordered at the same time under one order number.



mintclassics
Tables
customers
employees
offices
orderdetails
Columns
orderNumber
productCode
quantityOrdered
priceEach
orderLineNumber
Indexes
Foreign Keys
Triggers

Under one order number **10100** multiple products, meaning that 4 different products are ordered with one order.

```
4 • SELECT * FROM mintclassics.orderdetails;
```



orderNumber	productCode	quantityOrdered	priceEach	orderLineNumber
10100	S18_1749	30	136.00	3
10100	S18_2248	50	55.09	2
10100	S18_4409	22	75.46	4
10100	S24_3969	49	35.29	1
10101	S18_2325	25	108.06	4
10101	S18_2795	26	167.06	1
10101	S24_1937	45	32.53	3
10101	S24_2022	46	44.35	2
10102	S18_1342	39	95.55	2
10102	S18_1367	41	43.13	1
10103	S10_1949	26	214.30	11
10103	S10_4962	42	119.67	4
10103	S12_1666	27	121.64	8
10103	S18_1097	35	94.50	10
10103	S18_2432	22	58.34	2
10103	S18_2949	27	92.19	12
10102	S18_2957	35	61.84	14

Check:

- **orderdetails** table:

a. **orderNumber** column – what it actually represents?

b. **productCode** – is this unique for each product?

Q1: Under same orderNumber different productCode (i.e., products) appear in the table. This might mean that different products are sold to the same customer under the same orderNumber? => meaning that one order by a customer can contain multiple products:

	orderNumber	productCode
▶	10100	S18_1749
	10100	S18_2248
	10100	S18_4409
		

Q2: In the photo below, same products (productCode) have different orderNumber, which might mean the same product is ordered by different customers under different order number?

	orderNumber	productCode
▶	10107	S10_1678
	10121	S10_1678
	10134	S10_1678

c. **orderLineNumber** column – what is this column about?

MySQL Workbench

Local instance x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

Tables

- customers
- employees
- offices
- orderdetails
- orders
- Columns
 - orderNumber
 - orderDate
 - requiredDate
 - shippedDate
 - status
 - comments
 - customerNumber
- Indexes
- Foreign Keys
- Triggers
- payments
- productlines
- products
 - Columns
 - productCode

Administration Schemas

Information

Table: **warehouses**

Columns:

- warehouseCode** char(1) PK
- warehouseName varchar(45)
- warehousePctCap varchar(50)

Object Info Session

Query Completed

payments customers employees offices orderdetails orders payments productlines products

Limit to 1000 rows

```
31
32 • SELECT orderNumber, COUNT(productCode) AS product_count_per_order
33 FROM mintclassics.orderdetails
34 GROUP BY orderNumber
35 ORDER BY product_count_per_order DESC
```

Result Grid

	orderNumber	product_count_per_order
▶	10106	18
	10159	18
	10165	18
	10168	18
	10222	18
	10275	18
	10316	18
	10332	18
	10360	18
	10386	18
	10398	18
	10122	17
	10126	17
	10135	17
	10181	17
	10182	17
	10204	17
	10210	17
	10280	17
	10287	17

Result 43 x

Output

Action Output

#	Time	Action	Message
✓ 62	17:26:37	SELECT orderNumber, COUNT(productCode) AS product_count_per_order FROM mintclassics.order...	326 row(s) returned

I will try to answer these questions with SQL query in order to have a meaningful result.

A1: an **order (orderNumber)** can contain multiple **products (productCode)**, as I noticed in the orderdetails table. An order represents a **single transaction** where a customer buys one or more products. Let's check how many products each order contains using SQL query.

Observations:

Order number 10106 contains 18 products, which means a customer purchased 18 products under a single transaction/order. Q: Are all these products ordered on the same day? My intuition says yes. We need **orderDate** column to check that. This column is in other table called **orders**. We can use JOIN to merge the tables (Needs to be checked).

Key Takeaway:

When a customer places **one order**, they can buy **one or more products**.

We can find out **when** the **orders** were placed by a **customer**, i.e., which order placed when. This way we can know when each customer made the corresponding orders.

- And we know that **a customer can place multiple orders (a single transaction with 1 or more products) over time**.
- It looks like **multiple orders can be placed (most probably by different customers) on each date**. By grouping by the orderDates I can find out the number of orders (total_orders) placed on each date.

mintclassics

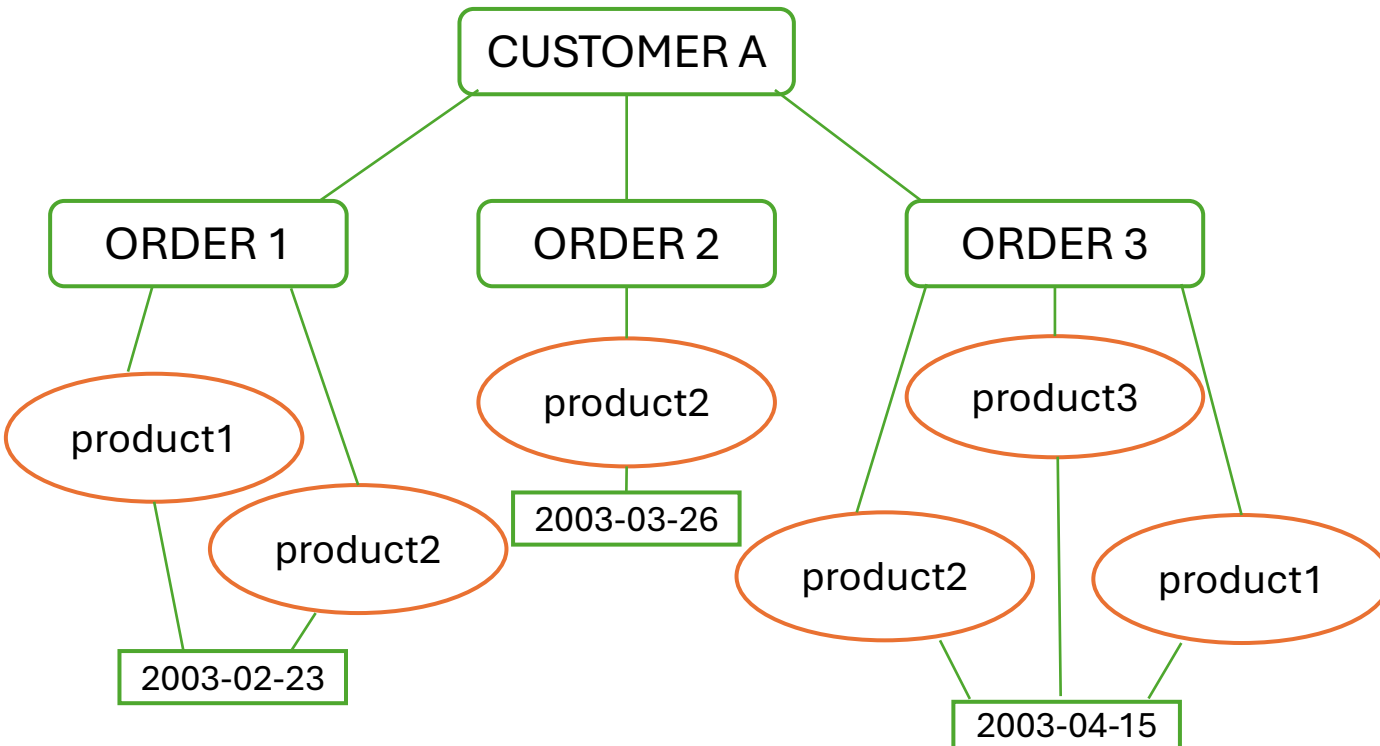
- Tables
 - customers
 - employees
 - offices
 - orderdetails
 - orders**
- Columns
 - orderNumber
 - orderDate
 - requiredDate
 - shippedDate
 - status
 - comments
 - customerNumber
- Indexes
- Foreign Keys
- Triggers

```
4 • SELECT * FROM mintclassics.orders;
```

	orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
▶	10138	2003-07-07	2003-07-16	2003-07-13	Shipped	NULL	496
	10179	2003-11-11	2003-11-17	2003-11-13	Cancelled	Customer cancelled due to urgent budgeting iss...	496
	10360	2004-12-16	2004-12-22	2004-12-18	Shipped	NULL	496
	10399	2005-04-01	2005-04-12	2005-04-03	Shipped	NULL	496
	10207	2003-12-09	2003-12-17	2003-12-11	Shipped	Check on availability.	495
	10243	2004-04-26	2004-05-03	2004-04-28	Shipped	NULL	495
	10186	2003-11-14	2003-11-20	2003-11-18	Shipped	They want to reevaluate their terms agreement...	489
	10213	2004-01-22	2004-01-28	2004-01-27	Shipped	Difficult to negotiate with customer. We need m...	489
	10149	2003-09-12	2003-09-18	2003-09-17	Shipped	NULL	487
	10219	2004-02-10	2004-02-17	2004-02-12	Shipped	NULL	487
	10109	2003-03-10	2003-03-19	2003-03-11	Shipped	Customer requested that FedEx Ground is used...	486
	10236	2004-04-03	2004-04-11	2004-04-08	Shipped	NULL	486
	10331	2004-11-17	2004-11-23	2004-11-23	Shipped	Customer requested special shippment. The inst...	486
	10184	2003-11-14	2003-11-22	2003-11-20	Shipped	NULL	484
	10303	2004-10-06	2004-10-14	2004-10-09	Shipped	Customer inquired about remote controlled mod...	484
	10199	2003-12-01	2003-12-10	2003-12-06	Shipped	NULL	475
	10215	2004-01-29	2004-02-08	2004-02-01	Shipped	Customer requested that FedEx Ground is used	475

orders 1 x

How can we explain the relationship between **a. a customer can place multiple orders (a single transaction with 1 or more products) over time** and **b. multiple orders can be placed (most probably by different customers) on each date** ? I will try to explain it with a simple tree example with my observations.



a. a customer can place multiple orders (a single transaction with 1 or more products) over time.

E.g. on each given dates in the tree, only one order is placed => one order at the exact same time.

I showed this claim with SQL query, see next slide.

Description of the example tree for a.: what you see here is nothing, but the simple structure of how **customer-order-product** relationship happens. We have a customer called CUSTOMER A who placed three orders (ORDER 1,2,3) with one or more products. Each order placed on different dates.

- On 2003-02-23, they ordered two products (product1 & product 2) under ORDER 1.
 - About a month later on 2003-03-26, they placed another order, ORDER 2, with one product (product2).
 - Finally, the last order, ORDER 3, is placed on 2003-04-15 with three products (product3, product2 & product1).
- So, we showed that the statement **a.** makes sense, and a customer has placed three orders over three months period.

Observations:

- A customer can order one specific product multiple times over a period of time. One possible reason for this could be that the customer actually likes this product a lot. In my example, it looks like the customer prefer **product2** over the other products; they ordered **product2** 3 times over time, whereas the others are ordered less.
- By one customer at most 1 order can be placed on each date (i.e., on the exact same time). But the orders can have multiple products. Is this true? Let's check it.



Local instance x

File Edit View Query Database Server Tools Scripting Help

Navigator: payments customers employees offices orderdetails orders payments productlines products warehouses Query 10

SCHEMAS

Filter objects

- mintclassics
 - Tables
 - customers
 - employees
 - offices
 - orderdetails
 - orders
 - Columns
 - orderNumber
 - orderDate
 - requiredDate
 - shippedDate
 - status
 - comments
 - customerNumber
 - Indexes
 - Foreign Keys
 - Triggers
- payments
- productlines
- products
- warehouses

Administration Schemas

Information

Column: **customerNumber**

Definition: **customerNumber** int

```
-- to show that my claim is true. My claim: by a customer at most 1 order can be placed on the exact same date:
SELECT orderDate, customerNumber, COUNT(orderNumber) as order_count
FROM mintclassics.orders
GROUP BY orderDate, customerNumber
ORDER BY orderDate DESC;
```

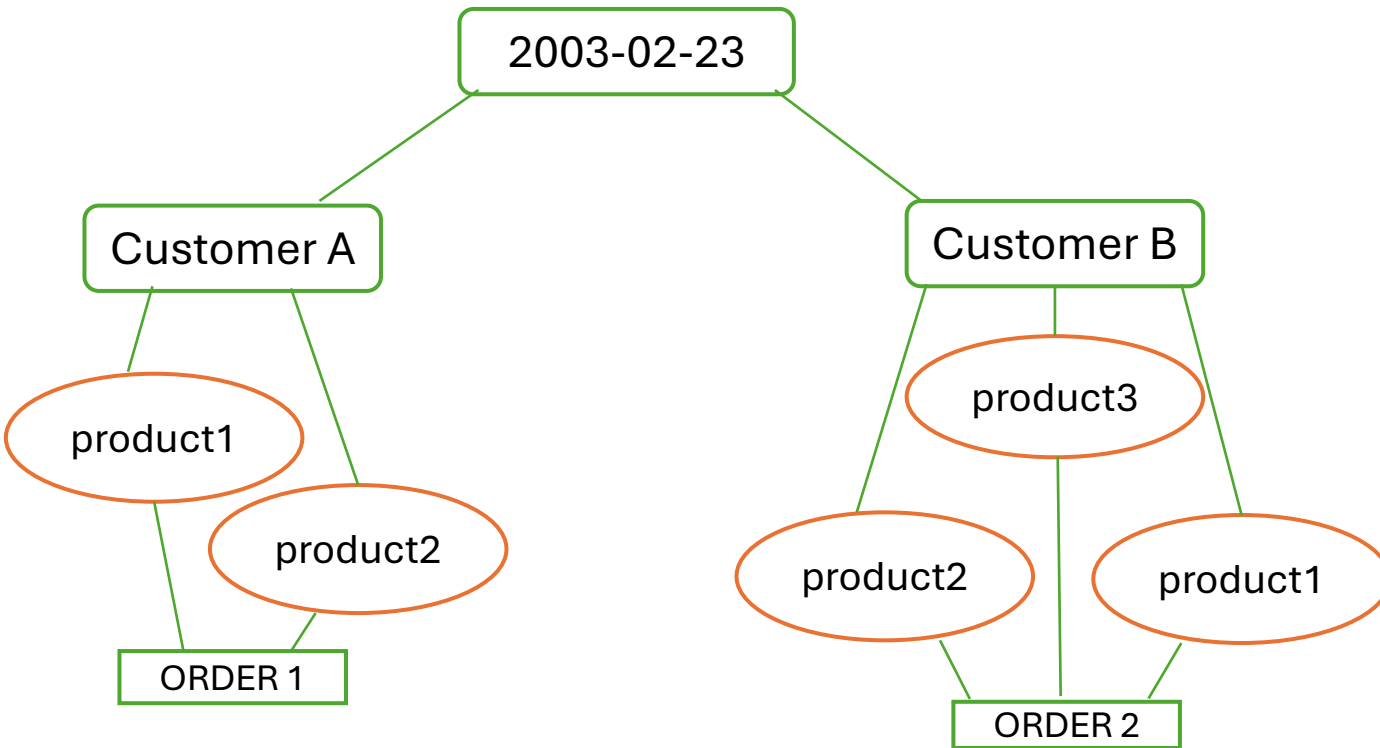
Result Grid

	orderDate	customerNumber	order_count
▶	2005-05-31	119	1
	2005-05-31	141	1
	2005-05-30	157	1
	2005-05-30	314	1
	2005-05-29	124	1
	2005-05-29	282	1
	2005-05-17	382	1
	2005-05-16	412	1
	2005-05-13	141	1
	2005-05-10	386	1
	2005-05-09	471	1
	2005-05-06	362	1
	2005-05-05	175	1
	2005-05-03	141	1
	2005-05-01	233	1
	2005-04-29	357	1
	2005-04-23	166	1
	2005-04-22	308	1

Result 69 x

For example, on date **2005-05-31** 2 orders were placed in total, but they are placed by two different customers with customer numbers **119** and **141**.

As claimed, only 1 order on each date by a customer. Note that this claim is not true when we consider the order count on the same date by different customers, because as we have shown, on the same day multiple orders can be placed by multiple/different customers (Statement b.).



b. multiple orders can be placed (most probably by different customers) on each date.

Now, we can move into the explanation of statement **b.** with a similar example.

Description of the example tree for b.: what you see here is nothing but the simple structure of **customer-order-product** relationship. Now the tree is slightly different, because what we investigate is different here. We have customers, Customer A & Customer B, who placed one order for each (ORDER 1 & 2, respectively) with one or more products. Here, each order by two different customers are placed on the same date.

- Customer A ordered two products (product1 & product2) under ORDER 1 on 2003-02-23.
- Customer B ordered three products (product1 & product2 & product3) under ORDER 2 on 2003-02-23.

So, 2 orders are placed on the same date, on 2003-02-23, but by two different customers. This proves our statement **b.**

Let's extract this information from our dataset with SQL query. See the next slide.

MySQL Workbench

Local instance x

File Edit View Query Database Server Tools Scripting Help

Navigator: symms customers employees offices orderdetails orders payments productlines products warehouses Query 10

SCHEMAS

Filter objects

mintclassics

Tables

customers

Columns

- customerNumber
- customerName
- contactLastName
- contactFirstName
- phone
- addressLine1
- addressLine2
- city
- state
- postalCode
- country
- salesRepEmployeeNumber
- creditLimit

Indexes

Foreign Keys

customers_ibfk_1

Triggers

Administration Schemas

Limit to 1000 rows

```
-- The query below just extract the data of which order placed when by whom;
-- To find out when the orders were placed by a customer, starting from the latest order date (i.e 2005-05-31):
SELECT orderNumber, orderDate, customerNumber
FROM mintclassics.orders
-- GROUP BY customerNumber, orderNumber
ORDER BY orderDate DESC;
```

Result Grid

orderNumber	orderDate	customerNumber
10424	2005-05-31	141
10425	2005-05-31	119
10422	2005-05-30	157
10423	2005-05-30	314
10420	2005-05-29	282
10421	2005-05-29	124
10419	2005-05-17	382
10418	2005-05-16	412
10417	2005-05-13	141
10416	2005-05-10	386
10415	2005-05-09	471
10414	2005-05-06	362

- The snapshot on the left shows which order (orderNumber) placed when (orderDate) by whom (customerNumber).
- It can be observed that multiple orders can be placed (most probably by different customers) on each date. I can find out the number of orders (total_orders) placed on each date (check the second snapshot below right corner).
- Moreover, instead of checking manually, you can find out how many orders have placed on the last date of our dataset by commenting out the WHERE clause in the query below (the answer is, 2 orders have been placed on 2005-05-31).

For example: multiple orders (10424 and 10425) on the same date (2005-05-31) by different customers (141 and 119). Others also continue accordingly.

This is the same what I stated above(in result1 with SQL slide), the only difference is this result contains information about order numbers (orderNumber) that placed by each customer, but the one above contains order dates (orderDate) of each customer.

MySQL Workbench

Local instance x

File Edit View Query Database Server Tools Scripting Help

Navigator: symms customers employees offices orderdetails orders payments productlines products warehouses

SCHEMAS

Filter objects

mintclassics

Tables

customers

Columns

- customerNumber
- customerName
- contactLastName
- contactFirstName
- phone
- addressLine1
- addressLine2
- city
- state
- postalCode
- country
- salesRepEmployeeNumber
- creditLimit

Indexes

Foreign Keys

customers_ibfk_1

Triggers

Administration Schemas

Limit to 1000 rows

```
-- It looks like multiple orders can be placed (most probably by different customers) on each date.
-- By grouping the orderDates I can find out the number of orders (total_orders) placed on each date:
SELECT orderDate, COUNT(orderNumber) as total_orders
FROM mintclassics.orders
-- WHERE orderDate = "2005-05-31" -- to check how many orders have placed on the last date
GROUP BY orderDate
ORDER BY total_orders DESC;
```

Result Grid

orderDate	total_orders
2004-11-24	4
2003-11-05	3
2003-11-06	3
2003-11-14	3
2003-12-01	3
2003-12-02	3
2004-11-05	3
2003-05-21	2
2003-06-16	2
2003-10-06	2
2003-10-10	2

I would like to check our database using SQL queries? I could:

1. ~~Check how many orders each customer has placed.~~
2. Check how many orders a specific customer has placed.
3. Check how many products are included in an order.
4. See if any customers placed multiple orders on the same day.

1. I already found out the total number of orders placed on each date. Now I want to know the total number of orders has been placed by each customer:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'mintclassics' expanded, showing 'Tables' and 'Columns'. The 'Columns' list includes customerNumber, customerName, contactLastName, contactFirstName, phone, addressLine1, addressLine2, city, state, postalCode, country, salesRepEmployeeNumber, and creditLimit. The main editor shows a SQL query to count orders per customer. The 'Result Grid' at the bottom displays the query results.

```
66
67 -- how many orders has each customer placed in total?
68 • SELECT orders.customerNumber, COUNT(orders.orderNumber) as total_orders, customers.customerName
69 FROM mintclassics.orders
70 INNER JOIN mintclassics.customers ON orders.customerNumber = customers.customerNumber
71 GROUP BY orders.customerNumber
72 ORDER BY total_orders DESC;
```

customerNumber	total_orders	customerName
141	26	Euro+ Shopping Channel
124	17	Mini Gifts Distributors Ltd.
114	5	Australian Collectors, Co.
353	5	Reims Collectables
145	5	Danish Wholesale Imports
148	5	Dragon Souvenirs, Ltd.
323	5	Down Under Souvenirs, Inc
381	4	Royale Belge
276	4	Anna's Decorations, Ltd
119	4	La Rochelle Gifts
121	4	Baane Mini Imports
128	4	Blauer See Auto, Co.
131	4	Land of Toys Inc.
144	4	Volvo Model Replicas, Co
496	4	Kelly's Gift Shop
151	4	Muscle Machine Inc
157	4	Diecast Classics Inc.
161	4	Technics Stores Inc.
166	4	Handji Gifts & Co
450	4	The Sharp Gifts Warehouse
282	4	Souvenirs And Things Co.
398	4	Tokyo Collectables, Ltd

1. Check how many orders each customer has placed.
2. ~~Check how many orders a specific customer has placed.~~
3. Check how many products are included in an order.
4. See if any customers placed multiple orders on the same day.

2. Check how many orders a specific customer has placed:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'mintclassics' expanded, showing 'Tables' and 'Columns'. The 'Columns' list includes customerNumber, customerName, contactLastName, contactFirstName, phone, addressLine1, addressLine2, city, state, postalCode, country, salesRepEmployeeNumber, and creditLimit. The main editor shows a SQL query to count orders for a specific customer. The 'Result Grid' at the bottom displays the query results.

```
75 -- how many orders has a specific customer placed?
76 • SELECT orders.customerNumber, COUNT(orders.orderNumber) as total_orders, customers.customerName
77 FROM mintclassics.orders
78 INNER JOIN mintclassics.customers ON orders.customerNumber = customers.customerNumber
79 WHERE customers.customerNumber = 186
80 ORDER BY total_orders DESC;
81
```

customerNumber	total_orders	customerName
186	3	Toys of Finland, Co.

Observations/Insights:

- In 1., we can observe that the **Euro + Shopping Channel** is the customer that placed the highest number of orders - in total 26 order.
- ⇒ We can check what products are ordered by this customer frequently. Where those products that are in demand by this customer are stored, etc.

1. Check how many orders each customer has placed.
2. Check how many orders a specific customer has placed.
3. ~~Check how many products are included in a specific order.~~
4. See if any customers placed multiple orders on the same day.

3. Check how many products are included in a specific order (including the name of the customer). To find how many products are included in a specific order **along with the name of the customer who placed the order**, we need to join the **orders**, **orderdetails**, and **customers** tables. The result is shown in the snapshot below. Why I wanted to include a customer name too? Because I observed when analysing the **1.** in the previous slide that the **Euro + Shopping Channel** has the highest total number of order. Which can be an interesting customer to analyse later on. That is why here, I chose an order number that belongs to that very customer. It looks like, this customer buy more than 1 product in each order. I will investigate it in next slide.

MySQL Workbench

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'mintclassics' database with tables 'customers' and 'orders'. The main editor shows a SQL query to count products per order for a specific customer. The result grid at the bottom shows one row for order 10424 with 6 products, placed by 'Euro+ Shopping Channel'.

```
-- How many products are included in a specific order? --> including the name of the customer who placed the order:
SELECT o.orderNumber, COUNT(productCode) AS total_products, customerName
FROM mintclassics.orders AS o
JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
WHERE o.orderNumber = 10424
GROUP BY o.orderNumber, c.customerName;
```

orderNumber	total_products	customerName
10424	6	Euro+ Shopping Channel

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

mintclassics

Tables

customers

Columns

- customerNumber
- customerName
- contactLastName
- contactFirstName
- phone
- addressLine1
- addressLine2
- city
- state
- postalCode
- country
- salesRepEmployeeNumber
- creditLimit

Indexes

Foreign Keys

Triggers

employees

Administration Schemas

Information

Schema: mintclassics

EDA_UsingSQL_Project

Limit to 1000 rows

```

98 -- Finding the total number of products for each order placed by a specific customer:
99 • SELECT o.orderNumber, COUNT(od.productCode) AS total_products, c.customerName
100 FROM mintclassics.orders AS o
101 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
102 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
103 -- WHERE c.customerName = 'Euro + Shopping Channel'
104 WHERE c.customerName LIKE '%Euro%'
105 GROUP BY o.orderNumber, c.customerName
106 ORDER BY total_products DESC;

```

Result Grid

orderNumber	total_products	customerName
10386	18	Euro+ Shopping Channel
10350	17	Euro+ Shopping Channel
10212	16	Euro+ Shopping Channel
10262	16	Euro+ Shopping Channel
10358	14	Euro+ Shopping Channel
10104	13	Euro+ Shopping Channel
10153	13	Euro+ Shopping Channel
10380	13	Euro+ Shopping Channel
10383	13	Euro+ Shopping Channel
10203	11	Euro+ Shopping Channel
10246	11	Euro+ Shopping Channel
10311	11	Euro+ Shopping Channel
10412	11	Euro+ Shopping Channel
10355	10	Euro+ Shopping Channel
10378	10	Euro+ Shopping Channel
10244	9	Euro+ Shopping Channel
10133	8	Euro+ Shopping Channel
10394	7	Euro+ Shopping Channel
10279	6	Euro+ Shopping Channel

Result 36

Q: What if I want to know the total number of products for each order placed by a specific customer?

As I stated, I found out that the customer **Euro + Shopping Channel** has placed the highest number of orders, with which I can claim that it orders more products than the others. To find out if this claim is true, I did the analysis considering only the name of this customer. This snapshot on the left shows the total number of products for each order (separately) by **Euro + Shopping Channel**.

I can also check, what is the sum of all products, to find out total number of products that is ordered by **Euro + Shopping Channel**. The snapshot below shows the result – it turned out the number of products that **Euro + Shopping Channel** ordered in total is 259.

Tools Scripting Help

EDA_UsingSQL_Project

Limit to 1000 rows

```

117
118 -- Summing total number of products in each order to get the total number of products for a specific customer:
119 • SELECT COUNT(od.productCode) AS total_products, c.customerName
120 FROM mintclassics.orders AS o
121 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
122 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
123 WHERE c.customerName = 'Euro\+ Shopping Channel'
124 GROUP BY c.customerName;
125

```

Result Grid

total_products	customerName
259	Euro+ Shopping Channel

Challenge: In the snapshots, you might have observed that I used different syntax in the **WHERE** clause to filter the result based on a customer name **Euro + Shopping Channel**. But the query wasn't running. The issue is most likely caused by the + sign in the name. In SQL, special characters (like +) can sometimes cause problems in queries, depending on the database system. I fixed the problem using two different methods:

1. **LIKE** with wildcards ('%'). % - represents any sequence of characters before or after the specified word: **WHERE c.customerName LIKE '%Euro%'**

2. escaping the + sign using backslashes (\):

WHERE c.customerName = 'Euro \+ Shopping Channel'

Pro tip: If there are multiple customers with "Euro" in their name, this method could return unintended results. But if we're sure that there's only **one** match, it's a quick and effective solution!

```
EDA_UsingSQL_Project x orders
Limit to 1000 rows
117
118 -- see if there is any customer who placed multiple orders on the same day:
119 • SELECT orderDate, customerNumber, COUNT(orderNumber) AS order_count
120 FROM mintclassics.orders
121 GROUP BY orderDate, customerNumber
122 HAVING COUNT(orderNumber) > 1;
123
124 -- The SQL HAVING clause is used to filter the results of a GROUP BY query based on the result of an aggregate function.
125 -- It is similar to the WHERE clause but is specifically applied after grouping and aggregation,
126 -- allowing you to filter on the results of aggregate functions like COUNT, SUM, AVG, and others.
127
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	orderDate	customerNumber	order_count
▶	2005-02-10	141	2

1. Check how many orders each customer has placed.
2. Check how many orders a specific customer has placed.
3. Check how many products are included in an order.
4. See if any customers placed multiple orders on the same day.

4. See if any customers placed multiple orders on the same day:
Interestiiinnngg..The result of this analysis shows that we have one customer (i.e. 141) that placed 2 orders on 2005-02-10. Interestingly, this customer is **Euro + Shopping Channel**. Now, the question is, why only this company placed multiple orders on the same date?

EDA_UsingSQL_Project x orders x
Limit to 1000 rows

1
2
3
4 • SELECT * FROM mintclassics.orders;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
10390	2005-03-04	2005-03-11	2005-03-07	Shipped	They want to reevaluate their terms agreement...	124
10388	2005-03-03	2005-03-11	2005-03-09	Shipped	HULL	462
10389	2005-03-03	2005-03-09	2005-03-08	Shipped	HULL	448
10387	2005-03-02	2005-03-09	2005-03-06	Shipped	We need to keep in close contact with their Mar...	148
10386	2005-03-01	2005-03-09	2005-03-06	Resolved	Disputed then Resolved on 3/15/2005, Custom...	141
10385	2005-02-28	2005-03-09	2005-03-01	Shipped	HULL	124
10384	2005-02-23	2005-03-06	2005-02-27	Shipped	HULL	321
10383	2005-02-22	2005-03-02	2005-02-25	Shipped	HULL	141
10381	2005-02-17	2005-02-25	2005-02-18	Shipped	HULL	321
10382	2005-02-17	2005-02-23	2005-02-18	Shipped	Custom shipping instructions sent to warehouse	124
10380	2005-02-16	2005-02-24	2005-02-18	Shipped	HULL	141
▶ 10378	2005-02-10	2005-02-18	2005-02-11	Shipped	HULL	141
10379	2005-02-10	2005-02-18	2005-02-11	Shipped	HULL	141
10377	2005-02-09	2005-02-21	2005-02-12	Shipped	Cautious optimism. We have happy customers h...	186
10376	2005-02-08	2005-02-18	2005-02-13	Shipped	HULL	219
10375	2005-02-03	2005-02-10	2005-02-06	Shipped	HULL	119
10374	2005-02-02	2005-02-09	2005-02-03	Shipped	HULL	333
10373	2005-01-31	2005-02-08	2005-02-06	Shipped	HULL	311
10372	2005-01-26	2005-02-05	2005-01-28	Shipped	HULL	398
10371	2005-01-23	2005-02-03	2005-01-25	Shipped	HULL	124
10369	2005-01-20	2005-01-28	2005-01-24	Shipped	HULL	379
10370	2005-01-20	2005-02-01	2005-01-25	Shipped	HULL	276
10368	2005-01-19	2005-01-27	2005-01-24	Shipped	Can we renegotiate this one?	124
10367	2005-01-12	2005-01-21	2005-01-16	Resolved	This order was disputed and resolved on 2/1/20...	205
10366	2005-01-10	2005-01-19	2005-01-12	Shipped	HULL	381

orders 1 x

About the Customer with Multiple Orders on the Same Day

That's an interesting insight! Here are a few possible explanations for why this customer placed multiple orders on the same day:

1. Large or Bulk Orders Split into Multiple Transactions

1. The customer might be ordering in large quantities, and the system (or the company's policy) might split orders into smaller chunks.
2. Maybe there's a limit per order, so they had to place multiple ones.

2. Separate Orders for Different Products or Departments

1. If this customer is a retailer or wholesaler, they might order different product categories separately.
2. Some companies have internal policies that require separate purchases for different items or departments.

3. Back-to-Back Orders Due to Stock Availability

1. Maybe they initially placed an order, then realized they needed more items and placed another one on the same day.

4. Technical or System Issues

1. There's also a small chance that duplicate or redundant orders were placed accidentally due to a system issue.

What to Do Next?

- a) Check **which products** were included in these multiple orders. Are they different, or are there duplicates?
- b) Compare **order amounts**—are they similar in size, or is one much larger?
- c) Check **time differences**—were the orders placed minutes apart or hours apart?
- d) Check the amount of money paid for each product.
- e) See if this customer is a **wholesaler or retailer** (if such information is available).

If this pattern is **consistent** (happens regularly), it could indicate a trend worth analysing further!

EDA_UsingSQL_Project x orders



Limit to 1000 rows



```
177 -- What to do next: a) Check which products were included in these multiple orders.
178 -- Are the products different, or are there duplicates?
179 • SELECT od.productCode, od.quantityOrdered AS orderQuantity,
180        c.customerName, o.customerNumber,
181        o.orderNumber, o.orderDate,
182        p.productName
183 FROM mintclassics.orders AS o
184 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
185 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
186 JOIN mintclassics.products AS p ON p.productCode = od.productCode
187 WHERE o.customerNumber = 141 AND o.orderDate = '2005-02-10'
188 ORDER BY o.orderNumber;
```

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	productCode	orderQuantity	customerName	customerNumber	orderNumber	orderDate	productName
▶	S18_1589	34	Euro+ Shopping Channel	141	10378	2005-02-10	1965 Aston Martin DB5
	S18_3278	22	Euro+ Shopping Channel	141	10378	2005-02-10	1969 Dodge Super Bee
	S18_3482	43	Euro+ Shopping Channel	141	10378	2005-02-10	1976 Ford Gran Torino
	S18_3782	28	Euro+ Shopping Channel	141	10378	2005-02-10	1957 Vespa GS150
	S18_4721	49	Euro+ Shopping Channel	141	10378	2005-02-10	1957 Corvette Convertible
	S24_2972	41	Euro+ Shopping Channel	141	10378	2005-02-10	1982 Lamborghini Diablo
	S24_3371	46	Euro+ Shopping Channel	141	10378	2005-02-10	1971 Alpine Renault 1600s
	S24_3856	33	Euro+ Shopping Channel	141	10378	2005-02-10	1956 Porsche 356A Coupe
	S24_4620	41	Euro+ Shopping Channel	141	10378	2005-02-10	1961 Chevrolet Impala
	S32_2206	40	Euro+ Shopping Channel	141	10378	2005-02-10	1982 Ducati 996 R
	S18_1749	39	Euro+ Shopping Channel	141	10379	2005-02-10	1917 Grand Touring Sedan
	S18_2248	27	Euro+ Shopping Channel	141	10379	2005-02-10	1911 Ford Town Car
	S18_2870	29	Euro+ Shopping Channel	141	10379	2005-02-10	1999 Indy 500 Monte Carlo SS
	S18_3685	32	Euro+ Shopping Channel	141	10379	2005-02-10	1948 Porsche Type 356 Roadster
	S24_1628	32	Euro+ Shopping Channel	141	10379	2005-02-10	1966 Shelby Cobra 427 S/C

Result 34 x

What to Do Next?

Let's analyse the points mentioned above from a) to e):

a) Check **which products** were included in these multiple orders. Are they different, or are there duplicates?

⇒ In the **WHERE** clause, I filtered the date and the customer number because, while analyzing whether multiple orders were placed on the same day, I discovered that a specific customer (customer number **141**) had placed **two** orders on **2005-02-10** (see the slide *Multiple Orders on the Same Day*). To efficiently analyze the **products** ordered by this customer on that date, I incorporated this information into my query.

⇒ Observations: Customer **Euro + Shopping Channel (141)** placed **2 orders** on **2005-02-10**. In total **15 products** are ordered. All products are distinct. Of these, **10 products** were included in order **#10378**, while the remaining **5 products** were part of order **#10379**.

⇒ So, maybe there's a limit in the number of products per order, that's why, they had to place second one (**#10379**) when this limit is exceeded.

Now, let's check the total value of each order by **Euro + Shopping Channel**. If the amounts are similar, it might indicate systematic multiple orders. If one is much larger, it could be a separate large purchase.

Result 60

EDA_UsingSQL_Project

Limit to 1000 rows

```

243 -- This query shows the price per product:
244 • SELECT c.customerName, o.customerNumber,
245         o.orderNumber, o.orderDate, p.productCode,
246         od.priceEach, od.quantityOrdered
247 FROM mintclassics.orders AS o
248 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
249 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
250 JOIN mintclassics.products AS p ON p.productCode = od.productCode
251 GROUP BY o.orderNumber, c.customerNumber, o.orderDate, c.customerName,
252         od.priceEach, od.quantityOrdered, p.productCode
253 ORDER BY o.orderDate;

```

Result Grid

	customerName	customerNumber	orderNumber	orderDate	productCode	priceEach	quantityOrdered
▶	Land of Toys Inc.	131	10107	2003-02-24	S10_1678	81.35	30
	Reims Collectables	353	10121	2003-05-07	S10_1678	86.13	34
	Lyon Souvenirs	250	10134	2003-07-01	S10_1678	90.92	41
	Toys4GrownUps.com	205	10145	2003-08-25	S10_1678	76.56	45
	Corporate Gift Ideas Co.	321	10159	2003-10-10	S10_1678	81.35	49
	Technics Stores Inc.	161	10168	2003-10-28	S10_1678	94.74	36
	Daedalus Designs Imports	171	10180	2003-11-11	S10_1678	76.56	29
	Herkku Gifts	167	10188	2003-11-18	S10_1678	95.70	48
	Mini Wheels Co.	129	10201	2003-12-01	S10_1678	82.30	22
	Baane Mini Imports	121	10103	2003-01-29	S10_1949	214.30	26
	Volvo Model Replicas, Co	144	10112	2003-03-24	S10_1949	197.16	29
	Corrida Auto Replicas, Ltd	458	10126	2003-05-28	S10_1949	205.73	38
	Technics Stores Inc.	161	10140	2003-07-24	S10_1949	186.44	37
	Dragon Souvenirs, Ltd.	148	10150	2003-09-19	S10_1949	182.16	45
	Classic Legends Inc.	424	10163	2003-10-20	S10_1949	212.16	21
	Australian Gift Network, Co	333	10174	2003-11-06	S10_1949	207.87	34
	Classic Gift Ideas, Inc	339	10183	2003-11-13	S10_1949	180.01	23

Result 6

d) Check the amount of money paid for each product.

Observation:

The price of each product fluctuates across different orders. Since each order is placed by a different customer, this suggests that product prices may vary depending on the customer. Two possible factors influencing these price variations could be the customer's location/country or the date of the order.

For instance, the product with the code **S10_1678** was priced at **\$81.35** on **February 24, 2003**. However, by **May 4, 2003**, its price had increased by approximately **\$5**, reaching **\$86.13**. While the product was ordered by different customers, the most likely factors driving the price change are the purchase date and possibly the region in which it was sold.

Find

```

242  -- This query shows the price per product:
243  •  SELECT c.customerName, o.customerNumber,
244         o.orderNumber, o.orderDate, p.productCode, p.productName,
245         od.priceEach -- , od.quantityOrdered
246         -- (od.quantityOrdered * od.priceEach) AS orderPricePerProduct,
247         -- c.country, c.city
248  FROM mintclassics.orders AS o
249  JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
250  JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
251  JOIN mintclassics.products AS p ON p.productCode = od.productCode
252  GROUP BY o.orderNumber, c.customerNumber, o.orderDate, c.customerName,
253         od.priceEach, p.productCode
254  ORDER BY od.priceEach DESC;

```

 Result Grid Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	customerName	customerNumber	orderNumber	orderDate	productCode	productName	priceEach
▶	Baane Mini Imports	121	10103	2003-01-29	S10_1949	1952 Alpine Renault 1300	214.30
	Cambridge Collectables Co.	173	10228	2004-03-10	S10_1949	1952 Alpine Renault 1300	214.30
	Mini Gifts Distributors Ltd.	124	10312	2004-10-21	S10_1949	1952 Alpine Renault 1300	214.30
	Classic Legends Inc.	424	10163	2003-10-20	S10_1949	1952 Alpine Renault 1300	212.16
	Scandinavian Gift Ideas	448	10291	2004-09-08	S10_1949	1952 Alpine Renault 1300	210.01
	Australian Gift Network, Co	333	10174	2003-11-06	S10_1949	1952 Alpine Renault 1300	207.87
	Corrida Auto Replicas, Ltd	458	10348	2004-11-01	S12_1108	2001 Ferrari Enzo	207.80
	Corrida Auto Replicas, Ltd	458	10126	2003-05-28	S10_1949	1952 Alpine Renault 1300	205.73
	Québec Home Shopping Network	233	10411	2005-05-01	S10_1949	1952 Alpine Renault 1300	205.73
	Amica Models & Co.	249	10280	2004-08-17	S10_1949	1952 Alpine Renault 1300	205.73
	West Coast Collectables Co.	475	10215	2004-01-29	S10_1949	1952 Alpine Renault 1300	205.73
	Danish Wholesale Imports	145	10105	2003-02-11	S12_1108	2001 Ferrari Enzo	205.72
	Lyon Souvenirs	250	10395	2005-03-17	S12_1108	2001 Ferrari Enzo	205.72
	Super Scale Inc.	455	10196	2003-11-26	S12_1108	2001 Ferrari Enzo	203.64

Result 10

EDA_UsingSQL_Project x orders

Limit to 1000 rows

```
222 -- This query shows the total order amount/price per order of each customer.
223 • SELECT o.orderNumber,
224        SUM(od.quantityOrdered * od.priceEach) AS totalOrderPrice
225 FROM mintclassics.orders AS o
226 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
227 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
228 GROUP BY o.orderNumber
229 ORDER BY totalOrderPrice;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	orderNumber	totalOrderPrice
▶	10408	615.45
	10144	1128.20
	10158	1491.38
	10116	1627.56
	10345	1676.14
	10242	1679.92
	10364	1834.56
	10286	1960.80
	10409	2326.18
	10317	2434.25
	10277	2611.84
	10132	2880.00
	10118	3101.40
	10376	3452.75
	10303	3474.66
	10387	3516.04
	10189	3879.96
	10294	4424.40
	10154	4465.85
	10385	4466.71

Result 65 x

EDA_UsingSQL_Project* x orders

```

230 -- we can count the number of orders per day for this customer:
231 • SELECT o.customerNumber, c.customerName, o.orderDate,
232        COUNT(o.orderNumber) AS numberOfOrders
233 FROM mintclassics.orders o
234 JOIN mintclassics.customers c ON o.customerNumber = c.customerNumber
235 WHERE c.customerName = 'Euro\+ Shopping Channel'
236 GROUP BY o.customerNumber, o.orderDate
237 -- HAVING COUNT(o.orderNumber) > 1
238 ORDER BY numberOfOrders DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customerNumber	customerName	orderDate	numberOfOrders
▶	141	Euro+ Shopping Channel	2005-02-10	2
	141	Euro+ Shopping Channel	2003-01-31	1
	141	Euro+ Shopping Channel	2003-06-06	1
	141	Euro+ Shopping Channel	2003-06-27	1
	141	Euro+ Shopping Channel	2003-09-28	1
	141	Euro+ Shopping Channel	2003-10-08	1
	141	Euro+ Shopping Channel	2003-11-19	1
	141	Euro+ Shopping Channel	2003-12-02	1
	141	Euro+ Shopping Channel	2003-12-03	1
	141	Euro+ Shopping Channel	2004-01-16	1
	141	Euro+ Shopping Channel	2004-04-29	1
	141	Euro+ Shopping Channel	2004-05-05	1
	141	Euro+ Shopping Channel	2004-06-24	1
	141	Euro+ Shopping Channel	2004-08-09	1
	141	Euro+ Shopping Channel	2004-10-16	1
	141	Euro+ Shopping Channel	2004-12-02	1
	141	Euro+ Shopping Channel	2004-12-07	1
	141	Euro+ Shopping Channel	2004-12-10	1
	141	Euro+ Shopping Channel	2005-02-16	1
	141	Euro+ Shopping Channel	2005-02-22	1

Result 70 x

c) Check Time Differences Between Orders on the Same Day

If we have timestamps for orders, we could check the exact times, but since we only have dates (orderDate), we can count the number of orders per day for this customer.

- ⇒ Uncommenting **HAVING** clause will show only the dates when more than 1 order has been placed by **Euro + Shopping Channel**.
- ⇒ As we can see, on other dates the customer placed only 1 order (with probably multiple products).
- ⇒ To show # of orders by each customer on each date, I just needed to uncomment **WHERE** clause where I filtered only one customer.

EDA_UsingSQL_Project* x orders

```

229
230 -- we can count the number of orders per day for this customer:
231 • SELECT o.customerNumber, c.customerName, o.orderDate,
232        COUNT(o.orderNumber) AS numberOfOrders
233 FROM mintclassics.orders o
234 JOIN mintclassics.customers c ON o.customerNumber = c.customerNumber
235 -- WHERE c.customerName = 'Euro\+ Shopping Channel'
236 GROUP BY o.customerNumber, o.orderDate
237 -- HAVING COUNT(o.orderNumber) > 1
238 ORDER BY numberOfOrders DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customerNumber	customerName	orderDate	numberOfOrders
▶	141	Euro+ Shopping Channel	2005-02-10	2
	363	Online Diecast Creations Co.	2003-01-06	1
	128	Blauer See Auto, Co.	2003-01-09	1
	181	Vitachrome Inc.	2003-01-10	1
	121	Baane Mini Imports	2003-01-29	1
	141	Euro+ Shopping Channel	2003-01-31	1
	145	Danish Wholesale Imports	2003-02-11	1
	278	Rovelli Gifts	2003-02-17	1
	131	Land of Toys Inc.	2003-02-24	1
	385	Cruz & Sons Co.	2003-03-03	1
	486	Motor Mint Distributors Inc.	2003-03-10	1
	187	AV Stores, Co.	2003-03-18	1
	129	Mini Wheels Co.	2003-03-25	1
	144	Volvo Model Replicas, Co	2003-03-24	1
	124	Mini Gifts Distributors Ltd.	2003-03-26	1
	172	La Corne D'abondance, Co.	2003-04-01	1
	424	Classic Legends Inc.	2003-04-04	1
	381	Royale Belge	2003-04-11	1
	148	Dragon Souveniers, Ltd.	2003-04-16	1

Result 71 x

Observation:

All the customer have 1 order on each date, except **Euro + Shopping Channel** which has 2, and we already showed this in the previous analysis.

EDA_UsingSQL_Project x orders orderdetails

Limit to 1000 rows

```

254 -- What To Do Next: e) See if this customer is a wholesaler or retailer (if such information is available):
255 • SELECT c.customerNumber, c.customerName,
256        COUNT(DISTINCT o.orderNumber) AS totalOrders,
257        SUM(od.quantityOrdered) AS totalProductsOrdered,
258        SUM(od.quantityOrdered * od.priceEach) AS totalSpent,
259        c.creditLimit
260 FROM mintclassics.customers c
261 JOIN mintclassics.orders o ON c.customerNumber = o.customerNumber
262 JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
263 -- WHERE c.customerName = 'Euro\+ Shopping Channel'
264 GROUP BY c.customerNumber, c.customerName, c.creditLimit
265 ORDER BY totalOrders DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customerNumber	customerName	totalOrders	totalProductsOrdered	totalSpent	creditLimit
▶	141	Euro+ Shopping Channel	26	9327	820689.54	227600.00
	124	Mini Gifts Distributors Ltd.	17	6366	591827.34	210500.00
	114	Australian Collectors, Co.	5	1926	180585.07	117300.00
	353	Reims Collectables	5	1433	126983.19	81100.00
	145	Danish Wholesale Imports	5	1315	129085.12	83400.00
	148	Dragon Souvenirs, Ltd.	5	1524	156251.03	103800.00
	323	Down Under Souvenirs, Inc	5	1691	154622.08	88000.00
	381	Royale Belge	4	278	29217.18	23500.00
	276	Anna's Decorations, Ltd	4	1469	137034.22	107800.00
	119	La Rochelle Gifts	4	1832	158573.12	118200.00
	121	Baane Mini Imports	4	1082	104224.79	81700.00
	128	Blauer See Auto, Co.	4	811	75937.76	59700.00
	131	Land of Toys Inc.	4	1631	149085.15	114900.00
	144	Volvo Model Replicas, Co	4	647	66694.82	53100.00
	496	Kelly's Gift Shop	4	1647	137460.79	110000.00
	151	Muscle Machine Inc	4	1775	177913.95	138500.00
	157	Diecast Classics Inc.	4	1111	104358.69	100600.00
	161	Technique Stores Inc	4	1170	104545.77	84600.00

Result 64 x

e) See if this customer is a **wholesaler or retailer** (if such information is available).

Check If This Customer is a Wholesaler or Retailer

We don't have a specific column that tells us whether a customer is a wholesaler, but we can check:

- If their **total order volume** is much higher than other customers.
- If they frequently place **large orders** (high **quantityOrdered**).
- If they have a **high credit limit** (usually assigned to big buyers).

Observation:

⇒ Interestingly, customer with the highest credit limit is **141 – Euro + Shopping Channel**

⇒ The **COUNT** function in Power BI is an aggregation method that calculates the total number of occurrences of a specific value or data point in a given dataset. This feature is very beneficial when understanding the frequency or distribution of certain values within your data.

EDA_UsingSQL_Project x

Limit to 1000 rows

```

369 -- There is multiple/different products under a single order(number):
370 • SELECT c.customerName,
371        o.orderNumber,
372        p.productCode
373 FROM mintclassics.customers c
374 JOIN mintclassics.orders o ON c.customerNumber = o.customerNumber
375 JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
376 JOIN mintclassics.products p ON p.productCode = od.productCode
377 ORDER BY c.customerName DESC;
378

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: IA

	customerName	orderNumber	productCode
▶	Euro+ Shopping Channel	10104	S12_3148
	Euro+ Shopping Channel	10104	S12_4473
	Euro+ Shopping Channel	10104	S18_2238
	Euro+ Shopping Channel	10104	S18_2319
	Euro+ Shopping Channel	10104	S18_3232
	Euro+ Shopping Channel	10104	S18_4027
	Euro+ Shopping Channel	10104	S24_1444
	Euro+ Shopping Channel	10104	S24_2840
	Euro+ Shopping Channel	10104	S24_4048
	Euro+ Shopping Channel	10104	S32_2509
	Euro+ Shopping Channel	10104	S32_3207
	Euro+ Shopping Channel	10104	S50_1392
	Euro+ Shopping Channel	10104	S50_1514
	Euro+ Shopping Channel	10128	S18_3140
	Euro+ Shopping Channel	10128	S18_3259
	Euro+ Shopping Channel	10128	S18_4522
	Euro+ Shopping Channel	10128	S700_1938
	Euro+ Shopping Channel	10133	S18_2581
	Euro+ Shopping Channel	10133	S24_1785

Result 2 x

Challenge

Without **DISTINCT**, just because one order can contain multiple products, the corresponding order numbers appear multiple times (in the table) for each product. This cause them to be counted multiple times. Solution: Using **DISTINCT** helps us count the number of different orders placed by each customer.

EDA_UsingSQL_Project* x orders orderdetails

Limit to 1000 rows

```

300 -- Using DISTINCT help us to count the number of different orders placed by each customer.
301 • SELECT c.customerName,
302        COUNT(o.orderNumber) AS totalOrder
303 FROM mintclassics.customers c
304 JOIN mintclassics.orders o ON c.customerNumber = o.customerNumber
305 JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
306 GROUP BY c.customerName
307 ORDER BY totalOrder DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: IA

	customerName	totalOrder
▶	Euro+ Shopping Channel	259
	Mini Gifts Distributors Ltd.	180
	Australian Collectors, Co.	55
	La Rochelle Gifts	53
	AV Stores, Co.	51
	Land of Toys Inc.	49
	Muscle Machine Inc	48
	Kelly's Gift Shop	48
	Rovelli Gifts	48
	Souvenirs And Things Co.	46
	Anna's Decorations, Ltd	46
	Down Under Souvenirs, Inc	46
	Dragon Souvenirs, Ltd.	43
	Reims Collectables	41
	Saveley & Henriot, Co.	41
	Corporate Gift Ideas Co.	41
	Salzburg Collectables	40
	The Sharp Gifts Warehouse	40
	L'ordine Souvenirs	39
	Scandinavian Gift Ideas	38
	Danish Wholesale Imports	36
	Handji Gifts& Co	36

Result 56 x

EDA_UsingSQL_Project x orders orderdetails

Limit to 1000 rows

```

309 -- Using DISTINCT help us to count the number of different orders placed by each customer.
310 • SELECT c.customerName,
311        COUNT(DISTINCT o.orderNumber) AS totalOrder
312 FROM mintclassics.customers c
313 JOIN mintclassics.orders o ON c.customerNumber = o.customerNumber
314 JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
315 GROUP BY c.customerName
316 ORDER BY totalOrder DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	customerName	totalOrder
▶	Euro + Shopping Channel	26
	Mini Gifts Distributors Ltd.	17
	Danish Wholesale Imports	5
	Australian Collectors, Co.	5
	Down Under Souvenirs, Inc	5
	Dragon Souvenirs, Ltd.	5
	Reims Collectables	5
	Technics Stores Inc.	4
	Land of Toys Inc.	4
	Anna's Decorations, Ltd	4
	Souvenirs And Things Co.	4
	Diecast Classics Inc.	4
	Salzburg Collectables	4
	Royale Belge	4
	Baane Mini Imports	4
	Muscle Machine Inc	4
	Blauer See Auto, Co.	4
	Kelly's Gift Shop	4
	Handji Gifts& Co	4
	Volvo Model Replicas, Co	4
	Tokyo Collectables, Ltd	4
	The Sharp Gifts Warehouse	4

Result 57 x

I used **DISTINCT** here to solve problem which might have led us to wrong insights about the business.

What to Visualize in Power BI?

Now, let's move on to visualizing these insights!

Total Orders per Customer → Bar Chart

- This will highlight how **Euro + Shopping Channel** compares to other customers.
- X-axis: customerName, Y-axis: totalOrders

Total Products Ordered per Customer → Bar Chart

- This will help confirm if Euro + Shopping Channel orders **way more products** than others.
- X-axis: customerName, Y-axis: totalProductsOrdered

Orders per Day for Euro + Shopping Channel → Line Chart

- To visualize the **ordering pattern over time**.
- X-axis: orderDate, Y-axis: orderCount

Most Ordered Products by Euro + Shopping Channel → Bar Chart

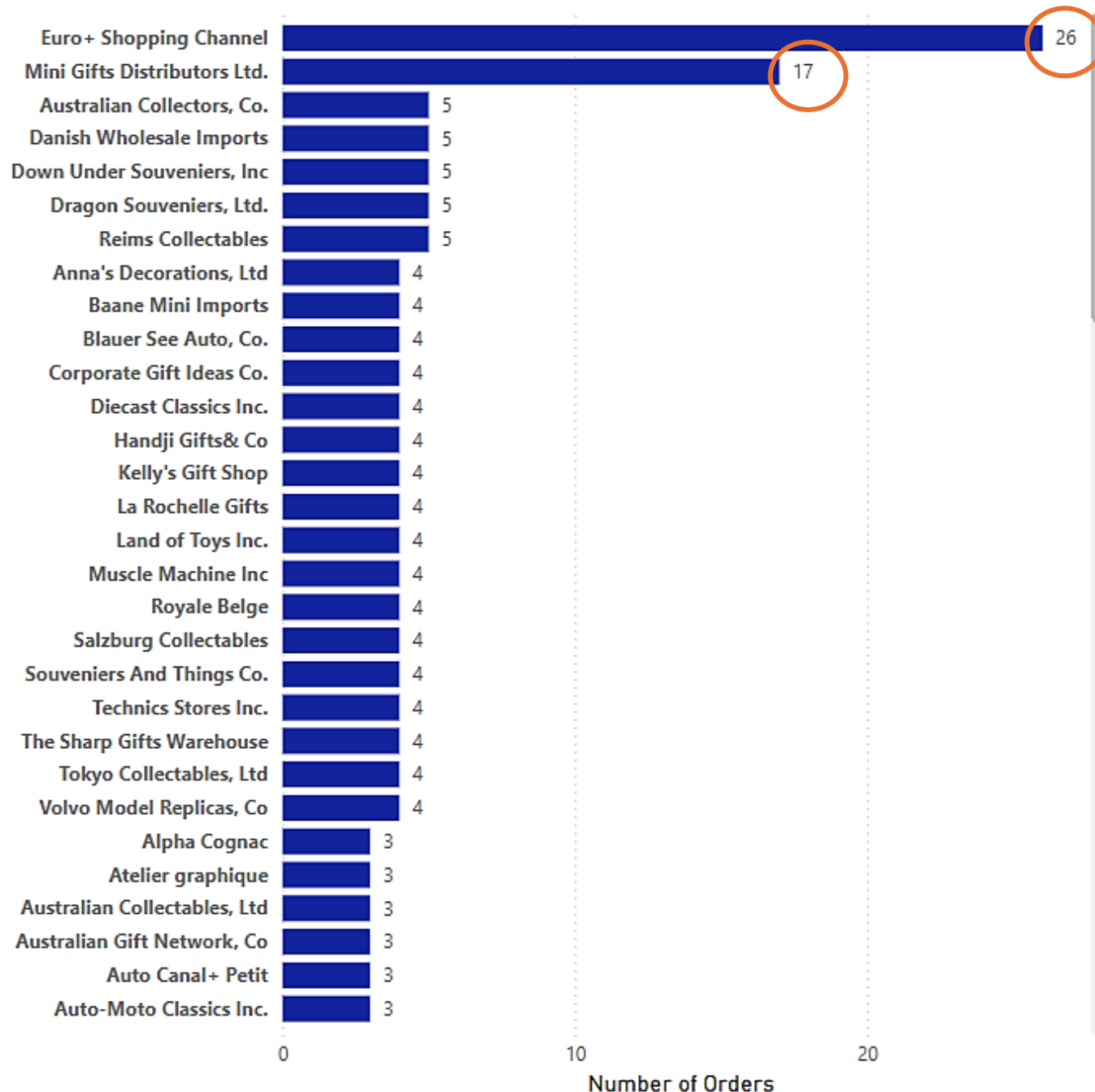
- To check **which products they frequently purchase**.
- This could reveal if they focus on specific items.

Customer Credit Limit vs. Total Orders → Scatter Plot

- X-axis: creditLimit, Y-axis: totalOrders
- This will show whether customers with **higher credit limits tend to order more**.

Total Orders per Customer

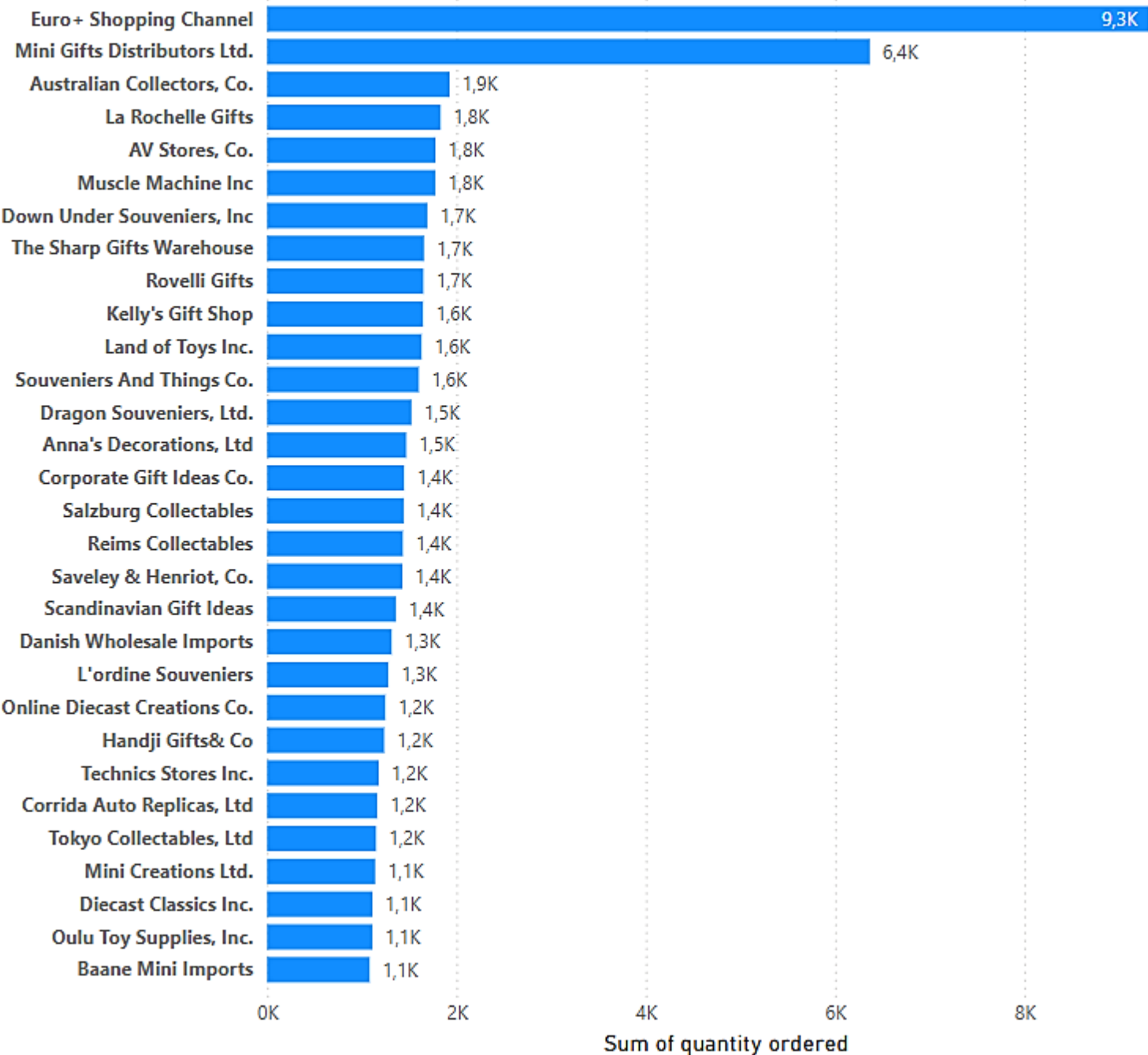
Customer Name



This will highlight how **Euro + Shopping Channel** compares to other customers. Sorted in **descending order** to highlight the customers with the most orders. It looks like **Mini Gifts Distributors Ltd.** might be a potential wholesaler, too.

Total Product Quantity per Customer

Customer name



customerName	Sum of quantityOrdered
Euro+ Shopping Channel	9327
Mini Gifts Distributors Ltd.	6366
Australian Collectors, Co.	1926
La Rochelle Gifts	1832
AV Stores, Co.	1778
Muscle Machine Inc	1775
Down Under Souvenirs, Inc	1691
The Sharp Gifts Warehouse	1656
Rovelli Gifts	1650
Kelly's Gift Shop	1647
Land of Toys Inc.	1631
Souvenirs And Things Co.	1601
Dragon Souvenirs, Ltd.	1524
Anna's Decorations, Ltd	1469
Corporate Gift Ideas Co.	1447
Salzburg Collectables	1442
Reims Collectables	1433
Saveley & Henriot, Co.	1428
Scandinavian Gift Ideas	1359
Danish Wholesale Imports	1315
L'ordine Souvenirs	1280
Online Diecast Creations Co.	1248
Handji Gifts& Co	1236
Technics Stores Inc.	1179
Corrida Auto Replicas, Ltd	1163
Tokyo Collectables, Ltd	1150
Mini Creations Ltd.	1140
Diecast Classics Inc.	1111
Oulu Toy Supplies, Inc.	1110
Baane Mini Imports	1082
Total	105516

This visualization and the corresponding table will help confirm if **Euro + Shopping Channel** orders **way more products** than the others.

EDA_UsingSQL_Project x orders orderdetails

Limit to 1000 rows

```

268 -- Most ordered products in general, i.e by each customer. To check which products they frequently purchase.
269 -- This way we can also find out top-selling product:
270 • SELECT p.productName, SUM(od.quantityOrdered) AS totalQuantitySold
271 FROM mintclassics.orders AS o
272 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
273 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
274 JOIN mintclassics.products AS p ON p.productCode = od.productCode
275 -- WHERE c.customerName = 'Euro+ Shopping Channel'
276 GROUP BY p.productName
277 ORDER BY totalQuantitySold DESC;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
productName	totalQuantitySold		
1992 Ferrari 360 Spider red	1808		
1937 Lincoln Berline	1111		
American Airlines: MD-11S	1085		
1941 Chevrolet Special Deluxe Cabriolet	1076		
1930 Buick Marquette Phaeton	1074		
1940s Ford truck	1061		
1969 Harley Davidson Ultimate Chopper	1057		
1957 Chevy Pickup	1056		
1964 Mercedes Tour Bus	1053		
1956 Porsche 356A Coupe	1052		
Corsair F4U (Bird Cage)	1051		
F/A 18 Hornet 1/72	1047		
1980s Black Hawk Helicopter	1040		
1913 Ford Model T Speedster	1038		
1997 BMW R 1100 S	1033		
1972 Alfa Romeo GTA	1030		
1962 Volkswagen Microbus	1029		
2002 Suzuki XREO	1028		
The USS Constitution Ship	1020		
2001 Ferrari Enzo	1019		

Note: to see the most ordered products by **Euro+ Shopping Channel**, simple uncomment the **WHERE** clause I've written above.

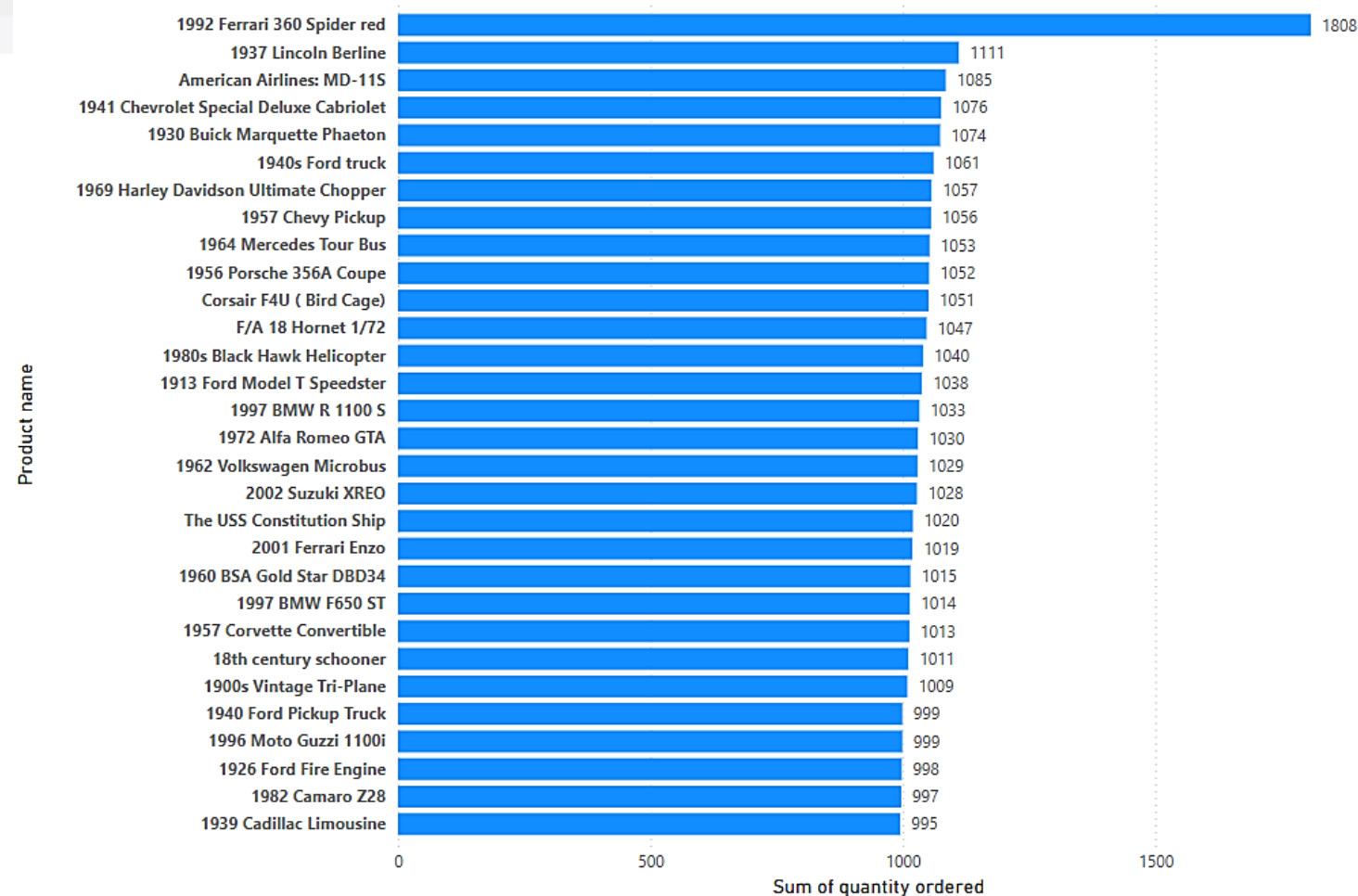
Most Ordered Products in general/by each customer

- To check **which products they frequently purchase**.
- This could reveal if they focus on specific items.

They indeed focus on a specific product: **1992 Ferrari 360 Spider red**

So, we can tell that this product is the most in-demand product among all customers.

Most Ordered Products



```
EDA_UsingSQL_Project x orders orderdetails
Limit to 1000 rows
278 -- How many specific product is ordered by a specific customer over time?
279 • SELECT p.productName, SUM(od.quantityOrdered) AS orderQuantity, o.orderDate
280 FROM mintclassics.orders AS o
281 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
282 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
283 JOIN mintclassics.products AS p ON p.productCode = od.productCode
284 WHERE c.customerName = 'Euro\+ Shopping Channel' AND
285        p.productName LIKE '%Ferrari%'
286 GROUP BY p.productName, o.orderDate
287 ORDER BY orderQuantity DESC;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

	productName	orderQuantity	orderDate
▶	1992 Ferrari 360 Spider red	60	2005-05-03
	1992 Ferrari 360 Spider red	48	2003-12-02
	1992 Ferrari 360 Spider red	47	2005-02-22
	1992 Ferrari 360 Spider red	40	2004-01-16
	1992 Ferrari 360 Spider red	36	2004-05-05
	1992 Ferrari 360 Spider red	32	2004-12-10
	1992 Ferrari 360 Spider red	23	2003-01-31
	1992 Ferrari 360 Spider red	22	2005-03-15
	2001 Ferrari Enzo	20	2003-09-28

Check:

I also wanted to check the total number of most in-demand product, **1992 Ferrari 360 Spider red**, that is ordered by **Euro + Shopping Channel** on each **date**.

Another thing to observe in the SQL query is that I filtered the name of the product using **LIKE** with wildcards (%), which is quick and simple, but problem can occur if there is other product names with '**Ferrari**' in it, for example, **2001 Ferrari Enzo**. I had that problem. It turned out there are two product names with 'Ferrari'. So, it might not be a perfect way of filtering in general, but in my situation, it did not hurt, so I left it as it is.

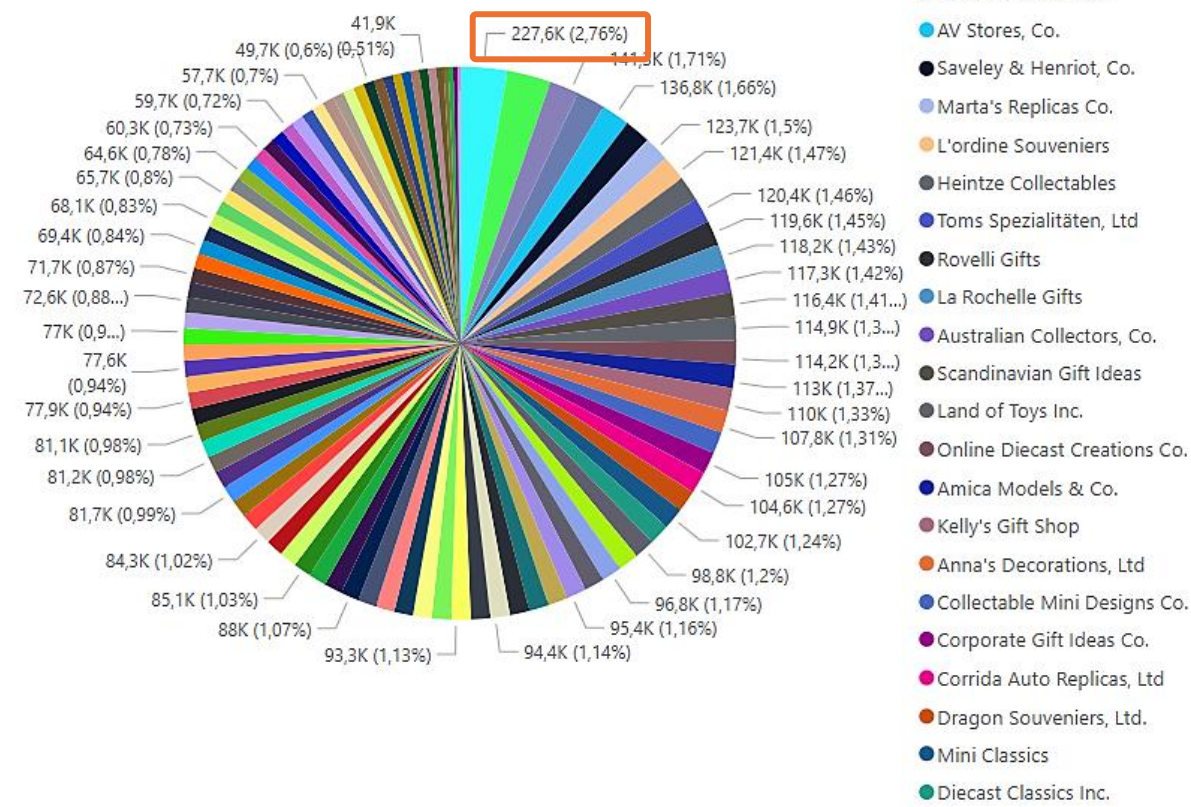
```

288 -- Credit limit vs Total orders:
289 • SELECT c.customerName, COUNT(o.orderNumber) AS totalOrders, c.creditLimit
290 FROM mintclassics.orders AS o
291 JOIN mintclassics.customers AS c ON c.customerNumber = o.customerNumber
292 GROUP BY c.customerName, c.creditLimit
293 ORDER BY c.creditLimit DESC;

```

customerName	totalOrders	creditLimit
Euro+ Shopping Channel	26	227600.00
Mini Gifts Distributors Ltd.	17	210500.00
Vida Sport, Ltd	2	141300.00
Muscle Machine Inc	4	138500.00
AV Stores, Co.	3	136800.00
Saveley & Henriot, Co.	3	123900.00
Marta's Replicas Co.	2	123700.00
L'ordine Souveniers	3	121400.00
Heintze Collectables	2	120800.00
Toms Spezialitäten, Ltd	2	120400.00
Rovelli Gifts	3	119600.00
La Rochelle Gifts	4	118200.00
Australian Collectors, Co.	5	117300.00
Scandinavian Gift Ideas	3	116400.00
Land of Toys Inc.	4	114900.00
Online Diecast Creations ...	3	114200.00
Amica Models & Co.	2	113000.00
Kelly's Gift Shop	4	110000.00
Anna's Decorations, Ltd	4	107800.00
Corporate Gift Ideas Co.	4	105000.00
Collectable Mini Designs ...	2	105000.00
Corrida Auto Replicas, Ltd	3	104600.00
Dragon Souveniers, Ltd.	5	103800.00

Credit Limit per Customer



Customer Credit Limit vs. Total Orders

This will show whether customers with **higher credit limits tend to order more.**

Two customers with the highest credit limit tend to place more orders than the other customers with less credit limit. See the pie chart.

EDA_UsingSQL_Project x orders orderdetails

Limit to 1000 rows

```
222 -- Total Revenue per Customer: This query shows the total order amount/price per order of each customer.
223 • SELECT c.customerName,
224         SUM(od.quantityOrdered * od.priceEach) AS totalOrderPrice
225 FROM mintclassics.orders AS o
226 JOIN mintclassics.customers AS c ON o.customerNumber = c.customerNumber
227 JOIN mintclassics.orderdetails AS od ON o.orderNumber = od.orderNumber
228 GROUP BY c.customerName
229 ORDER BY totalOrderPrice DESC;
230
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

customerName	totalOrderPrice
Euro+ Shopping Channel	820689.54
Mini Gifts Distributors Ltd.	591827.34
Australian Collectors, Co.	180585.07
Muscle Machine Inc	177913.95
La Rochelle Gifts	158573.12
Dragon Souvenirs, Ltd.	156251.03
Down Under Souvenirs, Inc	154622.08
Land of Toys Inc.	149085.15
AV Stores, Co.	148410.09
The Sharp Gifts Warehouse	143536.27
Salzburg Collectables	137480.07
Kelly's Gift Shop	137460.79
Anna's Decorations, Ltd	137034.22
Souvenirs And Things Co.	133907.12
Corporate Gift Ideas Co.	132340.78
Saveley & Henriot, Co.	130305.35
Danish Wholesale Imports	129085.12
Rovelli Gifts	127529.69
Reims Collectables	126983.19
L'ordine Souvenirs	125505.57
Continental Gift Ideas	120043.53

Result 35 x

- What this query does:
- Joins **customers**, **orders**, and **orderdetails** tables.
 - Calculates **total revenue**, (quantityOrdered * priceEach), for each customer.
 - Groups the data by customer name.
 - Sorts the result in **descending order** (highest revenue first)

Observations: as expected, **Euro Shopping Channel** and **Mini Gifts Distributors Ltd.** have the highest total revenue out of all customers. Our business earns a lot of money especially through these two customers. Because the total order placed by these very customers is also the highest.

EDA_UsingSQL_Project x orders orderdetails

Limit to 1000 rows

```

328
329  -- Total Revenue per Product:
330 • SELECT
331     p.productName,
332     SUM(od.quantityOrdered * od.priceEach) AS totalRevenue
333 FROM mintclassics.products p
334 JOIN mintclassics.orderdetails od ON p.productCode = od.productCode
335 GROUP BY p.productName
336 ORDER BY totalRevenue DESC;
337

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	productName	totalRevenue
▶	1992 Ferrari 360 Spider red	276839.98
	2001 Ferrari Enzo	190755.86
	1952 Alpine Renault 1300	190017.96
	2003 Harley-Davidson Eagle Drag Bike	170686.00
	1968 Ford Mustang	161531.48
	1969 Ford Falcon	152543.02
	1980s Black Hawk Helicopter	144959.91
	1998 Chrysler Plymouth Prowler	142530.63
	1917 Grand Touring Sedan	140535.60
	2002 Suzuki XREO	135767.03
	1956 Porsche 356A Coupe	134240.71
	1969 Corvair Monza	132363.79
	1928 Mercedes-Benz SSK	132275.98
	1957 Corvette Convertible	130749.31
	1972 Alfa Romeo GTA	127924.32
	1962 LanciaA Delta 16V	123123.01
	1970 Triumph Spitfire	122254.75
	1976 Ford Gran Torino	121890.60
	1948 Porsche Type 356 Roadster	121653.46
	1958 Setra Bus	119085.25
	1995 Honda Civic	119050.95

1992 Ferrari 360 Spider red is the product that our business receives the most revenue from. I was expecting this, because this very product is also the most frequently ordered product.

```
EDA_UsingSQL_Project x orders orderdetails
Limit to 1000 rows

283 -- Most Profitable Products:
284 • SELECT p.productName,
285       SUM(od.quantityOrdered * (od.priceEach - p.buyPrice)) AS totalProfit
286 FROM mintclassics.products p
287 JOIN mintclassics.orderdetails od ON p.productCode = od.productCode
288 GROUP BY p.productName
289 ORDER BY totalProfit DESC;
```

productName	totalProfit
1992 Ferrari 360 Spider red	135996.78
1952 Alpine Renault 1300	95282.58
2001 Ferrari Enzo	93349.65
2003 Harley-Davidson Eagle Drag Bike	81031.30
1968 Ford Mustang	72579.26
1969 Ford Falcon	72399.77
1928 Mercedes-Benz SSK	68423.18
2002 Suzuki XREO	67641.47
1980s Black Hawk Helicopter	64599.11
1948 Porsche Type 356 Roadster	62725.78
1917 Grand Touring Sedan	60945.00
1957 Corvette Convertible	59910.22
1962 Volkswagen Microbus	55655.47
1976 Ford Gran Torino	54647.25
1932 Model A Ford J-Coupe	54026.65
1999 Indy 500 Monte Carlo SS	52240.32
1903 Ford Model A	51219.92
1957 Chevy Pickup	51127.01
1940 Ford Pickup Truck	47291.45
1969 Corvair Monza	46521.97
1969 Dodge Charger	45950.46
ATA: B757-300	45058.29
1958 Setra Bus	43366.45

Most Profitable products

The query I wrote calculates the **total profit** for each product.

To determine the **most profitable products**, we need:
Total Profit → (quantityOrdered * (priceEach - buyPrice))
Observation: Again, as expected, **1992 Ferrari 360 Spider red** is the most profitable product for Mint Classics. To check the least profitable one, just remove **DESC** so we have ascending order; then the least profitable will appear first in the list.

EDA_UsingSQL_Project | orders | orderdetails | productlines | warehouses | products

Limit to 1000 rows

```

385 -- How many product lines does each warehouse have:
386 SELECT w.warehouseName,
387        p.productLine,
388        COUNT(p.productLine) AS productLineCount
389 FROM mintclassics.products p
390 JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode
391 GROUP BY w.warehouseName, p.productLine
392 ORDER BY productLineCount DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

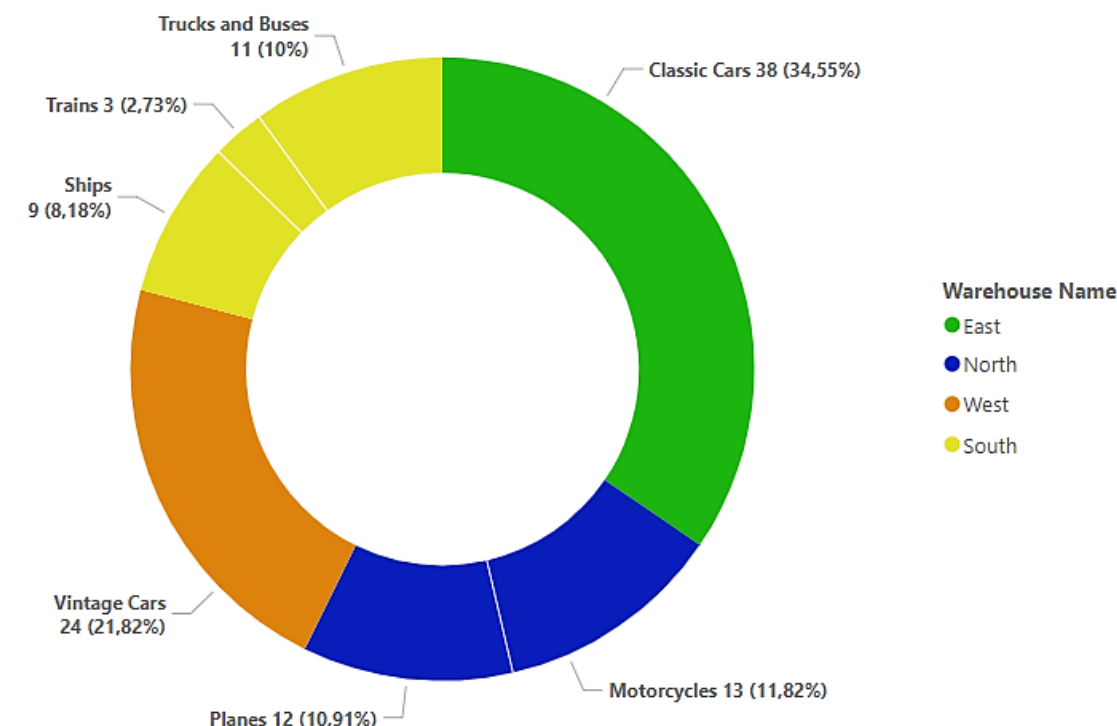
	warehouseName	productLine	productLineCount
	East	Classic Cars	38
	West	Vintage Cars	24
	North	Motorcycles	13
	North	Planes	12
	South	Trucks and Buses	11
	South	Ships	9
	South	Trains	3

With the SQL query got what I wanted – information about the amount of product lines in our warehouses. The aim was to understand the warehouse system of Mint Classics. I visualized this using Donut chart in Power BI. Each slice here represents the number of each product line and its percentage of total.

To get familiar with the dataset I want to analyse warehouses:

- How to analyse warehouses?
- What do warehouses contain?
- What kind of products?
- What information might be useful to solve the business problem we have?
- What insight we can potentially gain by knowing the number of product lines in each warehouse?

Number of Product Lines per Warehouse



EDA_UsingSQL_Project x orders orderdetails

Limit to

```
290
291 -- Warehouse Utilization:
292 • SELECT warehouseName, warehousePctCap
293    FROM mintclassics.warehouses;
294
```

Result Grid | Filter Rows: | Export:

	warehouseName	warehousePctCap
▶	South	75
	North	72
	East	67
	West	50

Warehouse Utilization

To analyse warehouse utilization, we need:

- **Percentage Capacity Used** → warehousePctCap (already available in warehouses table).

What this query does:

- Retrieves warehouse names and their **percentage capacity usage**.

```
EDA_UsingSQL_Project x orders orderdetails
Limit to 1000 rows
358 -- Stock levels: this query shows how many stocks each product line have. In total, there is 7 product lines.
359 -- Classic cars are the one with the highest stock level:
360 • SELECT w.warehouseName,
361         p.productLine,
362         SUM(p.quantityInStock) AS totalStock
363 FROM mintclassics.products p
364 JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode
365 GROUP BY w.warehouseName, p.productLine
366 ORDER BY totalStock DESC;
```

warehouseName	productLine	totalStock
East	Classic Cars	219183
West	Vintage Cars	124880
North	Motorcycles	69401
North	Planes	62287
South	Trucks and Buses	35851
South	Ships	26833
South	Trains	16696

What this query does:

- Retrieves stock levels for each **product line** in each **warehouse**.
- Groups data by **warehouse** and **product line**.
- Sorts by **warehouse name** and **total stock (descending order)**.

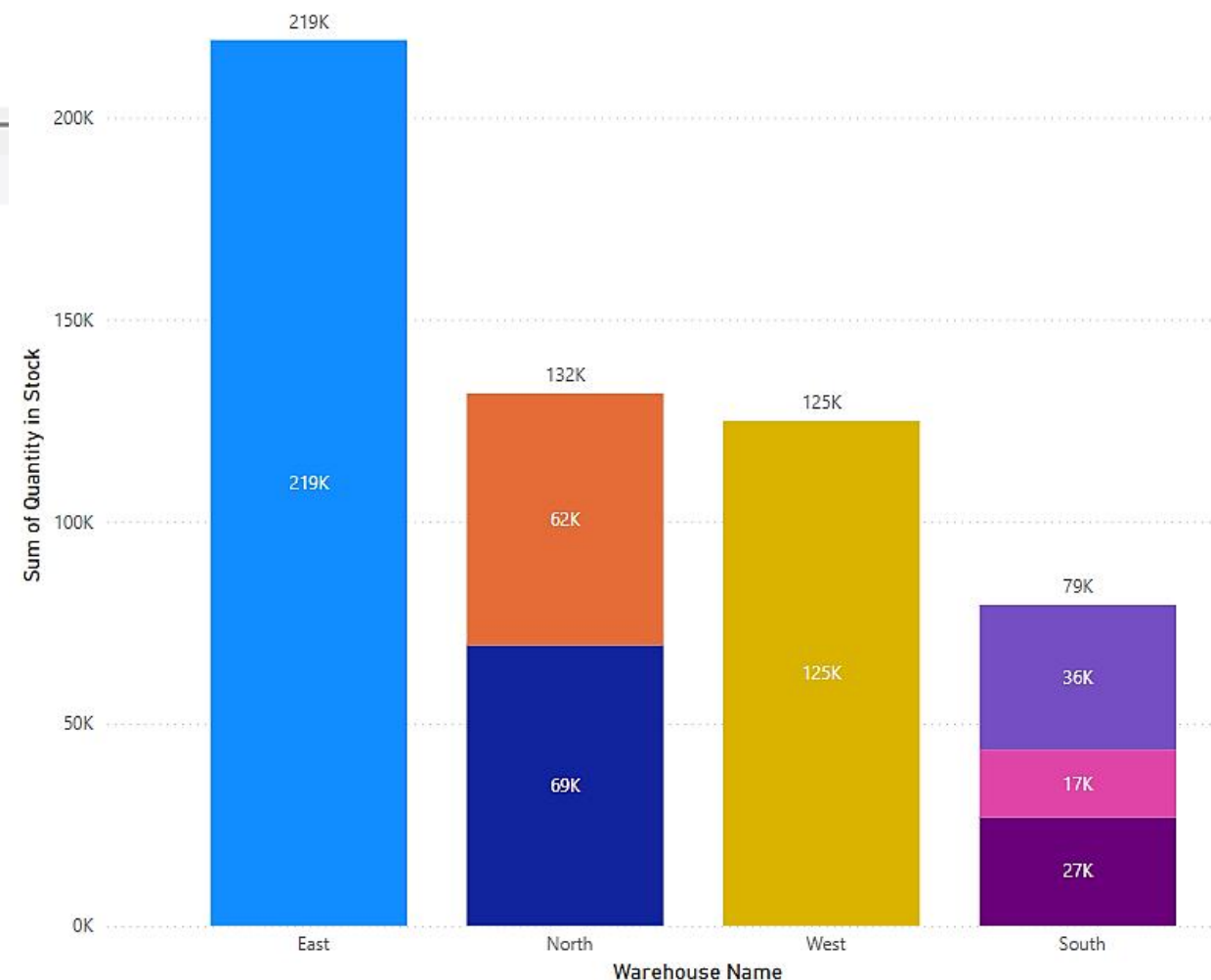
Stock levels: this query shows how many stocks each product line have. In total, there is 7 product lines. **Classic cars** are the one with the highest stock levels and based only in **East Warehouse**. The column chart on the right visualize what I wrote in the SQL query. **Trains** has the least stock level that is based in **South Warehouse**.

To analyze stock levels, we need:

- warehouseName (from warehouses)
- productLine (from productlines)
- quantityInStock (from products)

Sum of quantity In Stock by Warehouse Name and Product Line

Product Lines ● Classic Cars ● Motorcycles ● Planes ● Ships ● Trains ● Trucks and Buses ● Vintage Cars



```

EDA_UsingSQL_Project* x orders orderdetails productlines warehouses products SQL File 4*
Limit to 1000 rows
370 -- This query shows which product belongs to which product line,
371 -- where the products are stored, and how many stocks of each product we have in total:
372 • SELECT w.warehouseName,
373          w.warehouseCode,
374          w.warehousePctCap,
375          p.productLine,
376          p.productName,
377          SUM(p.quantityInStock) AS totalStock
378 FROM mintclassics.products p
379 JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode
380 JOIN mintclassics.productlines pl ON p.productLine = pl.productLine
381 -- WHERE productName LIKE '%red%'
382 -- WHERE p.productLine = 'Classic Cars'
383 GROUP BY w.warehouseName, p.productLine, p.productName, w.warehousePctCap, w.warehouseCode
384 ORDER BY totalStock DESC;

```

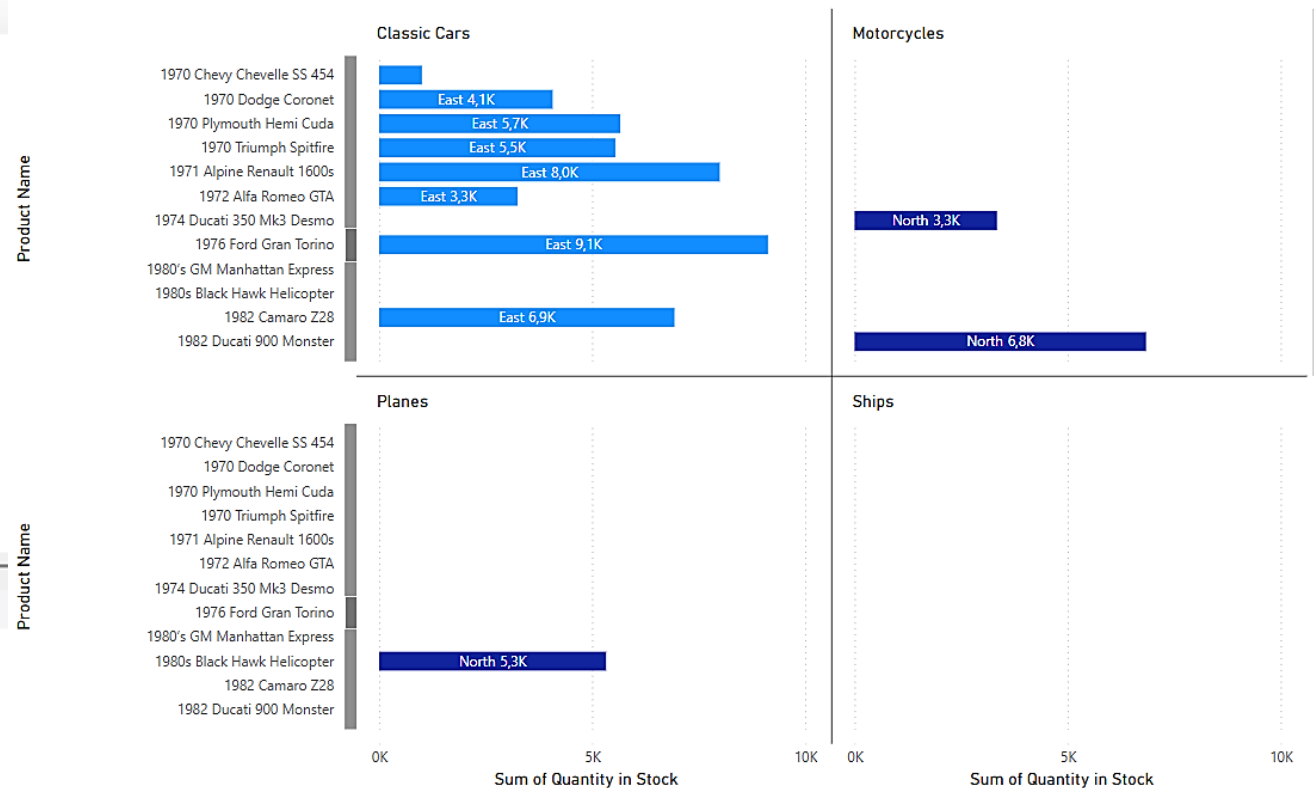
Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	warehouseName	warehouseCode	warehousePctCap	productLine	productName	totalStock
▶	North	a	72	Motorcycles	2002 Suzuki XREO	9997
	East	b	67	Classic Cars	1995 Honda Civic	9772
	North	a	72	Planes	America West Airlines B757-200	9653
	East	b	67	Classic Cars	2002 Chevy Corvette	9446
	West	c	50	Vintage Cars	1932 Model A Ford J-Coupe	9354
	North	a	72	Motorcycles	1982 Ducati 996 R	9241
	West	c	50	Vintage Cars	1912 Ford Model T Delivery Wagon	9173
	East	b	67	Classic Cars	1976 Ford Gran Torino	9127
	East	b	67	Classic Cars	1968 Dodge Charger	9123
	East	b	67	Classic Cars	1965 Aston Martin DB5	9042
	East	b	67	Classic Cars	1948 Porsche Type 356 Roadster	8990
	East	b	67	Classic Cars	1948 Porsche 356-A Roadster	8826
	North	a	72	Planes	American Airlines: MD-11S	8820

Result 95 x

Sum of Quantity in Stock by Product, Warehouse & Product Line

Warehouse names: ● East ● North ● South ● West



Tip: To see the low stock products, basically, remove DESC in the last line of the query.

Result: **2002 Suzuki XREO** is the product with high stock level of **9997**, whereas the **1960 BSA Gold Star DBD34** has the lowest stock level of **15**. **What should this mean? ->check it.**

Visualization: I preferred Stacked bar chart from Power BI to have information I want to analyse altogether. Only part of the chart is shown in the snapshot, the whole chart is available in my project folder. Basically, scroll down/up to see the details for each product.

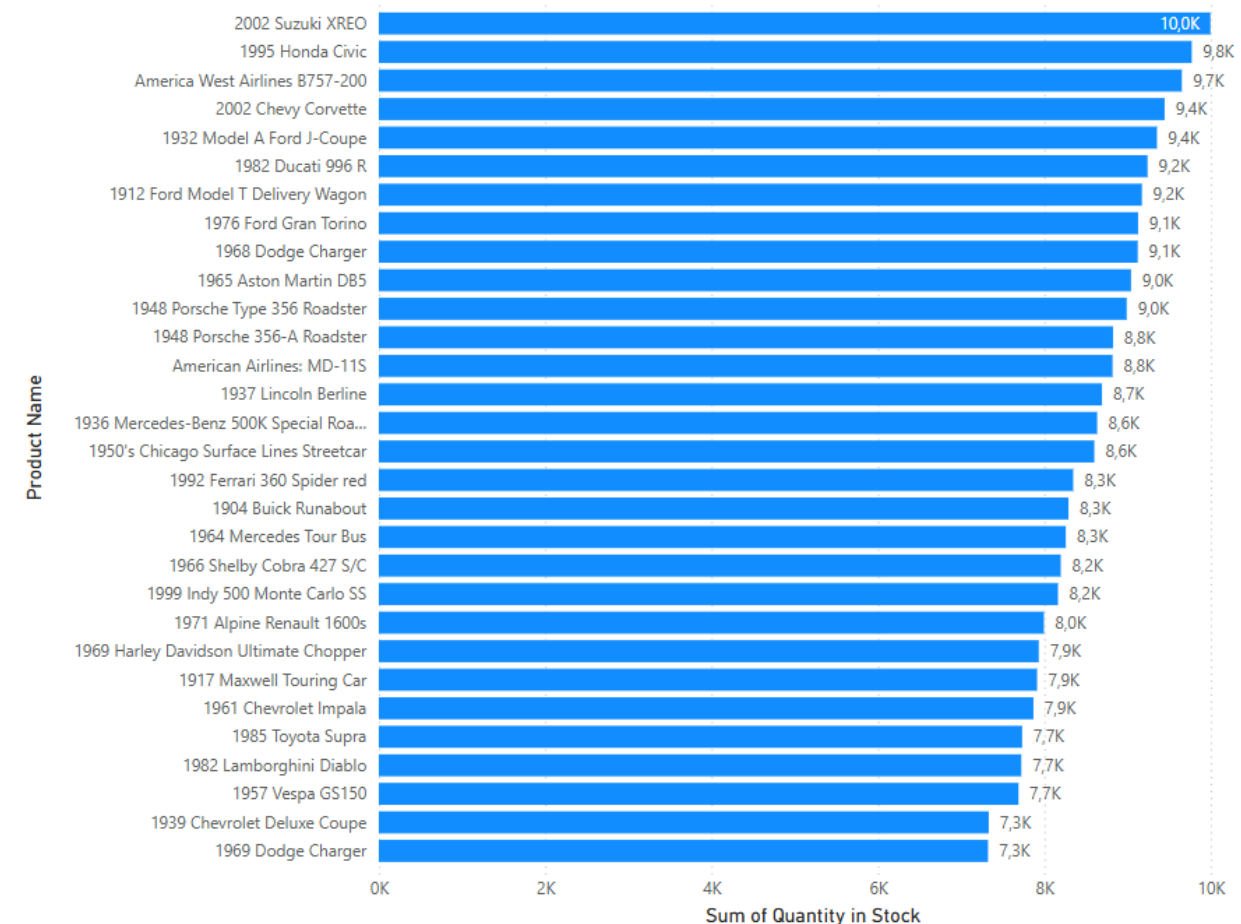
EDA_UsingSQL_Project x orders orderdetails productlin			EDA_UsingSQL_Project* x orders orderdetails productlin		
Limit to 1000			Limit to 1000 r		
<pre> 409 -- To find 10 Most Stocked Products: 410 SELECT productName, quantityInStock 411 FROM mintclassics.products 412 ORDER BY quantityInStock DESC 413 LIMIT 10; </pre>			<pre> 409 -- To find 10 Least Stocked Products: 410 SELECT productName, quantityInStock 411 FROM mintclassics.products 412 ORDER BY quantityInStock 413 LIMIT 10; </pre>		
Result Grid		Filter Rows:	Result Grid		Filter Rows:
productName	quantityInStock		productName	quantityInStock	
2002 Suzuki XREO	9997		1960 BSA Gold Star DBD34	15	
1995 Honda Civic	9772		1968 Ford Mustang	68	
America West Airlines B757-200	9653		1928 Ford Phaeton Deluxe	136	
2002 Chevy Corvette	9446		1997 BMW F650 ST	178	
1932 Model A Ford J-Coupe	9354		Pont Yacht	414	
1982 Ducati 996 R	9241		1911 Ford Town Car	540	
1912 Ford Model T Delivery Wagon	9173		1928 Mercedes-Benz SSK	548	
1976 Ford Gran Torino	9127		F/A 18 Hornet 1/72	551	
1968 Dodge Charger	9123		2002 Yamaha YZR M1	600	
1965 Aston Martin DB5	9042		The Mayflower	737	

Visualization: to see the low stock product in the clustered bar chart, basically, scroll down.

What this query does:

- Retrieves the **top 10 products** with the **highest and the lowest stock levels**.
- Orders them **from highest to lowest** and **lowest to highest** stock quantity.

Most & Least Stocked Products



MSRP, Profit Margin & Markup Percentage

What Does MSRP Mean?

- **MSRP = Manufacturer's Suggested Retail Price**
- It's the price that the manufacturer recommends selling the product for.
- The difference between **buyPrice** and **MSRP** could show **profit margins**.
- You can also calculate the **markup percentage** if you know cost price (buyPrice) and MSRP:

Formula for Markup Percentage

$$\text{Markup Percentage} = \left(\frac{\text{MSRP} - \text{buyPrice}}{\text{buyPrice}} \right) \times 100$$

This shows the percentage increase from buyPrice to MSRP.

Example Calculation

Let's say we have a product with:

- **buyPrice** = \$50
- **MSRP** = \$80

$$\text{Markup Percentage} = \left(\frac{80 - 50}{50} \right) \times 100 = 60\%$$

This means the MSRP is 60% higher than the buy price.

We can also check which warehouse the products with higher markup percentage belong to => this way we can get an idea about which warehouse is important and should not to be closed. We can also check the products with the less markup percentage, then we can see what can be done to increase the profit, or it can also give idea about if we can close the warehouses they belong to.

The **markup percentage** tells us how much a product's price is increased from its cost price (**buyPrice**) to its selling price (**MSRP**). We can calculate the markup percentage per product using SQL queries easily. In the query, I also included the warehouse code to know where these products belong to:

EDA_UsingSQL_Project* x orders orderdetails productlines warehouses products SQL File 4*

Limit to 1000 rows

```
415 -- Calculating markup percentage
416 -- to see how much a product's price is increased from its cost price (buyPrice) to its selling price (MSRP):
417 • SELECT productName, warehouseCode, buyPrice, MSRP,
418       (((MSRP - buyPrice)/ buyPrice)*100) AS markupPercentage
419 FROM mintclassics.products
420 ORDER BY markupPercentage DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	productName	warehouseCode	buyPrice	MSRP	markupPercentage
▶	1961 Chevrolet Impala	b	32.33	80.84	150.046397
	1970 Plymouth Hemi Cuda	b	31.92	79.80	150.000000
	1937 Horch 930V Limousine	c	26.30	65.75	150.000000
	1936 Harley Davidson El Knuckdehead	a	24.23	60.57	149.979364
	1926 Ford Fire Engine	d	24.92	60.77	143.860353
	2002 Yamaha YZR M1	a	34.17	81.36	138.103600
	Corsair F4U (Bird Cage)	a	29.34	68.24	132.583504
	1928 Mercedes-Benz SSK	c	72.56	168.75	132.566152
	1950's Chicago Surface Lines Streetcar	d	26.72	62.14	132.559880
	1999 Indy 500 Monte Carlo SS	b	56.76	132.00	132.558140
	1982 Lamborghini Diablo	b	16.24	37.76	132.512315
	1948 Porsche Type 356 Roadster	b	62.16	141.28	127.284427
	2002 Suzuki XREO	a	66.27	150.62	127.282330
	HMS Bounty	d	39.83	90.52	127.265880
	1958 Chevy Corvette Limited Edition	b	15.91	35.36	122.250157
	1936 Mercedes-Benz 500K Special Roadster	c	24.26	53.91	122.217642
	1939 Cadillac Limousine	c	23.14	50.31	117.415730
	1932 Model A Ford J-Coupe	c	58.48	127.13	117.390561
	1952 Alpine Renault 1300	b	98.58	214.30	117.386894
	2001 Ferrari Enzo	b	95.59	207.80	117.386756
	1982 Camaro Z28	b	46.53	101.15	117.386632
	1992 Ferrari 360 Spider red	b	77.90	169.34	117.381258
	1938 Cadillac V-16 Presidential Limousine	c	20.61	44.80	117.370209
	1957 Corvette Convertible	b	69.93	148.80	112.784213

Result 101 x

```
EDA_UsingSQL_Project x orders orderdetails productlines warehouses products SQL File 4*
Limit to 1000 rows
414 -- Calculating markup percentage
415 -- to see how much a product's price is increased from its cost price (buyPrice) to its selling price (MSRP):
416 • SELECT productName, warehouseCode, buyPrice, MSRP,
417       (MSRP - buyPrice) AS profitMargin,
418       CEILING(((MSRP - buyPrice)/ buyPrice)*100) + '%' AS markupPercentage
419 FROM mintclassics.products
420 ORDER BY markupPercentage DESC;
```

productName	warehouseCode	buyPrice	MSRP	profitMargin	markupPercentage
1961 Chevrolet Impala	b	32.33	80.84	48.51	151
1970 Plymouth Hemi Cuda	b	31.92	79.80	47.88	150
1936 Harley Davidson El Knucklehead	a	24.23	60.57	36.34	150
1937 Horch 930V Limousine	c	26.30	65.75	39.45	150
1926 Ford Fire Engine	d	24.92	60.77	35.85	144
2002 Yamaha YZR M1	a	34.17	81.36	47.19	139
1928 Mercedes-Benz SSK	c	72.56	168.75	96.19	133
1999 Indy 500 Monte Carlo SS	b	56.76	132.00	75.24	133
1982 Lamborghini Diablo	b	16.24	37.76	21.52	133
Corsair F4U (Bird Cage)	a	29.34	68.24	38.90	133
1950's Chicago Surface Lines Streetcar	d	26.72	62.14	35.42	133
2002 Suzuki XREO	a	66.27	150.62	84.35	128
1948 Porsche Type 356 Roadster	b	62.16	141.28	79.12	128
HMS Bounty	d	39.83	90.52	50.69	128
1936 Mercedes-Benz 500K Special Roadster	c	24.26	53.91	29.65	123
1958 Chevy Corvette Limited Edition	b	15.91	35.36	19.45	123
1952 Alpine Renault 1300	b	98.58	214.30	115.72	118
1939 Cadillac Limousine	b	85.50	207.00	121.50	110

Margins can never be more than 100 percent, but markups can be 200 percent, 500 percent, or 10,000 percent, depending on the price and the total cost of the offer. The higher your price and the lower your cost, the higher your markup.

For example, we marked the product **1961 Chevrolet Impala** up 151 %.

I couldn't make a percentage sign to appear in the result, will check it later.

CEILING aggregate function is used to round the results, because they were too long, and not reader friendly (it was containing too many number for stakeholders from different departments ☺).

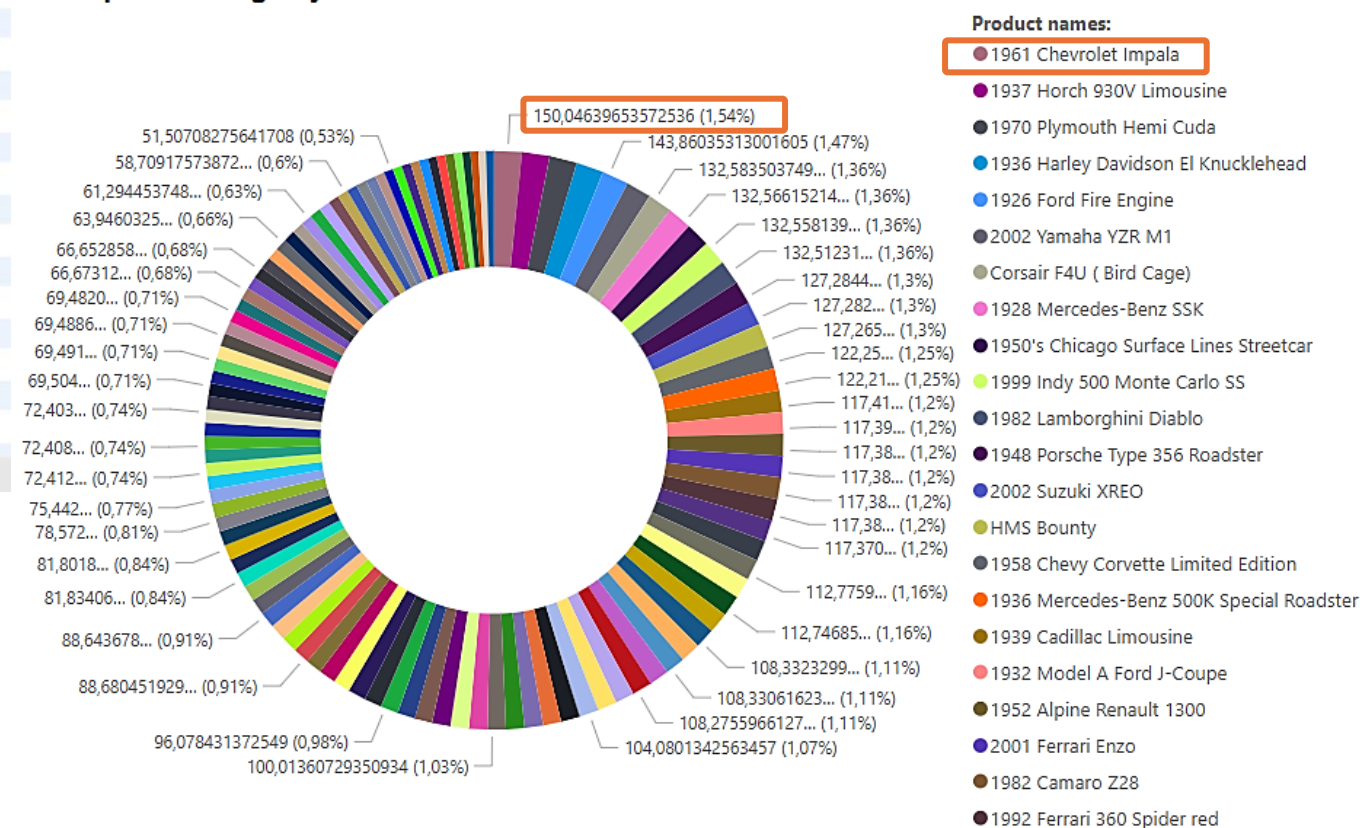
Visualization: I used donut chart for the visualization of markup percentage per product.

Challenge: Calculating markup percentage. The formula I used in **DAX (Data Analysis Expressions)** for Power BI was: $\text{MarkupPercentage} = ('mintclassics\ products'[MSRP] - 'mintclassics\ products'[buyPrice]) / 'mintclassics\ products'[buyPrice] * 100$.

I tried to do this calculation inside a measure. But it did not work, turned out that the error occurs because **Power BI requires explicit aggregation** when performing calculations inside a measure. Since 'mintclassics products'[MSRP] and 'mintclassics products'[buyPrice] contain multiple rows, Power BI doesn't know which single value to use.

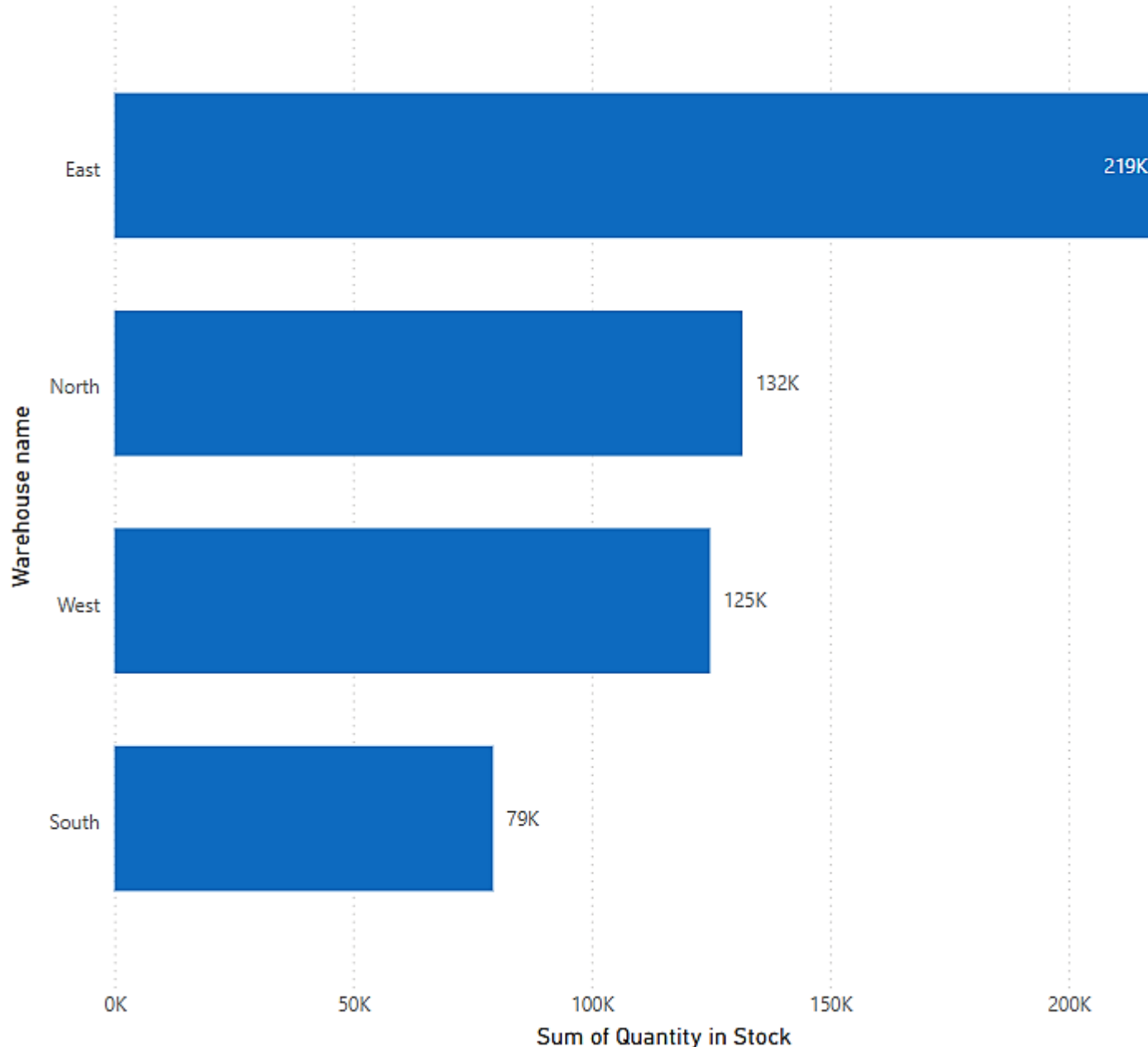
Solution: I solved the problem by using a **calculated column** instead of creating a **measure**. This works because calculated columns evaluate row by row and don't require aggregation.

Markup Percentage by Product



- The company wants to **ship orders within 24 hours**.
- If a warehouse is **understocked**, it might not be able to **fulfil orders quickly**, leading to **delays**.

Total Stock per Warehouse



What We Should Look For in the Chart?

I created the **Total Stock per Warehouse** chart in Power BI. Now we need to consider these:

- **Which warehouse holds the most inventory?** (Is it overloaded?)
- **Which warehouse has the least inventory?** (Is it underutilized?)
- **Are stocks evenly distributed, or is there an imbalance?**

If a warehouse has **very little stock**, it could indicate:

- It's **not needed** anymore.
- It's **not receiving enough shipments**.
- Products should be **redistributed** to other warehouses.

This insight will help us **decide if closing or reorganizing a warehouse is a good move**.

Trends that I noticed:

It turns out that the East warehouse holds the highest stock, with 219K units, while the South warehouse has the lowest, at 79K. The North and West warehouses have 132K and 125K units, respectively.

Now, let's **interpret** these findings and figure out what they mean for the business decision.

What does this tell us?

Total stock per warehouse summary:

- **East Warehouse** → **219K** (Highest stock)
- **North Warehouse** → **132K**
- **West Warehouse** → **125K**
- **South Warehouse** → **79K** (Lowest stock)

Key observations:

East warehouse has the most stock (219K), which could mean:

- It is a **primary hub** for inventory storage.
- It might handle **higher-demand products**.
- It could be **overloaded**, making logistics harder.

South warehouse has the least stock (79K), which raises questions:

- Is it **underutilized**?
- Does it hold **low-demand products**?
- Could we **redistribute its stock to other warehouses** and close it?

North & West warehouses have nearly equal stock (~130K), suggesting:

- They **play a balanced role** in inventory storage.
- They might **support different regions/customers**.

What we need to investigate next

Before recommending any warehouse closures, we should explore:

- **Stock Turnover by Warehouse** → Are some warehouses selling their stock faster than others? => if yes, then this warehouse is very efficient, and there's no need to close it.
- **Most & Least Sold Products by Warehouse** → Does the South warehouse store low-demand products?
- **Warehouse Efficiency (Capacity Usage %)** → Does South warehouse operate far below capacity?

Interpreting the results

- **High Turnover Ratio** (efficient warehouse):
→ Products sell quickly, suggesting **high demand & good inventory management**.
- **Low Turnover Ratio** (potential issue):
→ Products move slowly, meaning the warehouse might be **overstocked or dealing with low-demand items**.

If **South Warehouse has a low stock turnover ratio**, this could be a **strong reason to close it**. But before coming to that decision, let's check if this is indeed true.

Stock Turnover by Warehouse

We'll analyze **how fast products are being sold from each warehouse**. This will help us determine if South warehouse is just **low in stock but efficient** or if it is **truly underperforming and should be closed**.

Analysis: We need to calculate **Stock Turnover**, which measures how quickly inventory is being sold and replaced. A higher turnover indicates **efficient inventory movement**, while a lower turnover might suggest **overstocking or low demand**.

Stock Turnover can be calculated as:

$$\text{Stock Turnover} = \frac{\text{Total Products Sold from a Warehouse}}{\text{Total Stock in that Warehouse}}$$

Where:

- **Total Products Sold from a Warehouse** → Sum of **quantityOrdered** from **orderdetails**, grouped by **warehouseCode**.
- **Total Stock in that Warehouse** → **quantityInStock** from **products**, grouped by **warehouseCode**.

We first extract the necessary data from MySQL:

The screenshot shows the SQL Developer interface. At the top are tabs for 'orders', 'orderdetails', 'productlines', 'warehouses', and 'products'. The main area displays a SQL query in a window titled 'EDA_UsingSQL_Project'. The query is as follows:

```
-- Stock Turnover by Warehouse:
SELECT
    p.warehouseCode,
    w.warehouseName,
    SUM(od.quantityOrdered) AS totalProductsSold,
    SUM(p.quantityInStock) AS totalStock,
    (SUM(od.quantityOrdered) / NULLIF(SUM(p.quantityInStock), 0)) AS stockTurnoverRatio
FROM mintclassics.products p
JOIN mintclassics.orderdetails od ON p.productCode = od.productCode
JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode
GROUP BY p.warehouseCode, w.warehouseName
ORDER BY stockTurnoverRatio DESC;
```

Below the query editor is the 'Result Grid' tab, which displays the following data:

	warehouseCode	warehouseName	totalProductsSold	totalStock	stockTurnoverRatio
▶	d	South	22351	2186871	0.0102
	a	North	24650	3659553	0.0067
	c	West	22933	3439570	0.0067
	b	East	35582	5844033	0.0061

What to do next?

To make a final recommendation, we should investigate:

- **Shipping times** – Does closing a specific warehouse affect the 24-hour shipping goal?
- **Storage costs** – Which warehouse is the most expensive to operate?
- **Product demand** – Are some warehouses holding too much stock of low-demand products?

Key Observations:

1.Total Products Sold

- The **East warehouse (b)** has sold the highest number of products (**35,582**).
- The **South warehouse (d)** has sold the lowest number of products (**22,351**).

2.Total Stock

- The **East warehouse (b)** holds the highest stock (**5.84M**).
- The **South warehouse (d)** holds the lowest stock (**2.18M**).

3. Stock Turnover Ratio (Total Products Sold / Total Stock)

- The **South warehouse (d)** has the highest turnover ratio (**0.0102**), meaning it is selling its stock the fastest relative to its inventory.
- The **East warehouse (b)** has the lowest turnover ratio (**0.0061**), meaning it holds a lot of stock but sells at a slower rate.
- The **North (a) and West (c) warehouses** have the same turnover ratio (**0.0067**), slightly better than East but not as good as South.

What this means:

- The **South warehouse** is the most efficient in selling stock, which suggests it might not be the best candidate for closure.
- The **East warehouse** holds the most stock but has the lowest turnover, but it's not dealing with the low-demand products, so this could indicate **overstocking or inefficient distribution**.
- **North and West warehouses** have similar performance but are still not as efficient as South.

Next Steps:

- 1. Interpret the results – Which warehouse has the fastest/slowest shipping?
- 2. Visualize in Power BI – A bar chart of average shipping time per warehouse.
- 3. Decide how this affects our closure recommendation.

EDA_USingSQL_Project* | orders | orderdetails | productlines | warehouses | products

Limit to 1000 rows

```
456
457 -- Shipping time-Does closing a specific warehouse affect the 24-hour shipping goal?
458 -- This query calculates shipping time:
459 • SELECT p.warehouseCode, w.warehouseName,
460        ROUND(AVG(DATEDIFF(o.shippedDate, o.orderDate)), 2) AS avgShippingTime
461 FROM mintclassics.orders o
462 JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
463 JOIN mintclassics.products p ON od.productCode = p.productCode
464 JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode
465 WHERE o.shippedDate IS NOT NULL -- Ensures we only count shipped orders
466 GROUP BY p.warehouseCode, w.warehouseName
467 ORDER BY avgShippingTime;
468
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	warehouseCode	warehouseName	avgShippingTime
▶	c	West	3.52
	a	North	3.89
	b	East	3.96
	d	South	4.26

I can investigate why the South warehouse is slower → Does it have fewer employees, inefficient routes, or a bottleneck in stock?

Key takeaways from the results:

- West warehouse (3.52 days) has the fastest shipping time.
- South warehouse (4.26 days) has the slowest shipping time.
- The other two warehouses (North: 3.89 days, East: 3.96 days) are in the middle.

The differences between West and South aren't huge but could impact customer satisfaction.

What does this mean for the business?

Since timely shipping is a priority, closing a warehouse with a longer shipping time could be an option. However, we need to balance this with stock levels and demand before making a decision.

Next Steps:

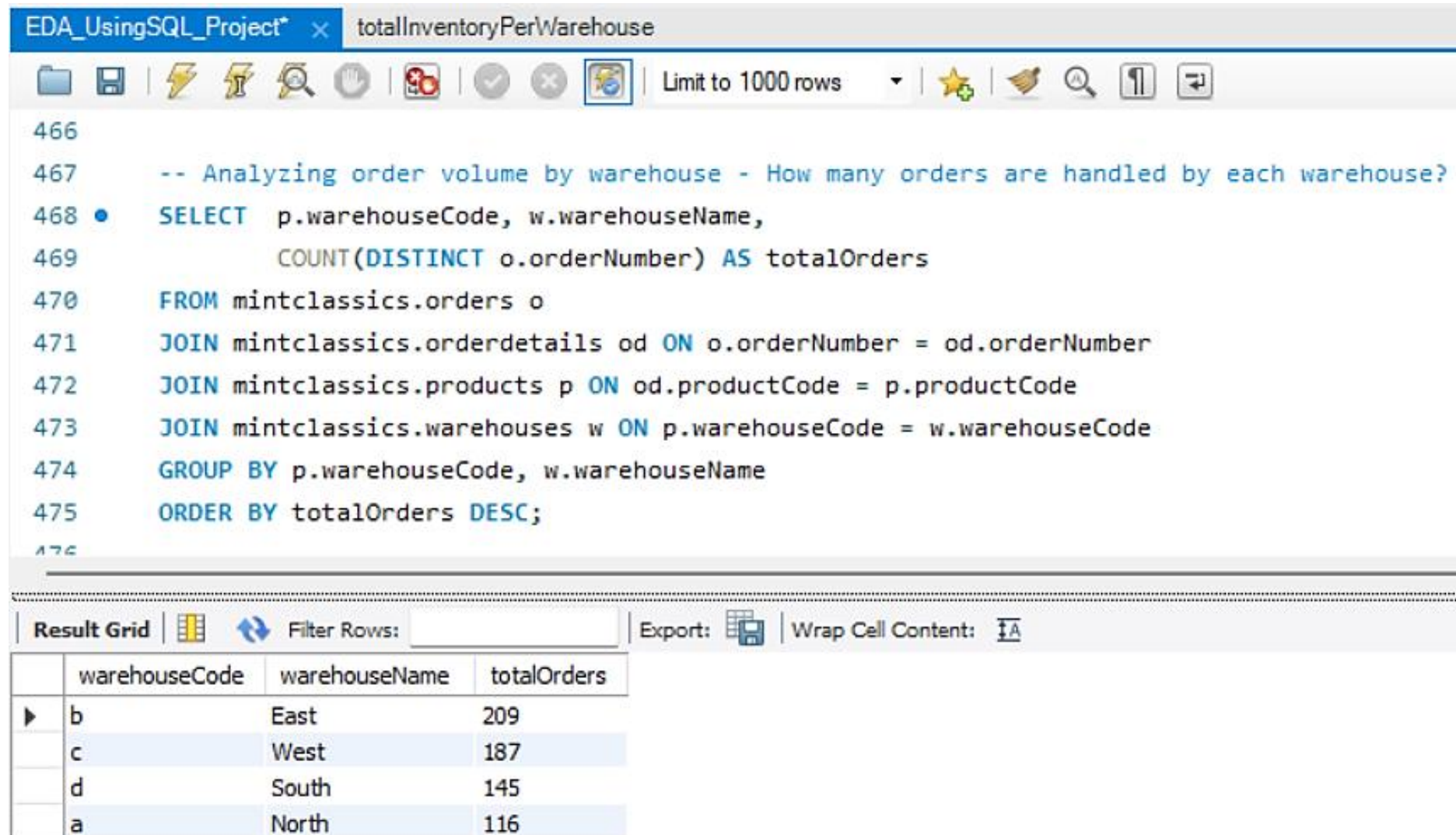
- 1. Visualize the data in Power BI
 - 1. Create a bar chart of average shipping time per warehouse.
 - 2. This will help us clearly see the differences.
- 2. Compare with inventory & sales
 - 1. We've already seen that South has the lowest stock, the lowest number of products sold and one of the lowest turnover ratios.
 - 2. Now, it also has the slowest shipping time.
 - 3. Could this be a strong case for closing the South warehouse?

NOTE: There are some orders with the status "on hold", "cancelled" and "in process" which have NULL shippedDate.

The DATEDIFF Issue (Negative Values)
Problem: The function calculates shippedDate - orderDate, but if shippedDate is before orderDate, you get negative values.
Why? This could happen if some orders haven't shipped yet (i.e., shippedDate is NULL) or if there are data entry errors. In our case, it is the status of the orders.

Analyse Order Volume by Warehouse

We already have **avgShippingTime per warehouse** from SQL. Let's move to **order volume per warehouse** next:



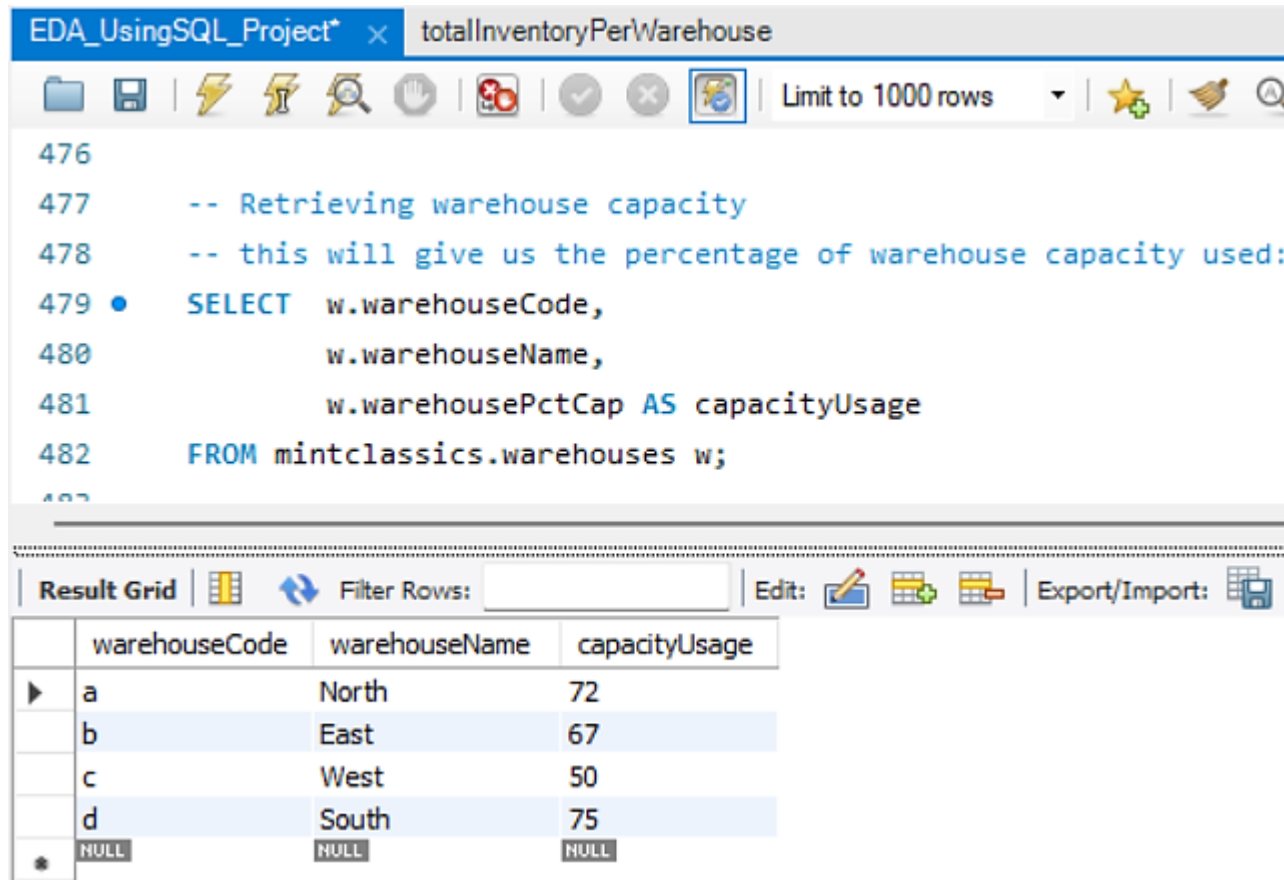
The screenshot shows a SQL IDE window titled "EDA_UsingSQL_Project" with a tab for "totalInventoryPerWarehouse". The query editor contains the following SQL code:

```
-- Analyzing order volume by warehouse - How many orders are handled by each warehouse?  
SELECT p.warehouseCode, w.warehouseName,  
       COUNT(DISTINCT o.orderNumber) AS totalOrders  
FROM mintclassics.orders o  
JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber  
JOIN mintclassics.products p ON od.productCode = p.productCode  
JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode  
GROUP BY p.warehouseCode, w.warehouseName  
ORDER BY totalOrders DESC;
```

Below the query editor, the "Result Grid" tab is active, displaying the results of the query in a table. The table has four columns: warehouseCode, warehouseName, and totalOrders. The results are ordered by totalOrders in descending order.

	warehouseCode	warehouseName	totalOrders
▶	b	East	209
	c	West	187
	d	South	145
	a	North	116

Compare Warehouse Capacity vs. Shipping Time



EDA_UsingSQL_Project* x totalInventoryPerWarehouse

Limit to 1000 rows

```
476
477 -- Retrieving warehouse capacity
478 -- this will give us the percentage of warehouse capacity used:
479 • SELECT w.warehouseCode,
480         w.warehouseName,
481         w.warehousePctCap AS capacityUsage
482 FROM mintclassics.warehouses w;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	warehouseCode	warehouseName	capacityUsage
▶	a	North	72
	b	East	67
	c	West	50
	d	South	75
*	NULL	NULL	NULL

Insights:

- Warehouses with higher capacity usage tend to have longer shipping times.
- West warehouse is operating efficiently with low capacity usage and faster shipping.
- East warehouse, despite handling the most orders, maintains moderate efficiency.
- South warehouse may need improvements (e.g., inventory redistribution, increased space, or process optimization).

Now we can analyze the **warehouse capacity vs. shipping time** to see if warehouses with higher capacity utilization have longer shipping times.

Now that we have **capacity usage** for each warehouse, let's analyze how it correlates with **average shipping time** and **total orders**.

Observations Based on the Data:

1. South warehouse has the highest capacity usage (75%) and the longest average shipping time (4.26 days).

⇒ This suggests that higher capacity usage might lead to delays.

2. North warehouse also has a high capacity usage (72%) and a relatively long shipping time (3.89 days).

⇒ This reinforces the idea that storage congestion could be slowing down shipping.

3. East warehouse has the highest number of orders (209) but a lower capacity usage (67%) and a moderate shipping time (3.96 days).

⇒ This suggests it handles high demand efficiently compared to others.

4. West warehouse has the lowest capacity usage (50%) and one of the shortest shipping times (3.52 days).

⇒ This indicates that having extra storage space may help speed up order processing.

Result Grid Filter Rows: Export: Wrap Cell Content:

Result 17

- **Low stock turnover ratio** (close to 0) → **Product is overstocked but not selling well**
- **High stock turnover ratio** (closer to 1) → **Product is selling efficiently**

Initial analysis of the results:

- **Low Turnover Ratio (<0.2):** Many products have very low sales relative to their stock, indicating overstocking or slow-moving inventory.
- **Moderate Turnover (0.2-0.5):** Some products sell at a steady pace, but stock levels may still be high.
- **High Turnover (>1.0):** These products are selling faster than they are restocked, indicating potential stock shortages.

Interesting observations:

- **Extremely Low Turnover (0.0000):** Example: 'S18_3233' - 1985 Toyota Supra has **7,733** units in stock but **0 sales**. This product might be obsolete or unwanted.

- Very High Turnover (>5.0):

- 'S24_2000' - 1960 BSA Gold Star DBD34' has only **15 units left** but **1,015 sold** → Demand is much higher than supply.
- 'S12_1099' - 1968 Ford Mustang' turnover is **13.72**, meaning it's selling out extremely fast.

- Overstocked Products:

- Products like 'S32_2206' - 1982 Ducati 996 R' have **9,241 in stock** but only **906 sold**.
- 'S700_2466' - America West Airlines B757-200' also has a very high stock-to-sales ratio.

Next Steps:

1. **Identify slow-moving inventory** (low turnover) and consider **discounts or promotions** to sell excess stock.
2. **Ensure sufficient stock for high-turnover products** to avoid lost sales.
3. **Check seasonal trends** to see if some products sell better in certain months.
4. **Analyse warehouse efficiency:** Compare turnover across warehouses to find underperforming locations.

1. Identify Slow-Moving Inventory (Low Turnover)

Products with low turnover may indicate overstocking. A common threshold is a turnover ratio below **0.5**:

EDA_UsingSQL_Project × totalInventoryPerWarehouse

 Limit to 1000 rows

```
509  -- 1. Identifying slow-moving inventory (low stock turnover):
510  •  SELECT  p.productCode, p.productName, p.quantityInStock,
511          COALESCE(SUM(od.quantityOrdered), 0) AS totalSold,
512          ROUND(COALESCE(SUM(od.quantityOrdered), 0) / NULLIF(p.quantityInStock, 0), 4) AS stockTurnoverRatio
513  FROM  mintclassics.products p
514  LEFT JOIN  mintclassics.orderdetails od ON p.productCode = od.productCode
515  GROUP BY  p.productCode, p.productName, p.quantityInStock
516  HAVING  stockTurnoverRatio < 0.5
517  ORDER BY  stockTurnoverRatio ASC;
```

Result Grid   Filter Rows: | Exports:  | Wrap Cell Content: 

	productCode	productName	quantityInStock	totalSold	stockTurnoverRatio
▶	S18_3233	1985 Toyota Supra	7733	0	0.0000
	S18_1984	1995 Honda Civic	9772	917	0.0938
	S24_3432	2002 Chevy Corvette	9446	894	0.0946
	S32_2206	1982 Ducati 996 R	9241	906	0.0980
	S18_3482	1976 Ford Gran Torino	9127	915	0.1003
	S18_1589	1965 Aston Martin DB5	9042	914	0.1011
	S12_3380	1968 Dodge Charger	9123	925	0.1014
	S700_2466	America West Airlines B757-200	9653	984	0.1019
	S18_2325	1932 Model A Ford J-Coupe	9354	957	0.1023
	S12_2823	2002 Suzuki XREO	9997	1028	0.1028
	S18_2870	1999 Indy 500 Monte Carlo SS	8164	855	0.1047
	S18_3685	1948 Porsche Type 356 Roadster	8990	948	0.1055
	S24_3151	1912 Ford Model T Delivery Wagon	9173	991	0.1080
	S32_3207	1950's Chicago Surface Lines Streetcar	8601	934	0.1086
	S18_1889	1948 Porsche 356-A Roadster	8826	972	0.1101

Result 24 ×

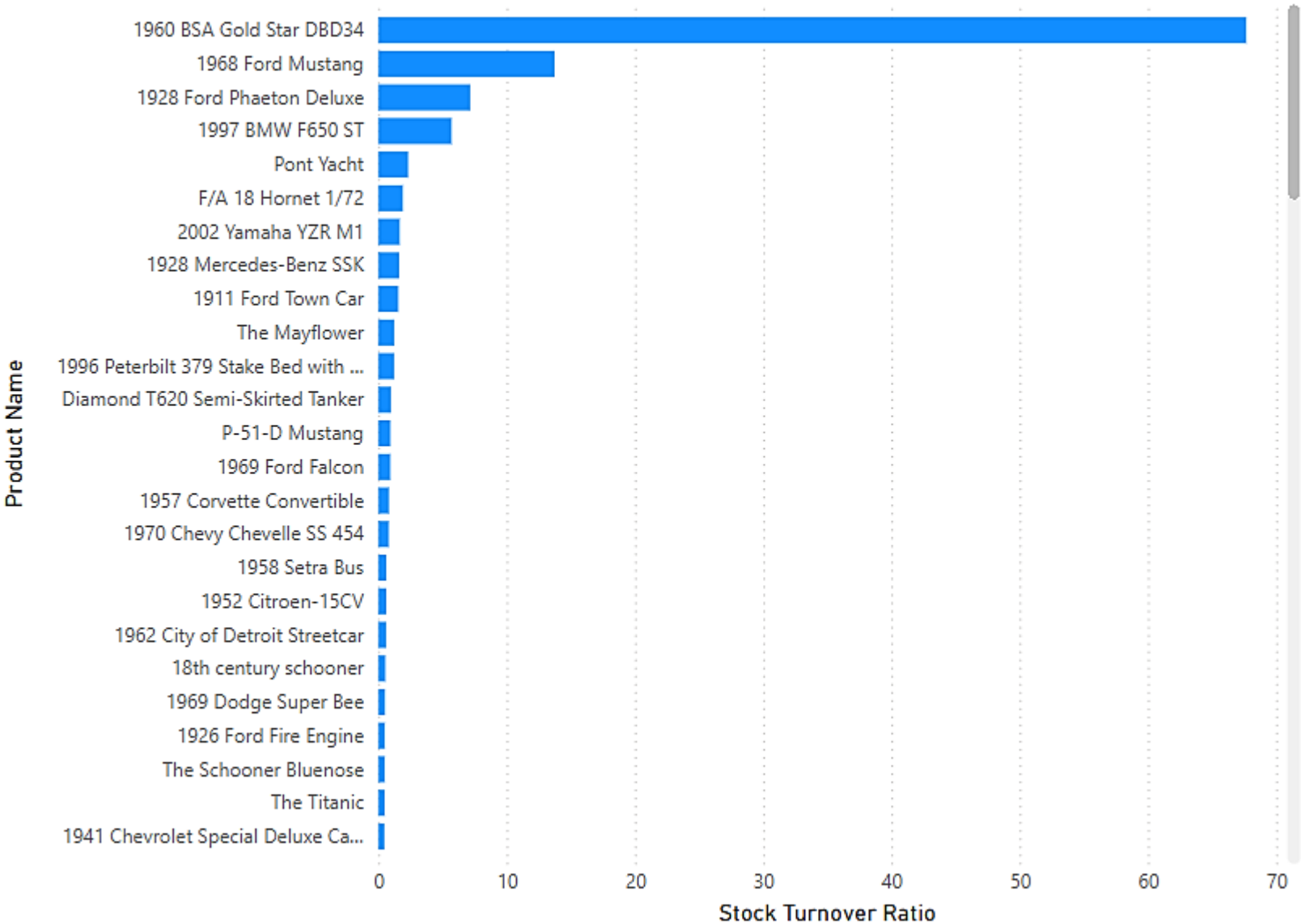
Analysis:

Products with **very low turnover** are candidates for promotions or discounts.

1. Identify Slow-Moving Inventory (Low Turnover)

Visualization: Bar Chart – Slow-Moving vs. Fast-Selling Products

Slow-Moving vs. Fast-Selling Products



- Analysis:**
- **Low-turnover products** (bars on the left) may need **discounts** or **promotions**.
 - **High-turnover products** (bars on the right) might need **higher stock levels**.

Scroll down to see products with low-turnover ratio.

2. Ensure Sufficient Stock for High-Turnover Products

If a product sells quickly, maintaining adequate stock prevents lost sales.

EDA_UsingSQL_Project x totalInventoryPerWarehouse

Limit to 1000 rows

```
520 -- 2. Ensuring sufficient stock for high-turnover products:
521 • SELECT p.productCode, p.productName, p.quantityInStock,
522         COALESCE(SUM(od.quantityOrdered), 0) AS totalSold,
523         ROUND(COALESCE(SUM(od.quantityOrdered), 0) / NULLIF(p.quantityInStock, 0), 4) AS stockTurnoverRatio
524 FROM mintclassics.products p
525 LEFT JOIN mintclassics.orderdetails od ON p.productCode = od.productCode
526 GROUP BY p.productCode, p.productName, p.quantityInStock
527 HAVING stockTurnoverRatio > 2 -- Considered high turnover
528 ORDER BY stockTurnoverRatio DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: IA

	productCode	productName	quantityInStock	totalSold	stockTurnoverRatio
▶	S24_2000	1960 BSA Gold Star DBD34	15	1015	67.6667
	S12_1099	1968 Ford Mustang	68	933	13.7206
	S32_4289	1928 Ford Phaeton Deluxe	136	972	7.1471
	S32_1374	1997 BMW F650 ST	178	1014	5.6966
	S72_3212	Pont Yacht	414	958	2.3140

Analysis: These products might need **higher stock levels** to prevent shortages.

Sufficient Stock for High-Turnover Products

productName	Sum of quantityInStock	StockTurnoverRatio	HighTurnoverFlag
1928 Ford Phaeton Deluxe	136	7,15	High Turnover
1960 BSA Gold Star DBD34	15	67,67	High Turnover
1968 Ford Mustang	68	13,72	High Turnover
1997 BMW F650 ST	178	5,70	High Turnover
Pont Yacht	414	2,31	High Turnover
18th century schooner	1898	0,53	Normal
18th Century Vintage Horse Carriage	5992	0,15	Normal
1900s Vintage Bi-Plane	5942	0,16	Normal
1900s Vintage Tri-Plane	2756	0,37	Normal
1903 Ford Model A	3913	0,23	Normal
1904 Buick Runabout	8290	0,12	Normal
1911 Ford Town Car	540	1,54	Normal
1912 Ford Model T Delivery Wagon	9173	0,11	Normal
1913 Ford Model T Speedster	4189	0,25	Normal
1917 Grand Touring Sedan	2724	0,34	Normal
1917 Maxwell Touring Car	7913	0,13	Normal
1926 Ford Fire Engine	2018	0,49	Normal
1928 British Royal Navy Airplane	3627	0,27	Normal
1928 Mercedes-Benz SSK	548	1,61	Normal
1930 Buick Marquette Phaeton	7062	0,15	Normal
1932 Alfa Romeo 8C2300 Spider Sport	6553	0,13	Normal
1932 Model A Ford J-Coupe	9354	0,10	Normal
1934 Ford V8 Coupe	5649	0,17	Normal
1936 Chrysler Airflow	4710	0,21	Normal
1936 Harley Davidson El Knucklehead	4357	0,22	Normal
1936 Mercedes Benz 500k Roadster	2081	0,40	Normal
1936 Mercedes-Benz 500K Special Roadster	8635	0,11	Normal
1937 Horch 930V Limousine	2902	0,30	Normal
1937 Lincoln Berline	8693	0,13	Normal
1938 Cadillac V-16 Presidential Limousine	2847	0,34	Normal
1939 Cadillac Limousine	6645	0,15	Normal
Total	555131	0,19	Normal

2. Ensure Sufficient Stock for High-Turnover Products

Analysis:

- Products with **High Turnover** should be **monitored closely** to avoid stockouts.
- Check supplier lead-times to **avoid lost sales**.

3. Check Seasonal Trends for Products

EDA_UsingSQL_Project x totalInventoryPerWarehouse

Limit to 1000 rows

```
-- 3. checking seasonal trends for products:
SELECT MONTH(o.orderDate) AS orderMonth,
       p.productCode, p.productName,
       SUM(od.quantityOrdered) AS totalSold
FROM mintclassics.orders o
JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
JOIN mintclassics.products p ON od.productCode = p.productCode
GROUP BY orderMonth, p.productCode, p.productName
ORDER BY totalSold DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	orderMonth	productCode	productName	totalSold
▶	11	S18_3232	1992 Ferrari 360 Spider red	354
	11	S700_4002	American Airlines: MD-11S	262
	11	S18_2432	1926 Ford Fire Engine	249
	11	S18_4721	1957 Corvette Convertible	242
	11	S18_3856	1941 Chevrolet Special Deluxe Cabriolet	242
	11	S18_4600	1940s Ford truck	241
	11	S24_2841	1900s Vintage Bi-Plane	241
	11	S12_3380	1968 Dodge Charger	240
	11	S700_2610	The USS Constitution Ship	240
	11	S18_1589	1965 Aston Martin DB5	237
	11	S32_1374	1997 BMW F650 ST	236
	11	S18_3685	1948 Porsche Type 356 Roadster	230
	11	S32_3522	1996 Peterbilt 379 Stake Bed with Outrigger	229
	11	S18_1129	1993 Mazda RX-7	228
	11	S18_1342	1937 Lincoln Berline	227

Result 28 x

My insights: I identified that product number S18_3232 sold well in November, which has the highest total sold of 354. Another example is, S24_3969 has the highest sell of 139 in January, etc. In general, trends show that November is the peak sales month (for most of the products).

Some products may have peak sales in specific months.

EDA_UsingSQL_Project x totalInventoryPerWarehouse

Limit to 1000 rows

```
-- 3. checking seasonal trends for products:
SELECT MONTH(o.orderDate) AS orderMonth,
       p.productCode, p.productName,
       SUM(od.quantityOrdered) AS totalSold
FROM mintclassics.orders o
JOIN mintclassics.orderdetails od ON o.orderNumber = od.orderNumber
JOIN mintclassics.products p ON od.productCode = p.productCode
GROUP BY orderMonth, p.productCode, p.productName
ORDER BY orderMonth, totalSold DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

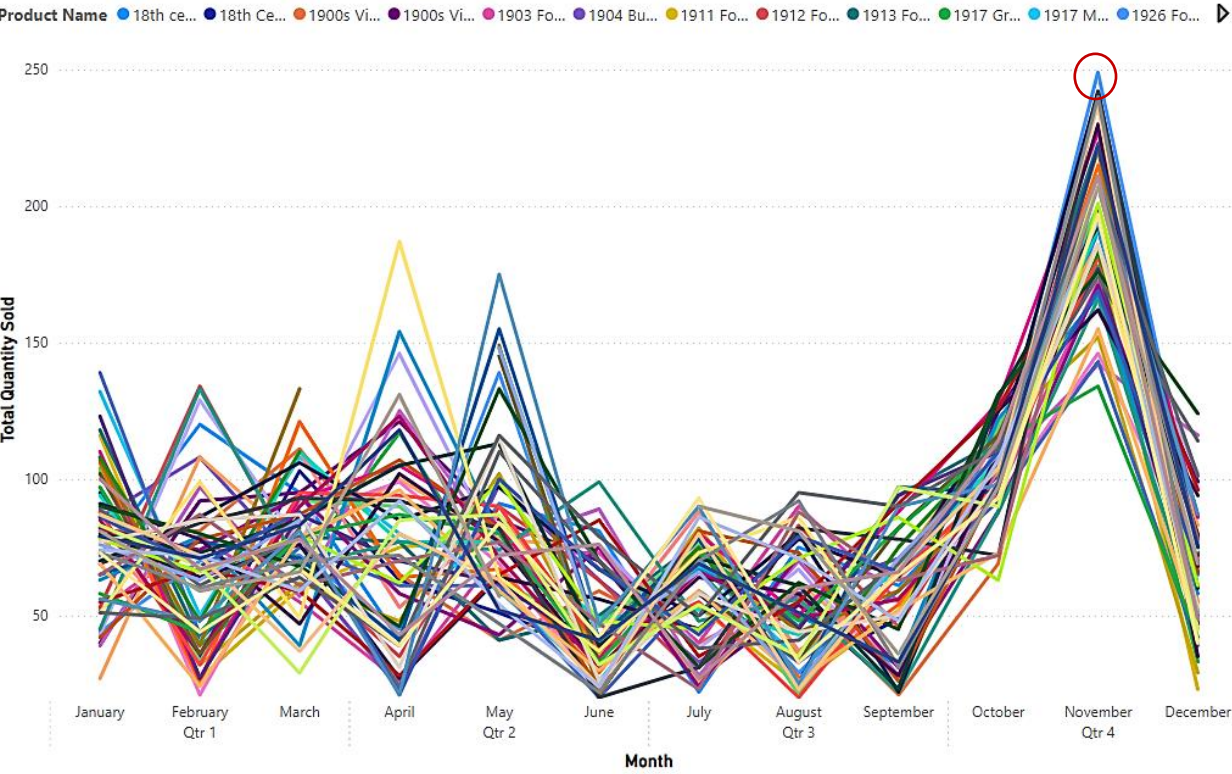
	orderMonth	productCode	productName	totalSold
▶	1	S24_3969	1936 Mercedes Benz 500k Roadster	139
	1	S18_3320	1917 Maxwell Touring Car	132
	1	S18_3232	1992 Ferrari 360 Spider red	124
	1	S24_2022	1938 Cadillac V-16 Presidential Limousine	123
	1	S18_2949	1913 Ford Model T Speedster	118
	1	S18_2248	1911 Ford Town Car	116
	1	S18_1342	1937 Lincoln Berline	110
	1	S18_2795	1928 Mercedes-Benz SSK	108
	1	S18_1097	1940 Ford Pickup Truck	108
	1	S18_4668	1939 Cadillac Limousine	107
	1	S18_1367	1936 Mercedes-Benz 500K Special Roadster	106
	1	S24_4258	1936 Chrysler Airflow	104
	1	S18_4409	1932 Alfa Romeo 8C2300 Spider Sport	103
	1	S10_1949	1952 Alpine Renault 1300	102
	1	S12_3148	1969 Corvair Monza	100

Result 31 x

Analysis:

- Identify products that **sell well in specific months** (e.g., holiday-related items).
- Plan **stock levels** accordingly to match peak demand periods.
- If possible, compare this **year-over-year** to confirm recurring seasonal trends.

Monthly Sales Trends by Product



Note: The message “Too many values” might pop up. Because number of products is too high, so Power BI is not showing all data. However, overall trend is similar, since great number of products sold in **November**.

Too many values

Too many 'productName' values. Not displaying all data. Filter the data or choose another field.

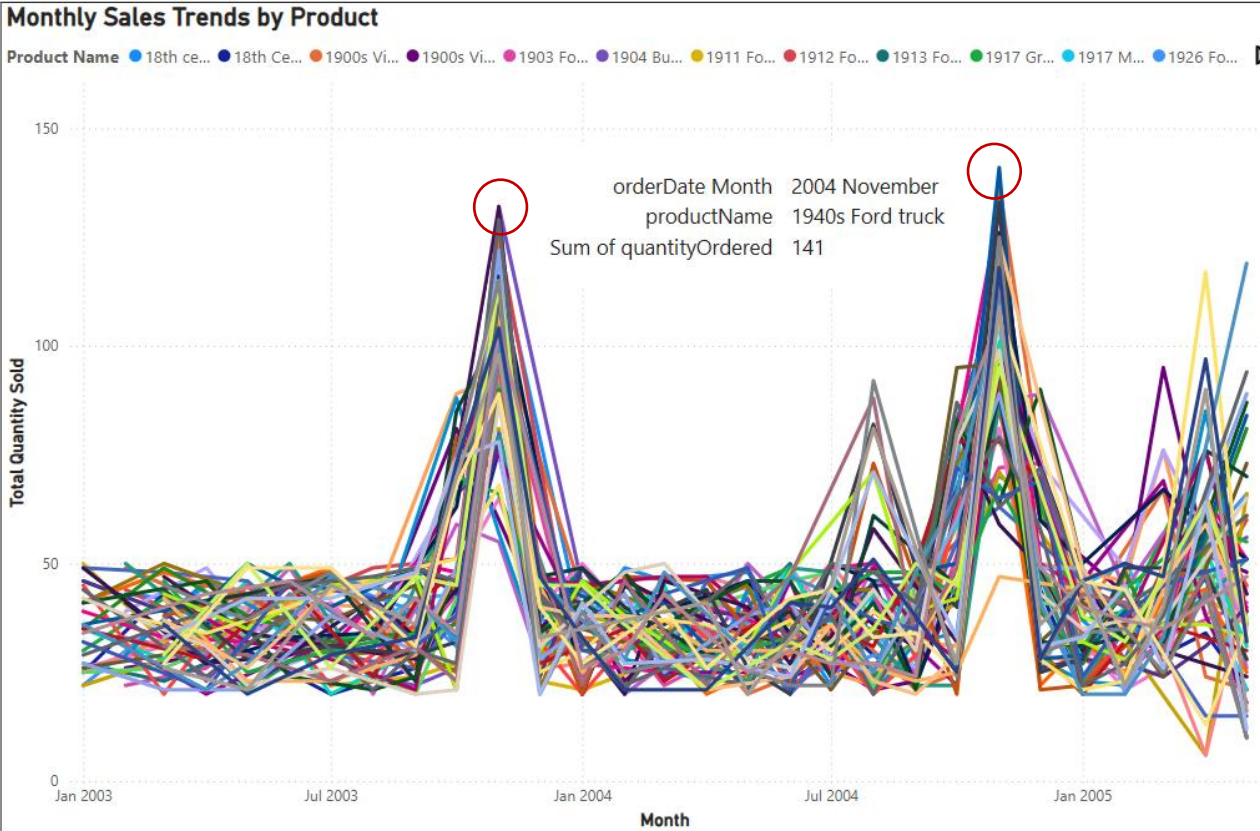
Close

3. Check Seasonal Trends for Products

Visualization: Line Chart – Monthly Sales Trends

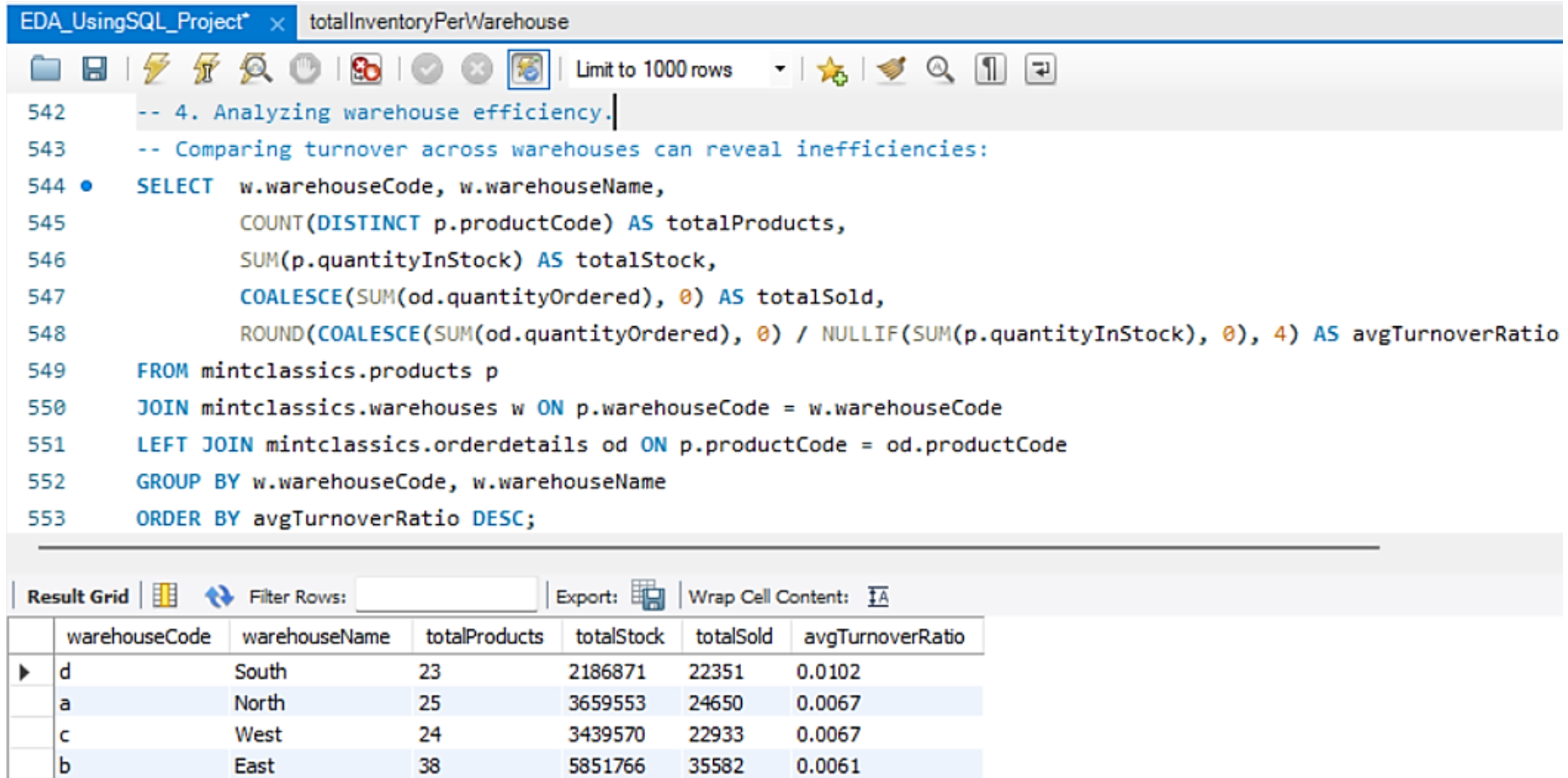
Analysis:

- Identify **peak sales months** for specific products.
 - Plan **restocking strategies** for seasonal demand.
 - Cross-check year-over-year trends for **long-term patterns**.
- ⇒ **when I checked this, it turned out November is the peak sales months for 2004 and 2005 (see chart below).**



4. Analyse Warehouse Efficiency

Comparing turnover across warehouses can reveal inefficiencies.



The screenshot shows a SQL IDE window titled "EDA_UsingSQL_Project" with a sub-tab "totalInventoryPerWarehouse". The query editor contains the following SQL code:

```
-- 4. Analyzing warehouse efficiency.
-- Comparing turnover across warehouses can reveal inefficiencies:
SELECT w.warehouseCode, w.warehouseName,
       COUNT(DISTINCT p.productCode) AS totalProducts,
       SUM(p.quantityInStock) AS totalStock,
       COALESCE(SUM(od.quantityOrdered), 0) AS totalSold,
       ROUND(COALESCE(SUM(od.quantityOrdered), 0) / NULLIF(SUM(p.quantityInStock), 0), 4) AS avgTurnoverRatio
FROM mintclassics.products p
JOIN mintclassics.warehouses w ON p.warehouseCode = w.warehouseCode
LEFT JOIN mintclassics.orderdetails od ON p.productCode = od.productCode
GROUP BY w.warehouseCode, w.warehouseName
ORDER BY avgTurnoverRatio DESC;
```

Below the query editor, the "Result Grid" tab is active, displaying the following data:

	warehouseCode	warehouseName	totalProducts	totalStock	totalSold	avgTurnoverRatio
▶	d	South	23	2186871	22351	0.0102
	a	North	25	3659553	24650	0.0067
	c	West	24	3439570	22933	0.0067
	b	East	38	5851766	35582	0.0061

Analysis:

- Warehouses with **low turnover ratios** may need optimization (e.g., stock redistribution).
- Check if underperforming warehouses have **longer shipping-times** that discourage purchases.
- Consider closing or repurposing low-efficiency storage locations (i.e., if a warehouse consistently has **low sales**) ⇒ South warehouse can be a candidate.

Final Step: Recommendations & Future Work

Now, let's summarize **recommendations** based on our findings and propose **future improvements** for Mint Classics' inventory management.

Final Conclusion & Decision (Updated with Shipping Time Insights)

Optimizing Warehouse Operations

- **The East Warehouse has the lowest efficiency** (low turnover and high inventory levels), making it a **potential candidate for closure or downsizing**.
- **However, the South Warehouse, while having the best turnover ratio, also has the longest average shipping time.** This suggests that it may not be the best option for absorbing inventory from a closed facility.
- Instead, the **North and West Warehouses**, which have similar turnover ratios but lower shipping times than the South Warehouse, should be considered for redistributing inventory.

Improving Inventory Management

- **Liquidate slow-moving stock** in warehouses with high inventory but low turnover (e.g., East Warehouse).
- **Ensure high-demand products** are stocked in warehouses that can fulfill orders quickly.

Balancing Shipping Efficiency vs. Inventory Performance

- The **South Warehouse has the highest stock turnover** but also the **longest average shipping time**.
- The company must **assess whether keeping South Warehouse fully operational** is worth the longer delivery time or whether **some inventory should be relocated** to a more central warehouse for faster shipping.

Warehouse Consolidation Strategy

- If a warehouse is to be closed, **the East Warehouse remains the most likely candidate**, but inventory must be strategically **reallocated to minimize shipping delays**.
- The **North and West Warehouses should be optimized** for absorbing stock without overloading their capacity.

Updated Next Steps & Future Work

- **Reevaluate the South Warehouse's role** – Despite its high efficiency, its **longer shipping times** may require **alternative logistics solutions**.
- **Optimize warehouse selection** – Instead of full closure, consider **redistributing stock based on both turnover and shipping efficiency**.
- **Customer Impact Analysis** – Investigate how **shipping times affect customer satisfaction and repeat purchases**.

Final Thought

By considering **both stock turnover and shipping performance**, Mint Classics can **close or optimize the right warehouse without negatively impacting delivery times**. This ensures a **cost-effective, efficient, and customer-friendly inventory strategy**.

Insights from the analysis

Euro + Shopping Channel has the highest credit limit:

- A high credit limit is typically assigned to **big buyers**, often wholesalers or large retailers.
- This suggests that **Euro + Shopping Channel** is not an average customer but possibly a key business partner.

They placed multiple orders on the same day (while others didn't):

- This could indicate **bulk ordering**, possibly in batches.
- Maybe they have an internal ordering system that splits large orders into separate transactions.
- Or, they could have **different departments** placing orders independently on the same day.

They ordered the highest number of products:

- This strengthens the idea that they might be **a wholesaler or a large distributor** rather than a regular retailer.
- It would be interesting to check **which products they order the most**—maybe they specialize in selling specific types of products.

Some Observations

MySQL Workbench

Local instance x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

Tables

customers

Columns

customerNumber

customerName

contactLastName

contactFirstName

phone

addressLine1

addressLine2

city

state

postalCode

country

salesRepEmployeeNumber

creditLimit

Indexes

Foreign Keys

customers_ibfk_1

Triggers

employees

offices

Administration Schemas

Information

Table: warehouses

Columns:

warehouseCode char(1) PK

warehouseName varchar(45)

warehousePctCap varchar(50)

payments customers employees offices orderdetails orders payments productlines products

Limit to 1000 rows

37

38 • SELECT orders.orderNumber, orders.orderDate, customers.customerName

39 FROM mintclassics.orders

40 INNER JOIN mintclassics.customers ON orders.customerNumber = customers.customerNumber

41 -- GROUP BY orders.orderDate

42 ORDER BY orderDate DESC;

Result Grid

orderNumber	orderDate	customerName
10424	2005-05-31	Euro + Shopping Channel
10425	2005-05-31	La Rochelle Gifts
10422	2005-05-30	Diecast Classics Inc.
10423	2005-05-30	Petit Auto
10420	2005-05-29	Souvenirs And Things Co.
10421	2005-05-29	Mini Gifts Distributors Ltd.
10419	2005-05-17	Salzburg Collectables
10418	2005-05-16	Extreme Desk Decorations, Ltd
10417	2005-05-13	Euro + Shopping Channel
10416	2005-05-10	L'ordine Souvenirs
10415	2005-05-09	Australian Collectables, Ltd
10414	2005-05-06	Gifts4AllAges.com
10413	2005-05-05	Gift Depot Inc.
10412	2005-05-03	Euro + Shopping Channel
10411	2005-05-01	Québec Home Shopping Network
10410	2005-04-29	GiftsForHim.com
10409	2005-04-23	Handji Gifts& Co
10407	2005-04-22	The Sharp Gifts Warehouse
10408	2005-04-22	Tokyo Collectables, Ltd
10406	2005-04-15	Danish Wholesale Imports
10405	2005-04-14	Mini Caravay
10403	2005-04-08	UK Collectables. Ltd.

Result 52 x

Note to myself: Understanding the dataset is indeed helpful. Through this process, I'm getting familiar with the data, and I feel like I actually understand what's going on here. Asking questions and then answering them with queries is surprisingly satisfying.

What I did here: My main goal was to see which order is placed by which customer on what date. To achieve this result, I needed to gather information from two tables (customers and orders), I did that using INNER JOIN and got the result here.

Observations:

- When I listed the order dates in descending order, I found out that the date that the last order placed is 2005-05-31. However, then I noticed that on that very date 10424 was not the only order, the order 10425 was also placed on the same date (by different customer). So, on the same date multiple orders can be placed by different customers.
- When we order by customerName, we can observe that most customers placed more than one separate orders on different dates. E.g. Alpha Cognac placed 3 different orders between the year 2003 and 2005.

MySQL Workbench

Local instance x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

Tables

customers

Columns

customerNumber

customerName

contactLastName

contactFirstName

phone

addressLine1

addressLine2

city

state

postalCode

country

salesRepEmployeeNumber

creditLimit

Indexes

Foreign Keys

customers_ibfk_1

Triggers

employees

offices

Administration Schemas

Information

Table: customers

Columns:

customerNumber int PK

customerName varchar(51)

contactLastName varchar(51)

contactFirstName varchar(51)

phone varchar(51)

addressLine1 varchar(51)

addressLine2 varchar(51)

city varchar(51)

state varchar(51)

payments customers employees offices orderdetails orders payments productlines products warehouses

Limit to 1000 rows

39

40

41 • SELECT orders.orderNumber, orders.orderDate, customers.customerName

42 FROM mintclassics.orders

43 INNER JOIN mintclassics.customers ON orders.customerNumber = customers.customerNumber

44 -- GROUP BY orders.orderDate

45 ORDER BY customerName;

Result Grid

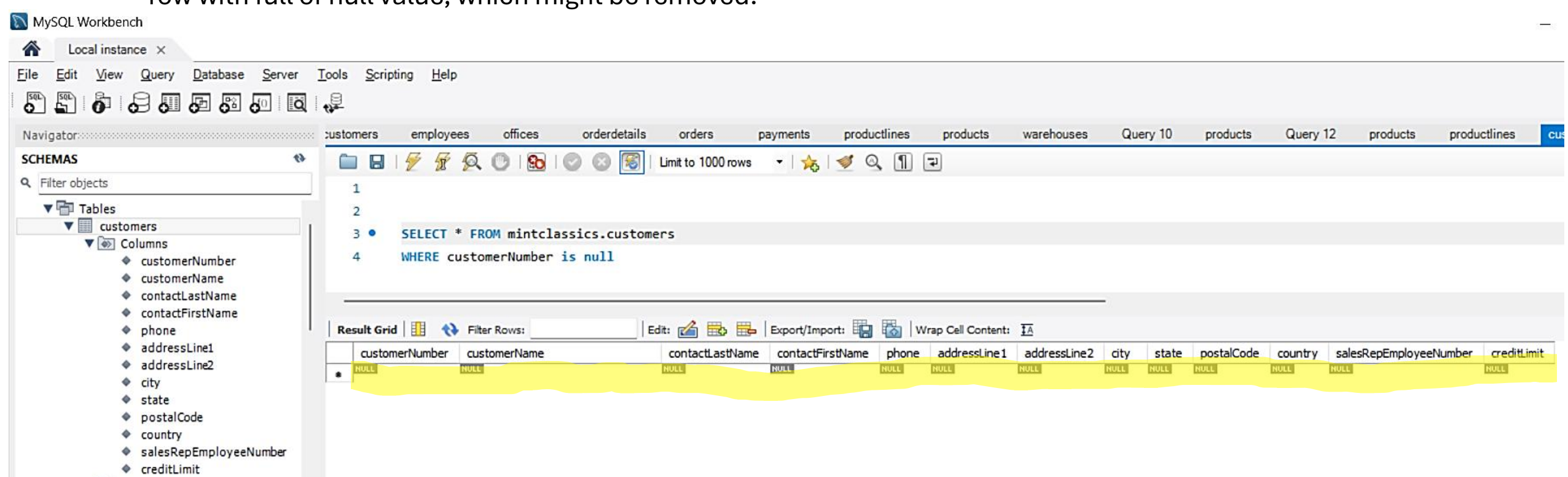
orderNumber	orderDate	customerName
10136	2003-07-04	Alpha Cognac
10178	2003-11-08	Alpha Cognac
10397	2005-03-28	Alpha Cognac
10280	2004-08-17	Amica Models & Co.
10293	2004-09-09	Amica Models & Co.
10148	2003-09-11	Anna's Decorations, Ltd
10169	2003-11-04	Anna's Decorations, Ltd
10370	2005-01-20	Anna's Decorations, Ltd
10391	2005-03-09	Anna's Decorations, Ltd
10123	2003-05-20	Atelier graphique
10298	2004-09-27	Atelier graphique
10345	2004-11-25	Atelier graphique
10193	2003-11-21	Australian Collectables, Ltd
10265	2004-07-02	Australian Collectables, Ltd
10415	2005-05-09	Australian Collectables, Ltd
10120	2003-04-29	Australian Collectors, Co.
10125	2003-05-21	Australian Collectors, Co.
10223	2004-02-20	Australian Collectors, Co.
10342	2004-11-24	Australian Collectors, Co.
10347	2004-11-29	Australian Collectors, Co.
10165	2003-06-16	Australian Gift Network, Co.

Result 55 x

Important note:

It is good to be careful when using INNER JOIN, because you might lose some information as it matches only the values in the columns that appears in both tables. In my dataset, if one customer doesn't have a customer number, then that customer won't show up in the result, because we combine information from both tables by matching rows where the customerNumbers column in the customers table matches the customerNumbers column in the orders table.

For this reason, I checked if there is a null value in the customerNumber column, it turned out there is one row with full of null value, which might be removed.



Key Takeaways (So Far)

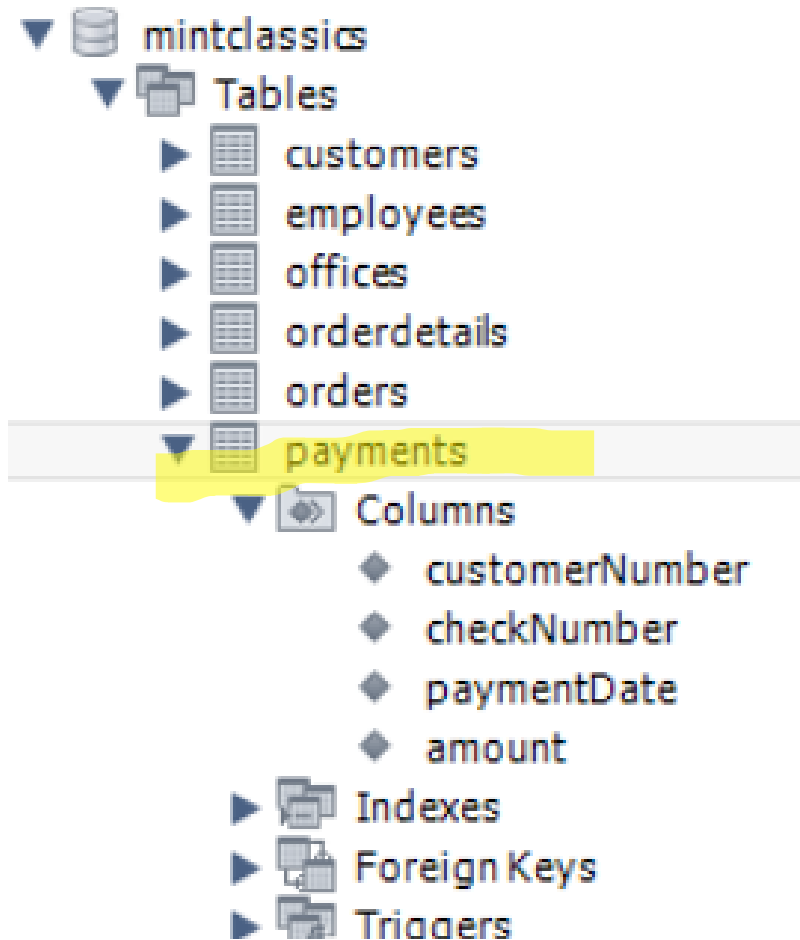
My primary focus has been on analyzing the **orders**, **orderdetails**, **products**, and **customers** tables in the database and exploring the relationships between them. As a result, my key findings so far include:

- A single order by a customer can include **multiple products**. In other words, when a customer places **one order**, they can buy **one or more products**. ⇒ The **orderdetails** table stores **which products** and **how many** of each product are included in the order.
- A customer can place **only one order per day** (i.e., on the same date). However, there is a single exception in our database - the customer with the highest number of orders. Why?
- Over time, a customer can make **multiple orders** containing different products.
- On the same date, **multiple orders** can be placed by different customers (only one exception, **Euro + Shopping Channel** with **customer number 141**).
- The **orderdetails** and **products** tables are linked via the **productCode** column. I used this relationship to join these tables for analysis.
- **orders** and **products** tables are connected through **orderdetails** table.
- **customers** and **orders** tables are connected with **customerNumber** column.

Challenges & Lessons Learned

One of the biggest challenges has been identifying relationships within the data, understanding how the business operates, and getting a clear picture of the database structure. What helped me most during this process was asking **a lot** of questions.

By running SQL queries (and occasionally using visualizations) to answer these questions, I became more familiar with the data, uncovered new insights, and even formulated **more** questions along the way. Having questions in mind is valuable, because in the process of solving them, you end up learning a lot along the way!



4 • `SELECT * FROM mintclassics.payments;`

Result Grid | Filter Rows: | Edit:

	customerNumber	checkNumber	paymentDate	amount
▶	103	HQ336336	2004-10-19	6066.78
	103	JM555205	2003-06-05	14571.44
	103	OM314933	2004-12-18	1676.14
	112	BO864823	2004-12-17	14191.12
	112	HQ55022	2003-06-06	32641.98
	112	ND748579	2004-08-20	33347.88
	114	GG31455	2003-05-20	45864.03
	114	MA765515	2004-12-15	82261.22
	114	NP603840	2003-05-31	7565.08
	114	NR27552	2004-03-10	44894.74
	119	DB933704	2004-11-14	19501.82
	119	LN373447	2004-08-08	47924.19
	119	NG94694	2005-02-22	49523.67
	121	DB889831	2003-02-16	50218.95
	121	FD317790	2003-10-28	1491.38
	121	KI831359	2004-11-04	17876.32
	121	MA302151	2004-11-28	34638.14

payments 1 x

Each product belongs to one product line and is stored in one warehouse.

The diagram shows a database named 'mintclassics' containing several tables: customers, employees, offices, orderdetails, orders, payments, and productlines. The 'productlines' table is highlighted, and its columns are listed below it: productLine, textDescription, htmlDescription, and image. Additionally, there are sections for Indexes, Foreign Keys, and Triggers, each with a plus icon to expand the view.

mintclassics
Tables
customers
employees
offices
orderdetails
orders
payments
productlines
Columns
productLine
textDescription
htmlDescription
image
Indexes
Foreign Keys
Triggers

9 • `SELECT * FROM mintclassics.productlines;`

The screenshot shows a database application interface. At the top, there's a toolbar with icons for 'Result Grid', 'Filter Rows', 'Edit', and 'Export/Import'. Below the toolbar is a table with the following data:

	productLine	textDescription	htmlDescription	image
▶	Classic Cars	Attention car enthusiasts: Make your wildest ca...	NULL	NULL
	Motorcycles	Our motorcycles are state of the art replicas of ...	NULL	NULL
	Planes	Unique, diecast airplane and helicopter replicas ...	NULL	NULL
	Ships	The perfect holiday or anniversary gift for exec...	NULL	NULL
	Trains	Model trains are a rewarding hobby for enthusi...	NULL	NULL
	Trucks and Buses	The Truck and Bus models are realistic replicas o...	NULL	NULL
	Vintage Cars	Our Vintage Car models realistically portray aut...	NULL	NULL
*	NULL	NULL	NULL	NULL

At the bottom, there's a tab labeled 'productlines 1' with a close button (X).

4 • `SELECT * FROM mintclassics.products;`

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	productCode	productName	productLine	productScale	productVendor	productDescription	quantityInStock	warehouseCode	buyPrice	MSF
▶	S10_1678	1969 Harley Davidson Ultimate Chopper	Motorcycles	1:10	Min Lin Diecast	This replica features working kickstand, front su...	7933	a	48.81	95.7
	S10_1949	1952 Alpine Renault 1300	Classic Cars	1:10	Classic Metal Creations	Turnable front wheels; steering function; detail...	7305	b	98.58	214.
	S10_2016	1996 Moto Guzzi 1100i	Motorcycles	1:10	Highway 66 Mini Classics	Official Moto Guzzi logos and insignias, saddle b...	6625	a	68.99	118.
	S10_4698	2003 Harley-Davidson Eagle Drag Bike	Motorcycles	1:10	Red Start Diecast	Model features, official Harley Davidson logos a...	5582	a	91.02	193.
	S10_4757	1972 Alfa Romeo GTA	Classic Cars	1:10	Motor City Art Classics	Features include: Turnable front wheels; steeri...	3252	b	85.68	136.
	S10_4962	1962 LanciaA Delta 16V	Classic Cars	1:10	Second Gear Diecast	Features include: Turnable front wheels; steeri...	6791	b	103.42	147.
	S12_1099	1968 Ford Mustang	Classic Cars	1:12	Autoart Studio Design	Hood, doors and trunk all open to reveal highly ...	68	b	95.34	194.
	S12_1108	2001 Ferrari Enzo	Classic Cars	1:12	Second Gear Diecast	Turnable front wheels; steering function; detail...	3619	b	95.59	207.
	S12_1666	1958 Setra Bus	Trucks and Buses	1:12	Welly Diecast Productions	Model features 30 windows, skylights & glare re...	1579	d	77.90	136.
	S12_2823	2002 Suzuki XREO	Motorcycles	1:12	Unimax Art Galleries	Official logos and insignias, saddle bags located ...	9997	a	66.27	150.
	S12_3148	1969 Corvair Monza	Classic Cars	1:18	Welly Diecast Productions	1:18 scale die-cast about 10" long doors open, ...	6906	b	89.14	151.
	S12_3380	1968 Dodge Charger	Classic Cars	1:12	Welly Diecast Productions	1:12 scale model of a 1968 Dodge Charger. Ho...	9123	b	75.16	117.
	S12_3891	1969 Ford Falcon	Classic Cars	1:12	Second Gear Diecast	Turnable front wheels; steering function; detail...	1049	b	83.05	173.
	S12_3990	1970 Plymouth Hemi Cuda	Classic Cars	1:12	Studio M Art Models	Very detailed 1970 Plymouth Cuda model in 1:1...	5663	b	31.92	79.8
	S12_4473	1957 Chevy Pickup	Trucks and Buses	1:12	Exoto Designs	1:12 scale die-cast about 20" long Hood opens, ...	6125	d	55.70	118.
	S12_4675	1969 Dodge Charger	Classic Cars	1:12	Welly Diecast Productions	Detailed model of the 1969 Dodge Charger. Thi...	7323	b	58.73	115.

products 1 x

Apply

mintclassics
Tables
customers
employees
offices
orderdetails
orders
payments
productlines
products
Columns
productCode
productName
productLine
productScale
productVendor
productDescription
quantityInStock
warehouseCode
buyPrice
MSRP

```

1
2
3 • SELECT * FROM mintclassics.orderdetails;

```

Result Grid			 Filter Rows:	Edit: 			Export/Im
	orderNumber	productCode	quantityOrdered	priceEach	orderLineNumber		
▶	10107	S10_1678	30	81.35	2		
	10121	S10_1678	34	86.13	5		
	10134	S10_1678	41	90.92	2		
	10145	S10_1678	45	76.56	6		
	10159	S10_1678	49	81.35	14		
	10168	S10_1678	36	94.74	1		
	10180	S10_1678	29	76.56	9		
	10188	S10_1678	48	95.70	1		
	10201	S10_1678	22	82.30	2		
	10103	S10_1949	26	214.30	11		
	10112	S10_1949	29	197.16	1		
	10126	S10_1949	38	205.73	11		
	10140	S10_1949	37	186.44	11		
	10150	S10_1949	45	182.16	8		
	10163	S10_1949	21	212.16	1		
	10174	S10_1949	34	207.87	4		
	10183	S10_1949	23	180.01	8		
	10194	S10_1949	42	203.59	11		
	10107	S10_2016	39	105.86	5		
	10120	S10_2016	29	118.94	3		
	10134	S10_2016	27	116.56	5		

orderdetails 1 ×

```

71 -- This query gives price for each product. But why each product has multiple price?
72 • SELECT products.productName AS productName, orderdetails.priceEach
73 FROM mintclassics.orderdetails
74 INNER JOIN mintclassics.products ON products.productCode = orderdetails.productCode
75 ORDER BY productName;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	productName	priceEach
▶	1952 Alpine Renault 1...	214.30
	1952 Alpine Renault 1...	197.16
	1952 Alpine Renault 1...	205.73
	1952 Alpine Renault 1...	186.44
	1952 Alpine Renault 1...	182.16
	1952 Alpine Renault 1...	212.16
	1952 Alpine Renault 1...	207.87
	1952 Alpine Renault 1...	180.01
	1952 Alpine Renault 1...	203.59
	1952 Alpine Renault 1...	203.59
	1952 Alpine Renault 1...	205.73
	1952 Alpine Renault 1...	214.30
	1952 Alpine Renault 1...	195.01
	1952 Alpine Renault 1...	177.87
	1952 Alpine Renault 1...	171.44
	1952 Alpine Renault 1...	205.73
	1950's Chicago Surfac...	56.55
	1950's Chicago Surfac...	60.28
	1950's Chicago Surfac...	60.90
	1950's Chicago Surfac...	60.90
	1950's Chicago Surfac...	57.17

Result 77

We can observe in the snapshots of both orderdetails table and the combination of orderdetails and products table that each product has multiple price.

Q: Why each product has different price? What affects the variation of price? Does the price of a product change depending on the location (country/state/warehouse) it is sold?

The screenshot shows the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. Below the menu is a toolbar with various icons. The left sidebar contains a 'Navigator' pane with a tree view of the database schema. The 'SCHEMAS' pane is active, showing a tree view of the database structure. The 'products' table is selected, and its columns are listed: productCode, productName, productLine, productScale, productVendor, and productDescription. The 'Administration' tab is selected, and the 'Schemas' sub-tab is active. The 'Information' pane at the bottom shows the column definition for 'productName':

Column: **productName**
Collation: latin1_swedish_ci
Definition: productName varchar(70)

The main query editor displays the following SQL query:

```
64  
65 • SELECT products.productName AS productName, SUM(orderdetails.priceEach) AS total_price  
66 FROM mintclassics.orderdetails  
67 INNER JOIN mintclassics.products ON products.productCode = orderdetails.productCode  
68 GROUP BY productName  
69 ORDER BY total_price DESC;
```

The 'Result Grid' pane shows the results of the query. The first row is highlighted in yellow:

productName	total_price
1992 Ferrari 360 Spider red	8074.17
1952 Alpine Renault 1300	5524.66
2001 Ferrari Enzo	5051.61
2003 Harley-Davidson Eagle Drag Bike	4824.07
1968 Ford Mustang	4656.05
1969 Ford Falcon	4271.91
1928 Mercedes-Benz SSK	4196.85
1998 Chrysler Plymouth Prowler	4036.00
1980s Black Hawk Helicopter	3894.95
1917 Grand Touring Sedan	3842.00
1969 Corvair Monza	3719.62
2002 Suzuki XREO	3700.73
1962 LanciaA Delta 16V	3690.57
1976 Ford Gran Torino	3577.71
1995 Honda Civic	3493.71
1957 Corvette Convertible	3490.85
1970 Triumph Spitfire	3484.26
1972 Alfa Romeo GTA	3478.88
1956 Porsche 356A Coupe	3463.00
1958 Setra Bus	3453.66

The 'Result Grid' pane also includes a 'Filter Rows' field and an 'Export' button. The 'Wrap Cell Content' checkbox is checked.

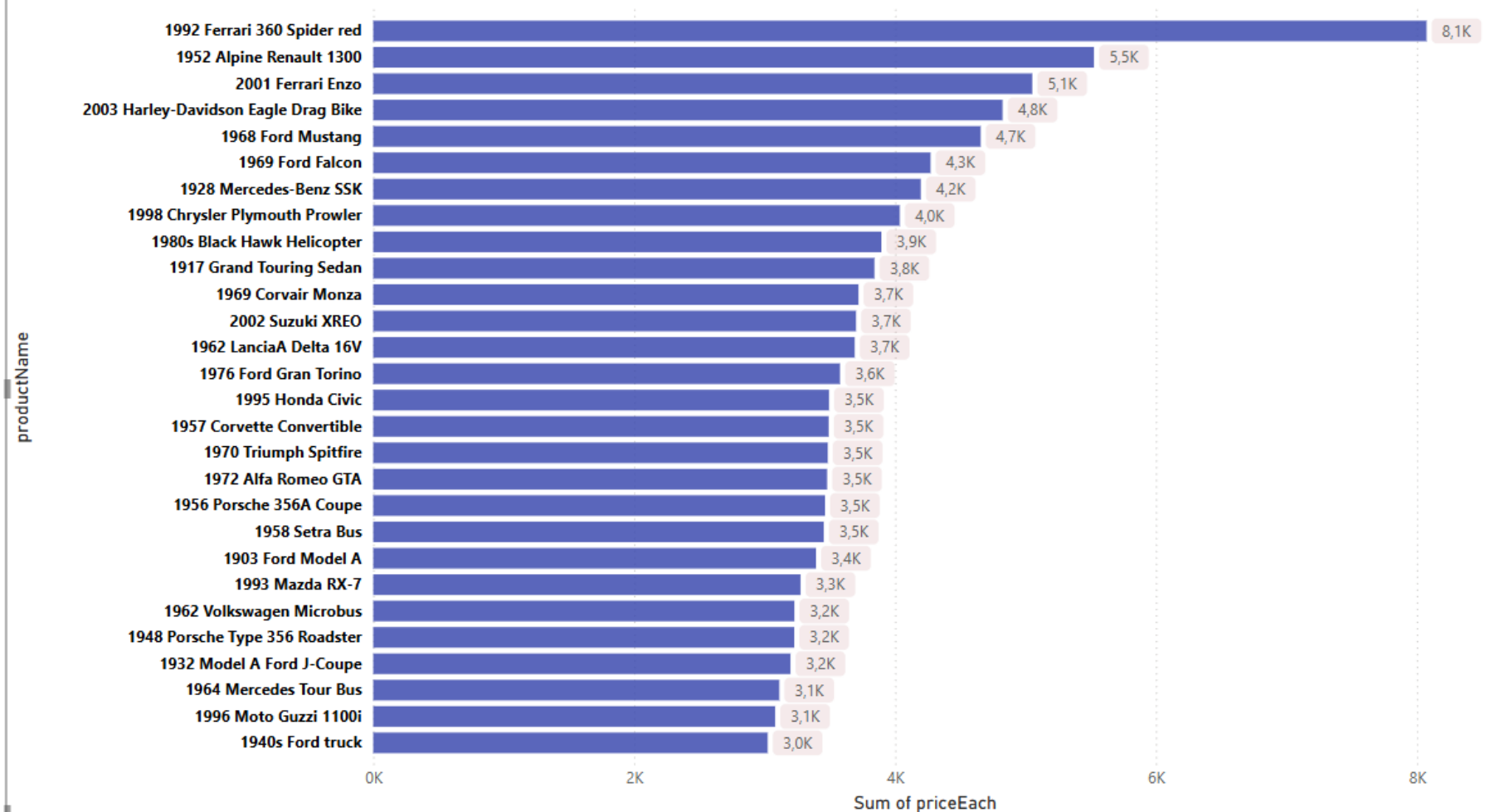
With this query I obtained the information about the total price of each product, this way I can find out which product is the most expensive and which is the least expensive one.

1992 Ferrari 360 Spider red is the product that makes the company the most money.

It might be interesting to check this very product further to get more insights.

Visualization on Power BI in the next slide demonstrate this with the help of bar chart.

Total Price of each Product



With the help of this visualization we can easily detect which product is the most expensive and which is the least expensive.

Data

Search

- mintclassics offices
- mintclassics orderdetails
 - ☐ orderLineNumber
 - ☐ orderNumber
 - ☒ priceEach
 - ☐ productCode
 - ☐ quantityOrdered
- mintclassics orders
- mintclassics payments
- mintclassics productlines
- mintclassics products
 - ☐ buyPrice
 - ☐ MSRP
 - ☐ productCode
 - ☐ productDescription
 - ☐ productLine
 - ☒ productName
 - ☐ productScale
 - ☐ productVendor
 - ☐ quantityInStock
 - ☐ warehouseCode
- mintclassics warehouses

TotalPrice_perProduct

TotalOrder_perDate



```

77
78 -- This query show the prices for the very product which makes the company the most money:
79 • SELECT products.productName AS productName, orderdetails.priceEach
80 FROM mintclassics.orderdetails
81 INNER JOIN mintclassics.products ON products.productCode = orderdetails.productCode
82 WHERE productName = '1992 Ferrari 360 Spider red'
83
84

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
productName	priceEach		
▶ 1992 Ferrari 360 Spider red	165.95		
1992 Ferrari 360 Spider red	160.87		
1992 Ferrari 360 Spider red	169.34		
1992 Ferrari 360 Spider red	137.17		
1992 Ferrari 360 Spider red	149.02		
1992 Ferrari 360 Spider red	169.34		
1992 Ferrari 360 Spider red	143.94		
1992 Ferrari 360 Spider red	143.94		
1992 Ferrari 360 Spider red	167.65		
1992 Ferrari 360 Spider red	140.55		
1992 Ferrari 360 Spider red	154.10		
1992 Ferrari 360 Spider red	165.95		
1992 Ferrari 360 Spider red	150.71		
1992 Ferrari 360 Spider red	147.33		
1992 Ferrari 360 Spider red	165.95		
1992 Ferrari 360 Spider red	137.17		
1992 Ferrari 360 Spider red	150.71		
1992 Ferrari 360 Spider red	157.49		

Investigating this product might give us some useful insights for the business problem we want to solve.

It might be interesting to check:

- Which warehouse stocks this product
- In which country this is ordered; price might change depending on the location.
- When it is ordered mostly
- Prices do not vary a lot. I can also find the average price for this product. See the snapshot below, **the average value for the price of 1992 Ferrari 360 Spider red is ~152.4**, which is plausible when you check the prices manually.

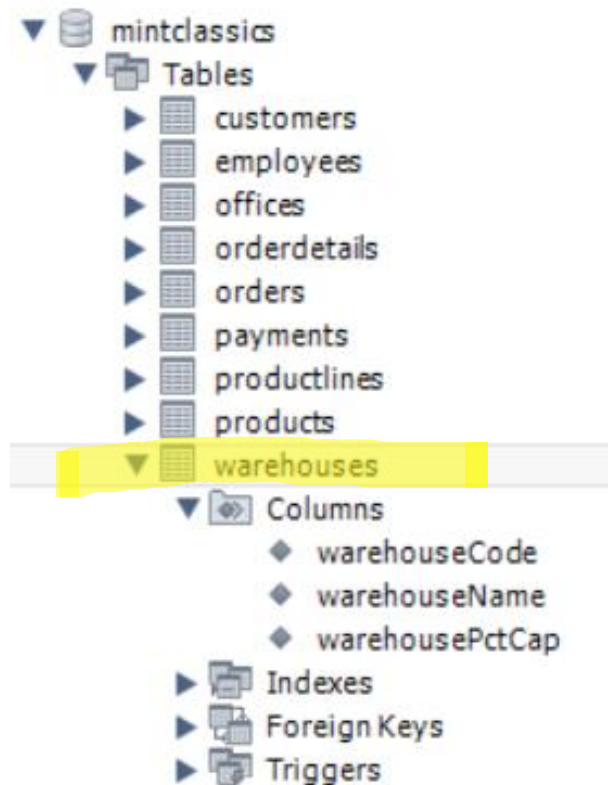
```

83
84 -- This query computes the average price for 1992 Ferrari 360 Spider red:
85 • SELECT products.productName AS productName, AVG(orderdetails.priceEach) AS average_price
86 FROM mintclassics.orderdetails
87 INNER JOIN mintclassics.products ON products.productCode = orderdetails.productCode
88 WHERE productName = '1992 Ferrari 360 Spider red'
89
90

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
productName	average_price		
▶ 1992 Ferrari 360 Spider red	152.342830		

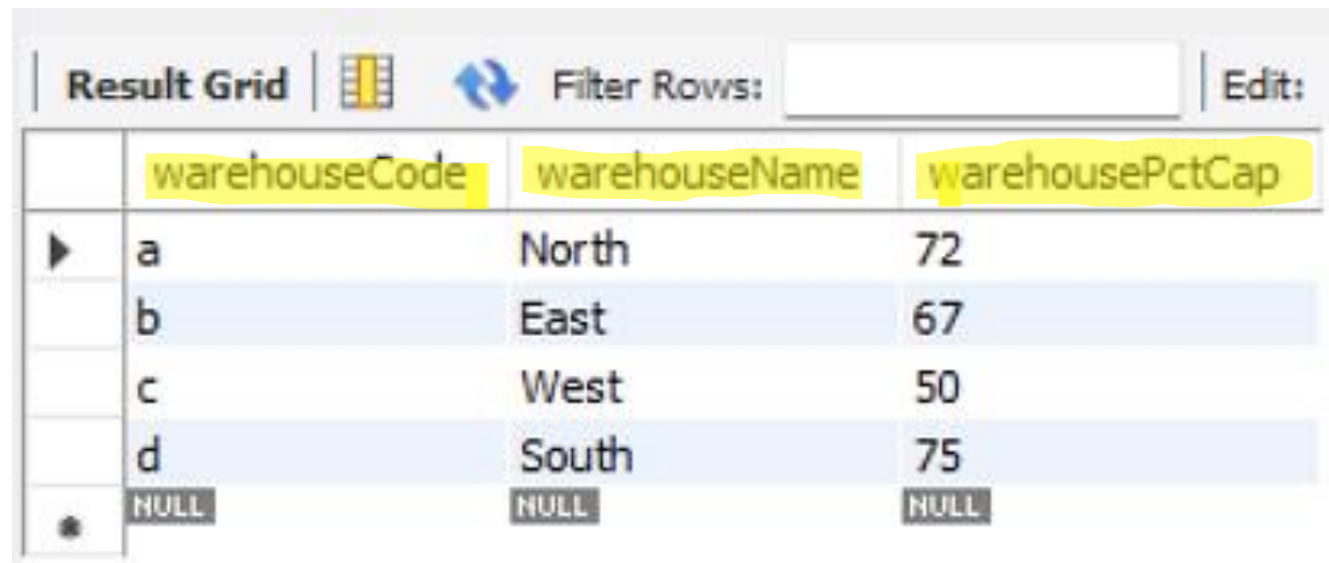
Warehouses - the smallest table of all; containing 3 columns and 5 rows.



The image shows a database schema diagram for a database named 'mintclassics'. Under the 'Tables' section, the 'warehouses' table is highlighted. The 'Columns' section for 'warehouses' lists three columns: 'warehouseCode', 'warehouseName', and 'warehousePctCap'. Other tables listed include 'customers', 'employees', 'offices', 'orderdetails', 'orders', 'payments', 'productlines', and 'products'.

Columns
warehouseCode
warehouseName
warehousePctCap

Each product is stored in one warehouse.



The image shows a 'Result Grid' displaying the data for the 'warehouses' table. The grid has four columns: 'warehouseCode', 'warehouseName', and 'warehousePctCap'. The data is as follows:

	warehouseCode	warehouseName	warehousePctCap
▶	a	North	72
	b	East	67
	c	West	50
	d	South	75
•	NULL	NULL	NULL

Step 3: Questions to be asked

After understanding the data and the business problem, it is natural to ask questions in order to come up with the solution for the business. I will list the questions popped up when I was working, which I believe will be helpful later:

- Who are the regular customers? Where they locate?
 - Who pays more money?
- ⇒ we can think of this as which customer helps our business to earn more money, this way we can also check which products they usually buy (i.e., which is their favourite products). From here we can ask another question: where these products are stored/stocked?
- Who does more sales?
 - Which warehouse has more popular products?
- ⇒ Then we can decide not to close that very warehouse. I don't know yet how I can investigate this, but we will see.
- What **products** are popular, i.e. the products that are ordered frequently?
- ⇒ Which warehouses handling orders of those products?
- ⇒ Are some warehouses handling more **orders** than others?
- We can later investigate in which **country** does the **warehouse** we wan to close is located.
- ⇒ We need to check **offices** dataset for this analysis.
- My **goal** is to analyse their **inventory, sales, and warehouse efficiency** to suggest:
- Whether the inventory/stock should be **moved to other location**?
- ⇒ How this will affect order fulfilment and customer satisfaction.
- Which warehouse(s) could be closed/reorganized?

Key Tables for the Analysis

Look specifically at values for common fields/columns that relate tables. For example, the warehouseCode field is present in both the warehouses table and in the products table.

To analyze **inventory management and warehouse efficiency**, these tables will be especially important:

- **products** → Contains quantityInStock and warehouseCode, which show where inventory is stored.
- **warehouses** → Shows the storage facilities and their capacity usage (warehousePctCap).
- **orders & orderdetails** → Help analyze sales trends and fulfillment times (shippedDate, orderDate, quantityOrdered).
- **customers** → Useful for understanding where demand is coming from (location-based analysis).

Business Recommendations

- Which products should be prioritized for storage?
- Can inventory be consolidated into fewer warehouses?
- Are there opportunities to reduce costs without affecting service time?